Brighter Blue

Core Testing>Basic Testing>Day 1

Instructor Guide

IBM

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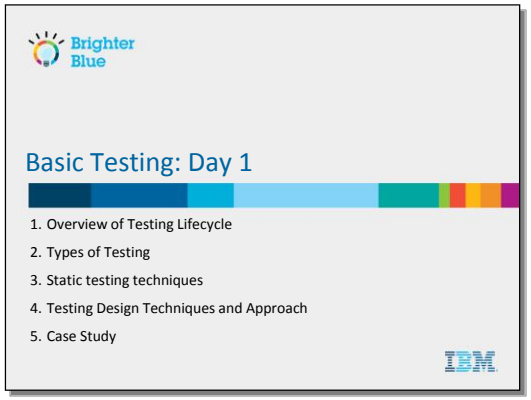
MODULE 3: TYPES AND LEVELS OF TESTING 115

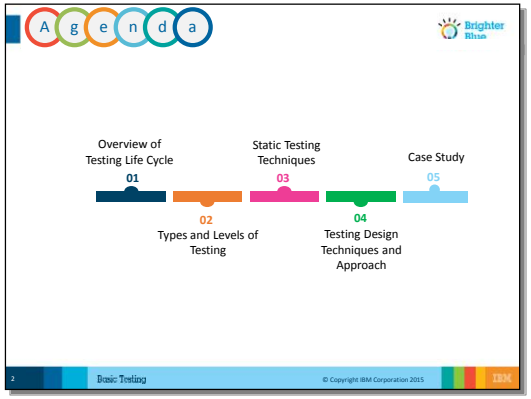
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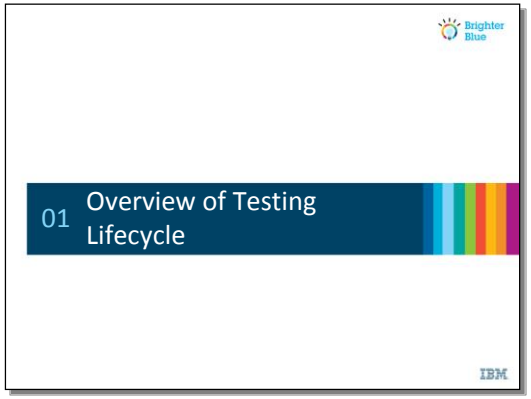
Module 01: Overview of Testing Life Cycle

Slide Content	Instructor Guide	Use this space for your notes
Slide 1 		

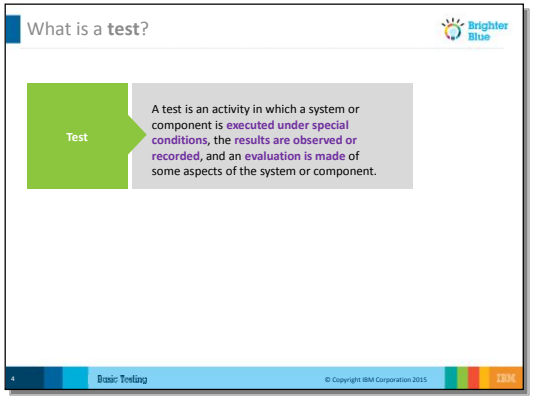
Slide Content	Instructor Guide	Use this space for your notes
Slide 2  	**Purpose:** To go over the course structure and tell the participants the module names **Approximate Duration:** 5 minutes (depending on length of program) **What to cover:** • Distribute the Participant guides and Program attendance sheets • Go over the agenda for each of the modules. Briefly describe the subject areas that will be covered and the activities that will be performed like exercises, case studies and so on. Address the expectations that the participants listed in their introductions. Include information about breaks, lunch and so on. While describing the agenda.	

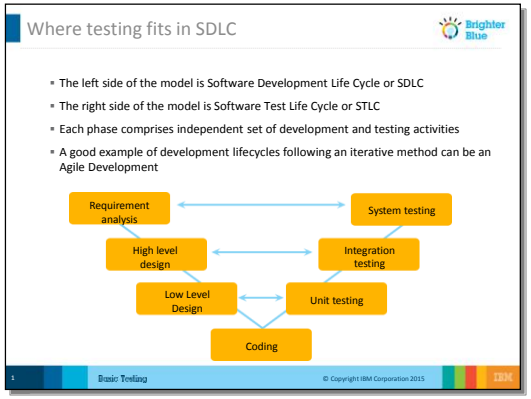
Slide Content	Instructor Guide	Use this space for your notes
	• Inform the participants about the evaluation criteria and end of course assessments. • Inform the participants that you expect highly interactive sessions and that you welcome any questions that they might have. • Ask that all participants turn off their phones or put them in silent mode. Inform them that if they are found to be using their phones or any mobile devices during the sessions that it will be confiscated and returned only at the end of the day. • Inform the participants of the location of exits (in case of fires or any other emergencies). • Go over the format of the Participant guide. Encourage them to take notes as the	

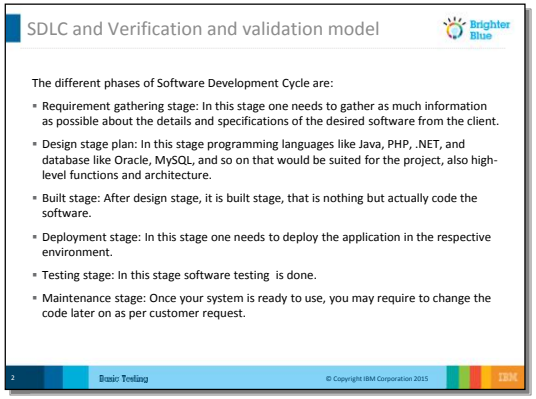
Slide Content	Instructor Guide	Use this space for your notes
	course using the Notes pages provided in the guide.	

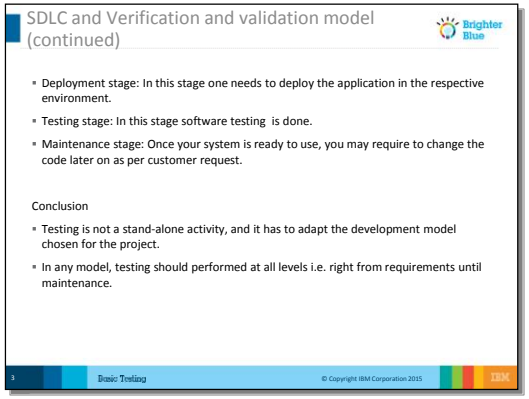
Slide Content	Instructor Guide	Use this space for your notes
Slide 3  **At the end of this module, you should be able to:** • **Give an overview of testing lifecycle** • **Describe the various test strategies** • **Describe the varied facets of test planning** • **Recall the process of Test Case creation** • **Recall the process of test execution** • **List the testing types** • **Describe default workflow**	**Purpose:** To go over the objectives of this module **Approximate Duration:** 3 mins **What to cover:** • Explain the module objectives.	

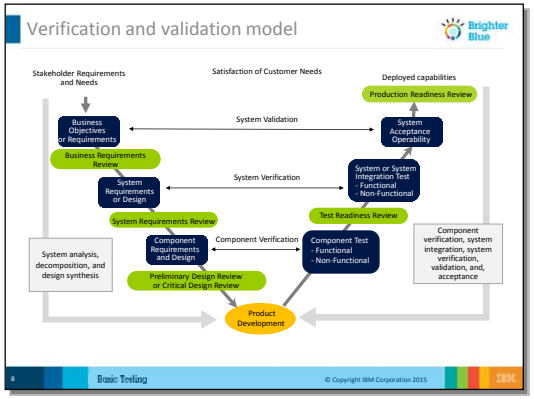
Slide Content	Instructor Guide	Use this space for your notes

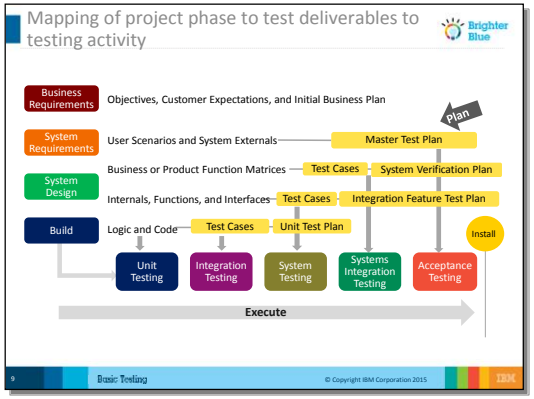
Slide Content	Instructor Guide	Use this space for your notes
Slide 4  	**Purpose:** To define a test. **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants that we begin from the basics. Before getting into test design techniques, let us define a test. ▪ Now refer to the slide and define what a test to the participants is.	

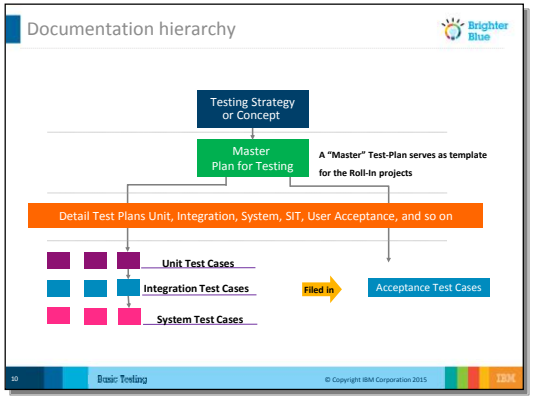
Slide Content	Instructor Guide	Use this space for your notes
Slide 5  	**Purpose:** Describe where testing fits in SDLC **Approximate Duration:** 3 mins **What to cover:** • Explain the phases of SDLC • Illustrate how testing is an integral part of this flow	

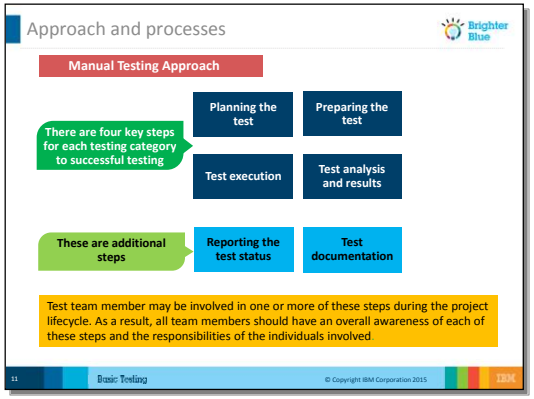
Slide Content	Instructor Guide	Use this space for your notes
Slide 6  	**Purpose:** To go over the SDLC and verification and validation model **Approximate Duration:** 5 mins **What to cover:** • List the different phases of Software Development Life Cycle	

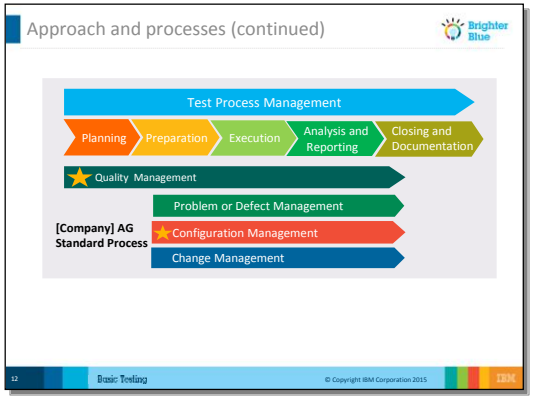
Slide Content	Instructor Guide	Use this space for your notes
Slide 7  	**Purpose:** To go over the SDLC and verification and validation model (continued from last slide) **Approximate Duration:** 3 mins **What to cover:** • List the different phases of SDLC • Explain the importance of testing phase in a SDLC	

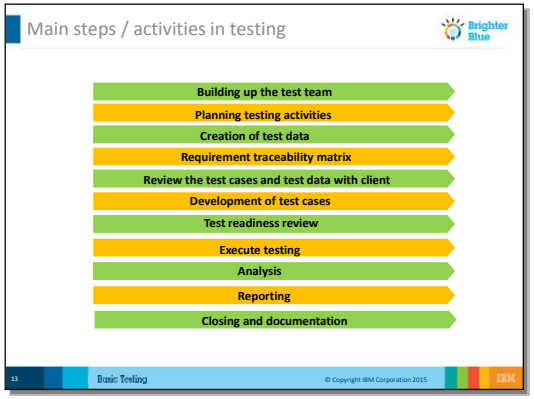
Slide Content	Instructor Guide	Use this space for your notes
Slide 8  	**Purpose:** To illustrate verification and validation model **Approximate Duration:** 5 mins **What to cover:** • Illustrate the verification and validation model • Describe how the product development process is directly affected by the verification process	

Slide Content	Instructor Guide	Use this space for your notes
Slide 9  	**Purpose:** To illustrate how project phase is mapped to test deliverables to testing activity **Approximate Duration:** 2 mins **What to cover:** • Illustrate how project phase is mapped to test deliverables to testing activity. • Describe how each of features are related to the various types and stages of testing	

Slide Content	Instructor Guide	Use this space for your notes
Slide 10  	**Purpose:** To illustrate the documentation hierarchy **Approximate Duration:** 2 mins **What to cover:** • Describe how the documentation hierarchy concept works through a clear illustration	

Slide Content	Instructor Guide	Use this space for your notes
Slide 11  	**Purpose:** To describe the manual testing approach **Approximate Duration:** 5 mins **What to cover:** • Describe the manual testing approach and its key steps involved. • Inform the participants about the additional steps needed as well	

Slide Content	Instructor Guide	Use this space for your notes
Slide 12  	**Purpose:** To describe the test process management process **Approximate Duration:** 5 mins **What to cover:** • Describe how the test process management process works • Illustrate the standard process of quality management	

Slide Content	Instructor Guide	Use this space for your notes
Slide 13  **Building up the test team:** • **Recruit and Onboard team members** • **Assign team members to roles (Business and IT)** • **Coach testing team members (Key User on Application** **Planning testing activities:** • **Analyze business requirements**	**Purpose:** To describe the various phases or steps in the planning phase of testing **Approximate Duration:** 8 mins **What to cover:** • Describe the various phases or steps in testing including planning, preparation, execution, analysis, and reporting and closing and documentation. • Inform the participants about the role that they play in the entire planning phase	

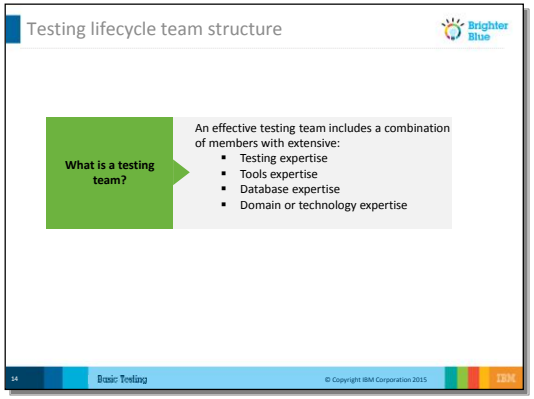
Slide Content	Instructor Guide	Use this space for your notes
• Estimate test effort • Create Project Plan • Prepare test plan or test strategy document for various types of testing • Select test tool • Plan resource and determine roles and responsibilities Creation of test data: • Align with business user and business analyst to define demand for test data • Create test data Requirement traceability matrix: • Prepare the requirement traceability verification matrix or RTVM by mapping each requirement with test cases Review the test cases and test data with client: • Send the initial version of test cases for review		

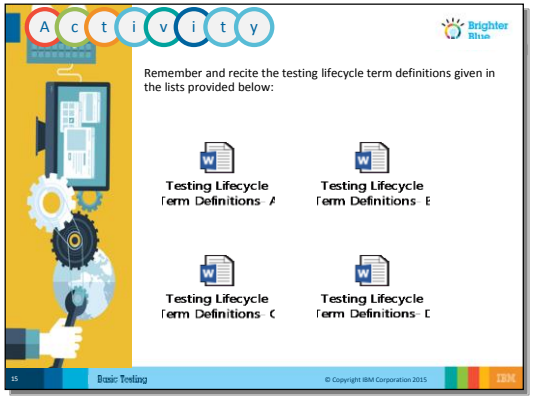
Slide Content	Instructor Guide	Use this space for your notes
• Close all the review comments by updating the test cases • Get signed off test cases and test data • Archive approved test cases and test data Development of test cases: • Identify test cases • Evaluate test cases regarding risk categories • Define test entry and exit criteria • Describe test cases • Peer review of test cases Test Readiness Review • Verify test environment is setup and stable for execution		

Slide Content	Instructor Guide	Use this space for your notes
• Verify unit test results pass the entry criteria of testing phase • Client-approved test cases and test data • Perform Smoke test Execute Testing • Execute tests as per plan • Document test results, and log defects for failed cases • Update test plans or test cases, if necessary • Retest the defect fixes and tack the defects to closure • Do regression testing of application Analysis • Analyze the Test results		

Slide Content	Instructor Guide	Use this space for your notes
• Prepare Test Metrics:Time, Test coverage, Cost, Software Quality, and Critical Business Objectives • Perform Causal analysis– prepare Causal Analysis Report Reporting • Publish Test Metrics • Publish Test reports: Test Summary report, Causal Analysis Report, and Audit Report Closing and Documentation • Analyze the Test results • Prepare Test Metrics- Time, Test coverage, Cost, Software Quality, and Critical Business Objectives • Perform Causal analysis– Prepare Causal Analysis Report		

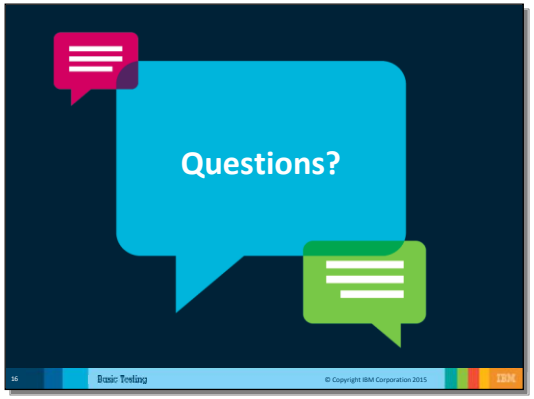
Slide Content	Instructor Guide	Use this space for your notes

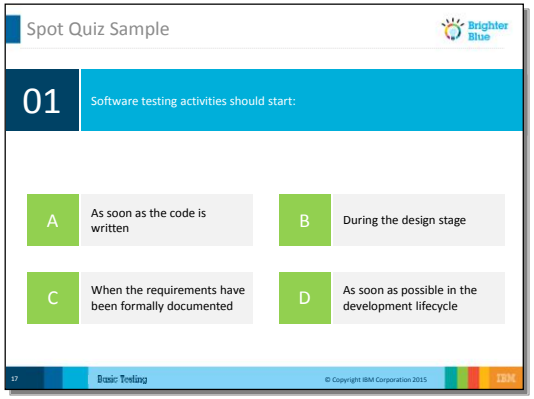
Slide Content	Instructor Guide	Use this space for your notes
Slide 14  	**Purpose:** To define a testing team **Approximate Duration:** 2 mins **What to cover:** • Describe what is a typical testing team and its structure	

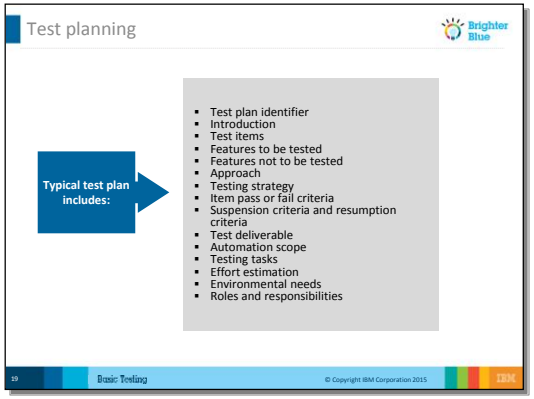
Slide Content	Instructor Guide	Use this space for your notes
Slide 15  	**Approximate Duration:** 16 mins **Additional Materials or Pre-session prep tasks**: • Divide the participants into four groups. • Distribute one list each to each group. • For the embedded documents, ask the participants to refer to the **Supporting Documents_Day 1** folder. **What to cover:** • Ask the participants to form four groups • Distribute one list each to each group • You can ask the groups to read the definitions given within two minutes	

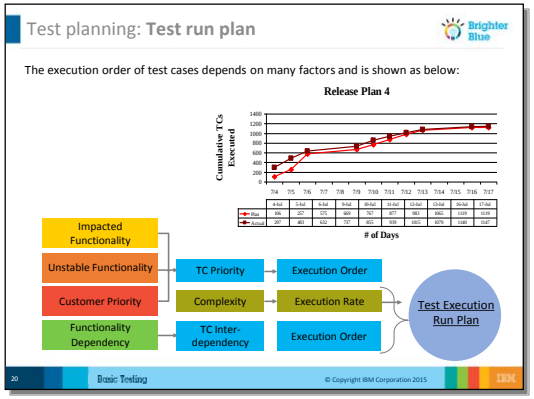
Slide Content	Instructor Guide	Use this space for your notes
	• Discuss the list from each group by asking them to recall as much as possible, the definitions read within two minutes • Interchange the lists among groups and repeat the activity until all the teams have covered all the lists **Purpose:** To see if the participants can remember all the testing lifecycle term definitions **Instructions for the participants:** • Ask the participants to form four groups • Distribute one list each to each group • You can ask the groups to read the definitions given within two minutes	

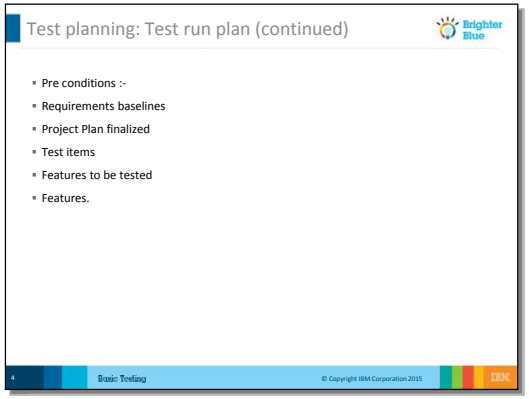
Slide Content	Instructor Guide	Use this space for your notes
	• Discuss the list from each group by asking them to recall as much as possible, the definitions read within two minutes • Interchange the lists among groups and repeat the activity until all the teams have covered all the lists **Debrief:** • Ask the participants about their experience of this activity and if the exercise has made some basic concepts clear or not **Note to SME:** Please validate above instruction.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 16 	**Approximate Duration:** 2 mins **What to cover:** • Ask the participants if they have any questions. • Include any questions that will be addressed later in the course as parking lot items. • Use this activity to recap the key takeaways from this module	

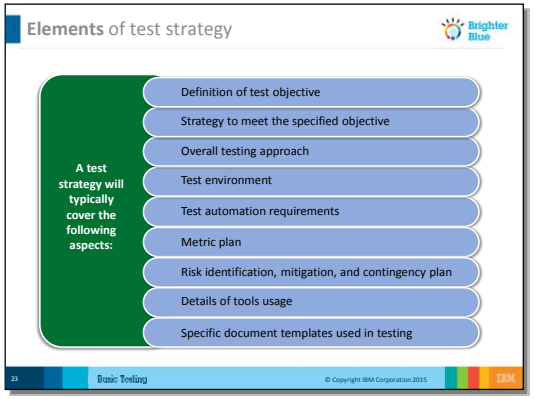
Slide Content	Instructor Guide	Use this space for your notes
Slide 17  **The correct answer is A. As soon as possible in the development lifecycle**	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** • Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. • The correct answer is **A. As soon as possible in the development lifecycle**	

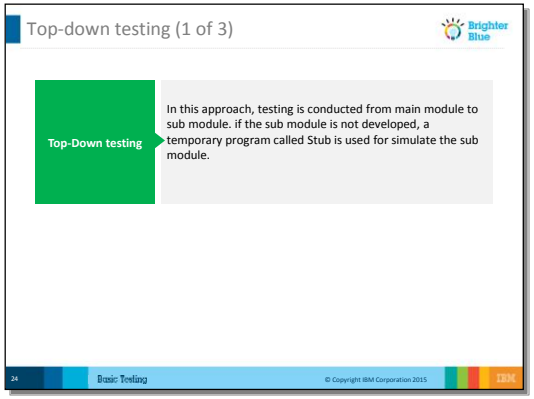
Slide Content	Instructor Guide	Use this space for your notes
Slide 18  	**Purpose:** To explain the test planning process. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 3 mins **What to cover:** • List what a typical test plan includes	

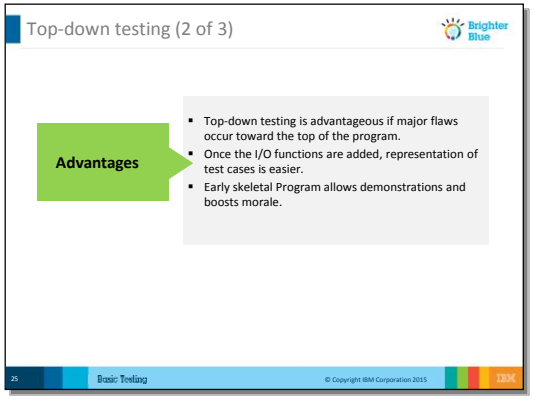
Slide Content	Instructor Guide	Use this space for your notes
Slide 19  	**Purpose:** To describe a test run plan. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 3 mins **What to cover:** • Describe how the execution order of test cases depends on multiple factors	

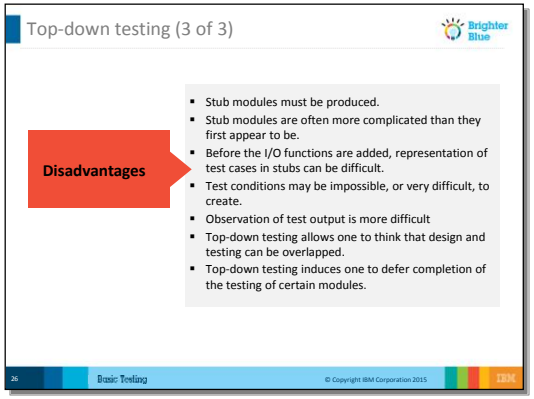
Slide Content	Instructor Guide	Use this space for your notes
Slide 20  	**Purpose:** To describe the preconditions for a test run plan. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** • Describe the preconditions for a test run plan	

Slide Content	Instructor Guide	Use this space for your notes
Slide 21  	**Purpose:** To describe the features of test strategy. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** • Give the participants a brief description regarding the features of test strategy	

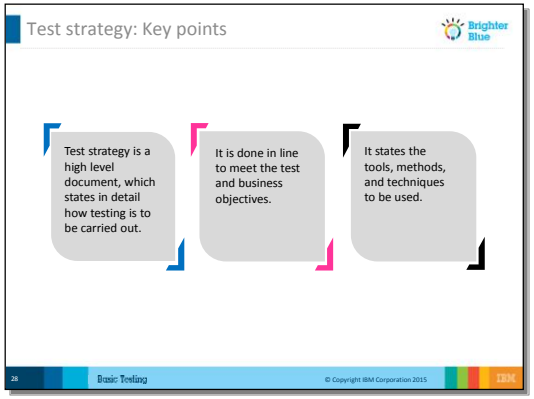
Slide Content	Instructor Guide	Use this space for your notes
Slide 22  	**Purpose:** To describe the elements of test strategy. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** • Describe the aspects that are typically covered by a test strategy	

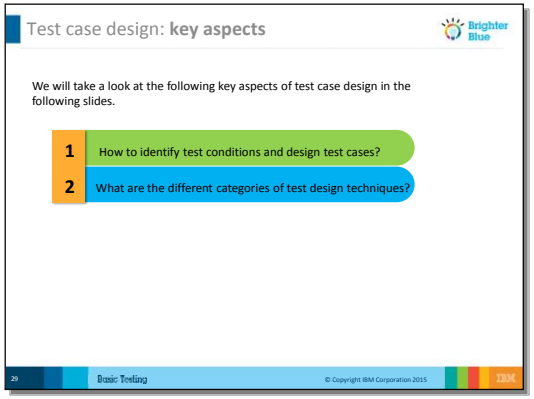
Slide Content	Instructor Guide	Use this space for your notes
Slide 23  	**Purpose:** To understand Top-down testing **Approximate Duration:** 8 mins **What to cover:** ▪ Provide definition and purpose of Top-down testing. ▪ Enumerate and discuss the advantages and disadvantages of Top-down testing. ▪ Cite relevant examples.	

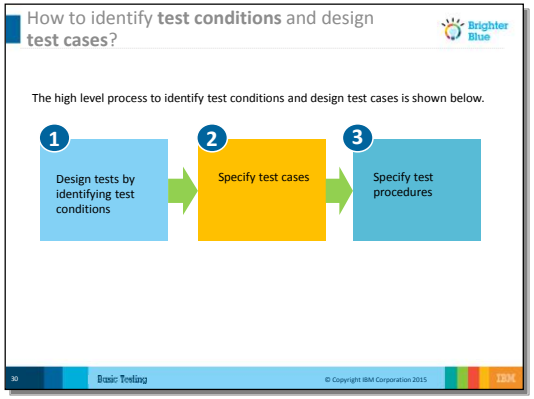
Slide Content	Instructor Guide	Use this space for your notes
Slide 24  	**Purpose:** To recall an example of testing strategy **Approximate Duration:** 2 mins **What to cover:** • Describe some of the facets of testing strategy including User Acceptance Testing, Single Testing Repository and Testing Sign-Off	

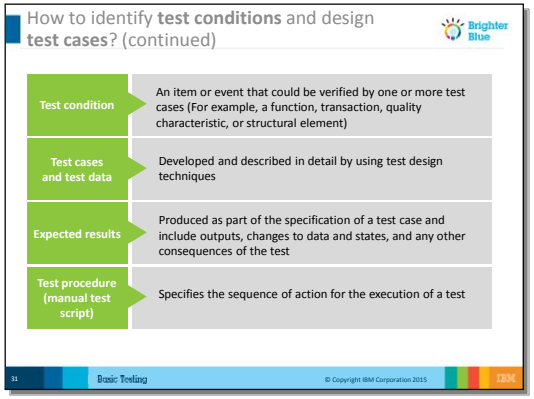
Slide Content	Instructor Guide	Use this space for your notes
Slide 25  Example: Suppose Sales Order Printing program uses another unit which calculates Sales discounts by some complex calculations. Then call to this unit will be replaced by a Stub, which will simply return fix discount data.	**Purpose:** To describe the key points of test strategy **Approximate Duration:** 2 mins **What to cover:** • Describe what is test strategy and the key points associated with a typical test strategy.	

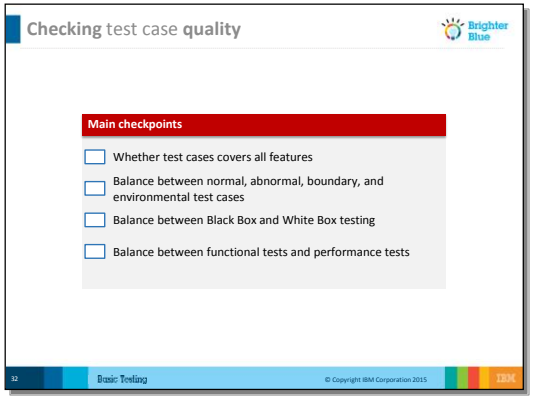
Slide Content	Instructor Guide	Use this space for your notes
Slide 26  	**Purpose:** To point out the main aspects of the test design techniques. **Approximate Duration:** 2 mins **What to cover:** ▪ State the two key aspects of the test design techniques. ▪ Tell the participants in the following slides, we will take up the each of the aspects in details.	

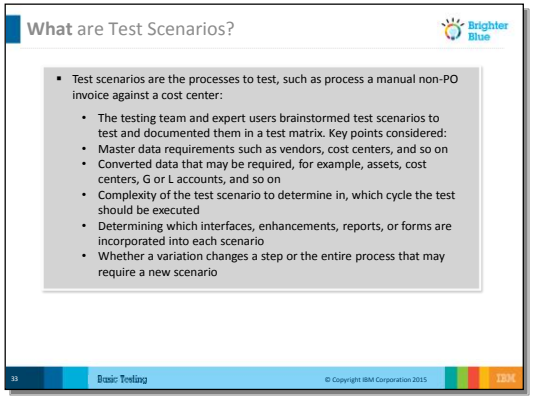
Slide Content	Instructor Guide	Use this space for your notes
Slide 27  	**Purpose:** To list the steps for identifying test conditions and designing test cases. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants the three steps used to identify test conditions and designing test cases. ▪ Once you have completed, ask the participants to move to the next slide where we will further discuss on the process.	

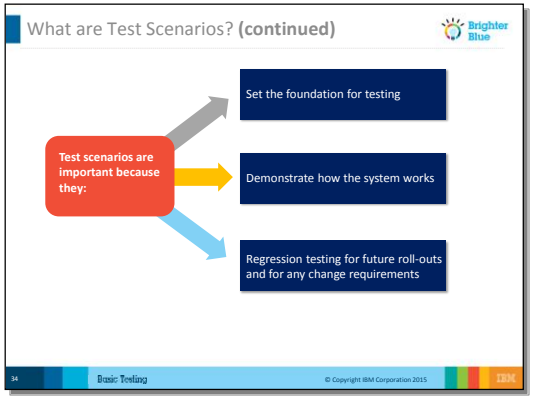
Slide Content	Instructor Guide	Use this space for your notes
Slide 28  	**Purpose:** To explain important terms related to identify test conditions and design test cases. **Approximate Duration:** 5 mins **What to cover:** ▪ Tell the participants that in this slide we discuss the important terms that are associated with identifying the test conditions and designing the test cases.	

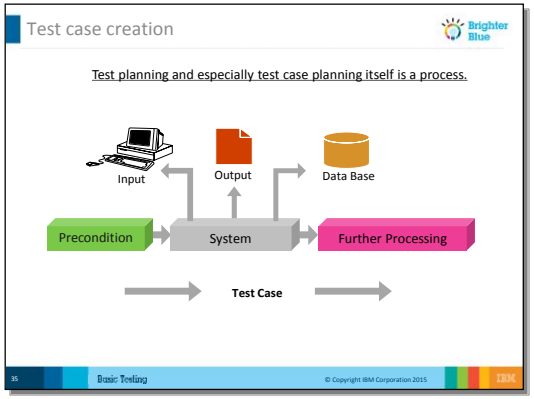
Slide Content	Instructor Guide	Use this space for your notes
Slide 29  	**Purpose:** To list the main checkpoints for checking test case quality. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** Point out to the participants the main checklists that they need to keep in mind for checking the quality of test cases.	

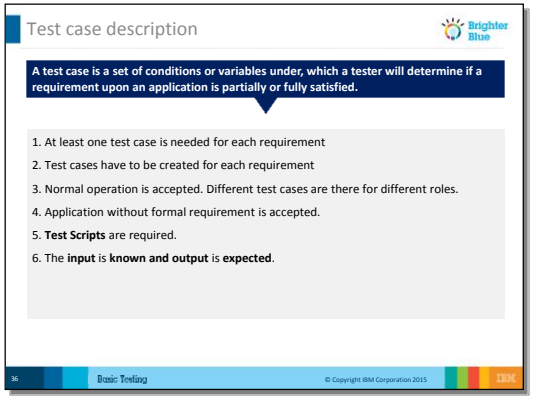
Slide Content	Instructor Guide	Use this space for your notes
Slide 30 	**Purpose:** To describe what are test scenarios. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** • Describe what test scenarios are. • Inform the participants about the features of test scenarios	

Slide Content	Instructor Guide	Use this space for your notes
Slide 31  	**Purpose:** To describe what are test scenarios. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** • Describe why test scenarios are important	

Slide Content	Instructor Guide	Use this space for your notes
Slide 32  	**Purpose:** To illustrate a test case creation process **Approximate Duration:** 5 mins **What to cover:** • Describe test case planning and creation process through illustrations	

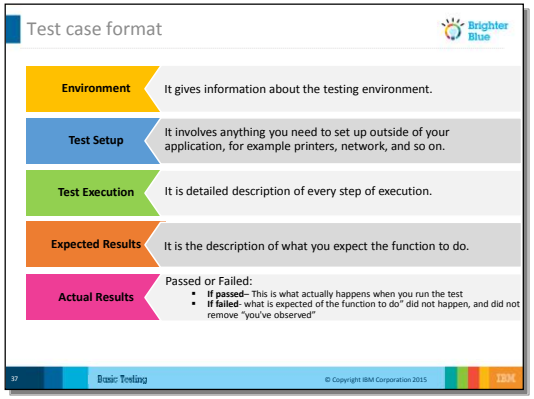
Slide Content	Instructor Guide	Use this space for your notes
Slide 33  	**Purpose:** To describe what is test case **Approximate Duration:** 4 mins **What to cover:** • Describe what is test case • Describe the factors and features associated with a typical test case • Cover the points detailed in the notes section of the slide	

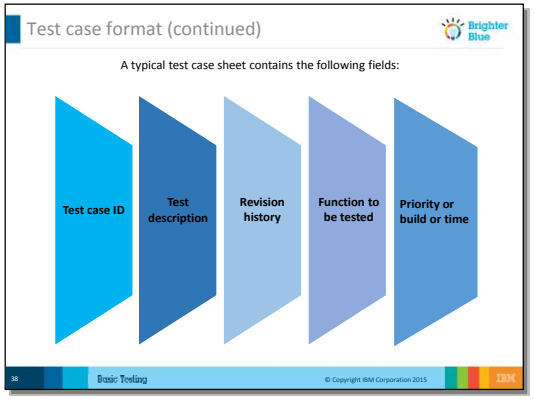
Slide Content	Instructor Guide	Use this space for your notes
Slide 34  	**Purpose:** To describe the format of a test case **Approximate Duration:** 2 mins **What to cover:** • Describe the factors constituting a test case	

Slide Content	Instructor Guide	Use this space for your notes
Slide 35  At least one test case for each requirement - It may take many test cases to determine that a requirement is fully satisfied. In order to fully test that all the requirements of an application are met, there must be at least one test case for each requirement unless a requirement has sub requirements. Create test cases for each requirement - In that situation, each sub requirement must have at least one test case, or we should be creating at least two	**Purpose:** To describe the format of a test case. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** • Describe the fields that a typical test case contains	

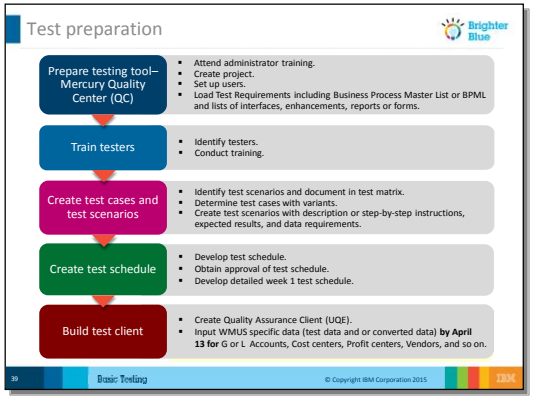
Slide Content	Instructor Guide	Use this space for your notes
test cases for each requirement. One of them should perform positive testing of requirement and other should perform negative testing. Create Test cases for different roles (for example, manager view, user view, group, and so on. Accepted normal operation - If the application is created without formal requirements, then test cases are written based on the accepted normal operation of programs of a similar class. Test cases for different roles - Create Test cases for different roles (for example, manager view, user view, group, and so on. Application without formal requirement - If the application is created without formal requirements, then test cases are written based on the accepted normal operation of programs of a similar class. Test Scripts - A test script is the executable form of a test.		

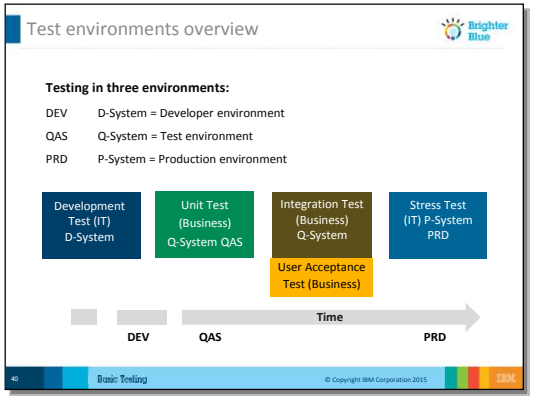
Slide Content	Instructor Guide	Use this space for your notes
Known input and expected output - What characterizes a formal, written test case is that there is a known input and an expected output, which is worked out before the test is executed. The known input should test a precondition and the expected output should test a post condition.		

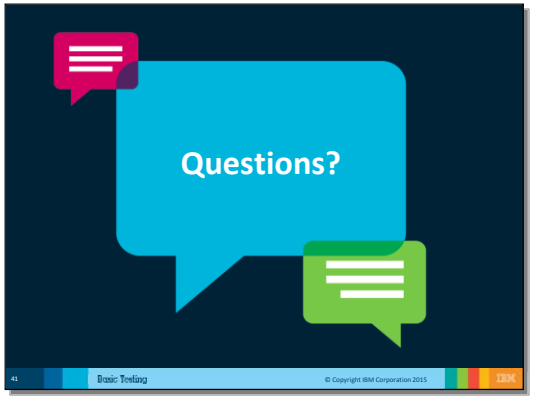
Slide Content	Instructor Guide	Use this space for your notes
Slide 36  	**Notes** **Purpose:** To describe the test preparation process. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** • Describe the steps in the test preparation process. • Give an elaborate description of the activities that must be done in each of those steps	

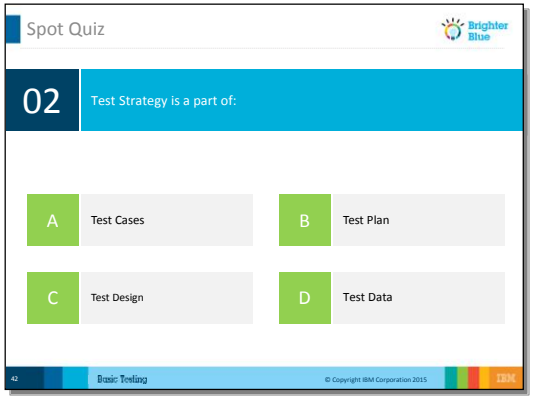
Slide Content	Instructor Guide	Use this space for your notes
Slide 37  A typical test case sheet contains the following fields … Test case ID: It is a unique number given to test case in order to be identified. Test description: This is the description of test case you are going to test. Revision history: Each test case has to have its revision history in order to know when and by whom it is created or modified.	**Purpose:** To recall an overview of test environments. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** • Recall an overview of test environments through an illustration to make the participants understand in a lucid manner	

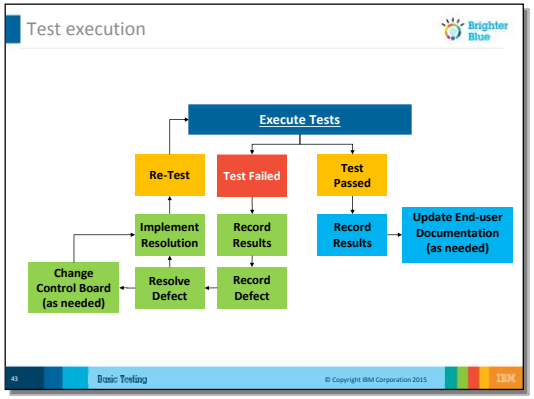
Slide Content	Instructor Guide	Use this space for your notes
Function to be tested: This is the name of function to be tested. Priority or build or time: The priority of the test case, build version of the test case and the minimum time-to-execute the test case are also mentioned.		

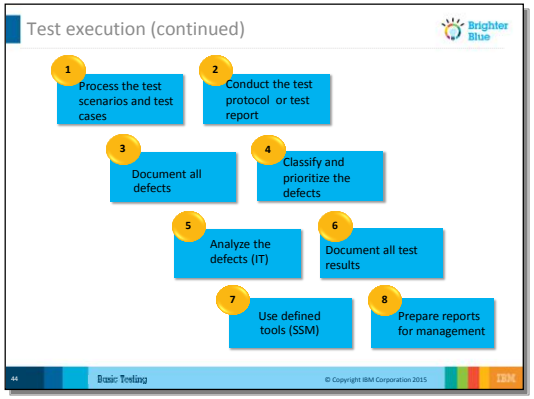
Slide Content	Instructor Guide	Use this space for your notes
Slide 38 	**Approximate Duration:** 2 mins **What to cover:** • Ask the participants if they have any questions. • Include any questions that will be addressed later in the course as parking lot items. • Use this activity to recap the key takeaways from this module	

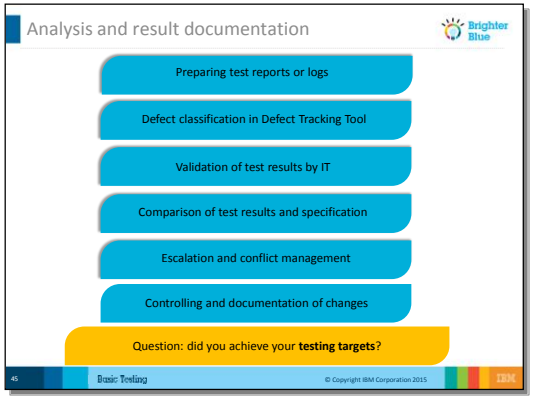
Slide Content	Instructor Guide	Use this space for your notes
Slide 39  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** • Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. • The correct answer is **B**. Test plan	

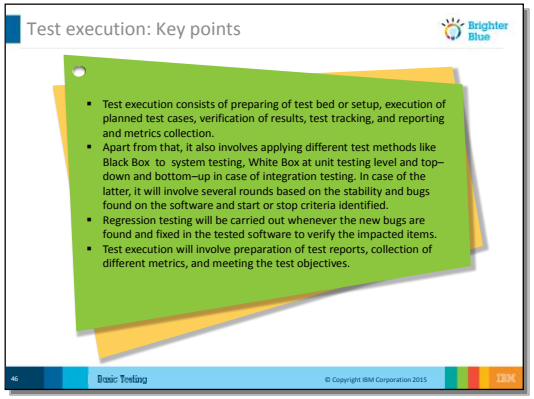
Slide Content	Instructor Guide	Use this space for your notes
Slide 40 	**Purpose:** To illustrate the test execution process **Approximate Duration:** 3 mins **What to cover:** • Describe the test execution process in an illustrative manner	

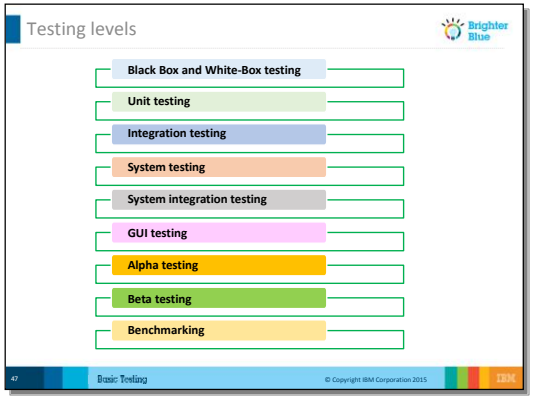
Slide Content	Instructor Guide	Use this space for your notes
Slide 41  **The correct answer is B. Test plan**	**Purpose:** To illustrate the test execution process **Approximate Duration:** 2 mins **What to cover:** • Describe the exact steps involved in the test execution process	

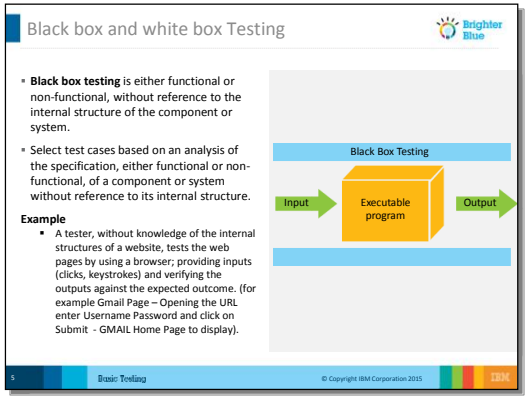
Slide Content	Instructor Guide	Use this space for your notes
Slide 42  	**Purpose:** To describe some of the key steps in analysis and result documentation. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** • Describe the key steps in the process of analysis and result documentation.	

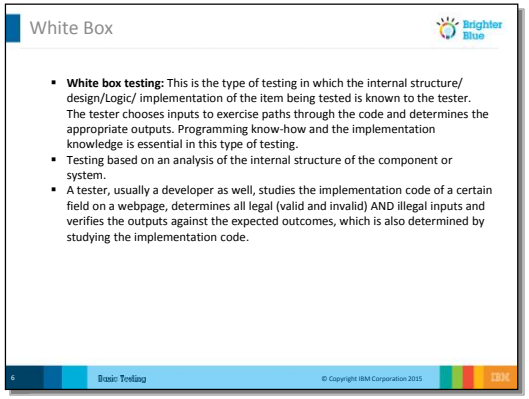
Slide Content	Instructor Guide	Use this space for your notes
Slide 43  	**Purpose:** To describe some key points in test execution. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** • Describe some key points and features of test execution	

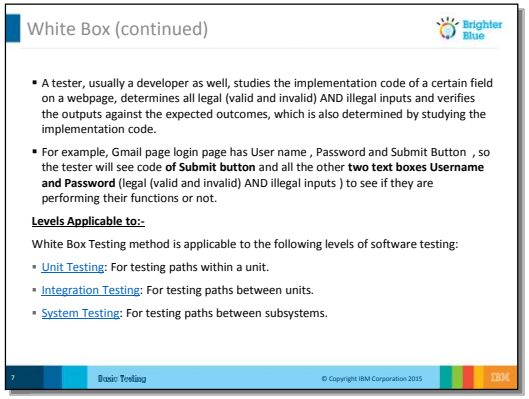
Slide Content	Instructor Guide	Use this space for your notes
Slide 44  	**Purpose:** To name the various testing levels **Approximate Duration:** 2 mins **What to cover:** • Name the various testing levels • Inform the participants about their relevance	

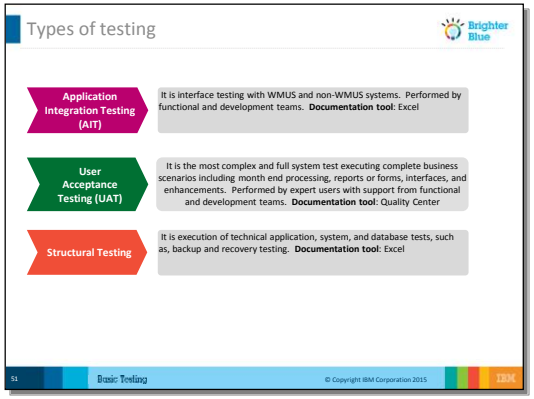
Slide Content	Instructor Guide	Use this space for your notes
Slide 45  	**Purpose:** To define black box and white box testing. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 3 mins **What to cover:** • Explain the process of Black Box testing • Provide an example of when this is applicable	

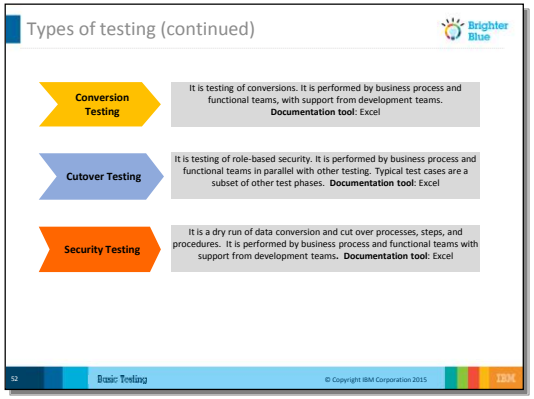
Slide Content	Instructor Guide	Use this space for your notes
Slide 46  	**Purpose:** To define black box and white box testing **Approximate Duration:** 3 mins **What to cover:** • Explain the process of white box testing	

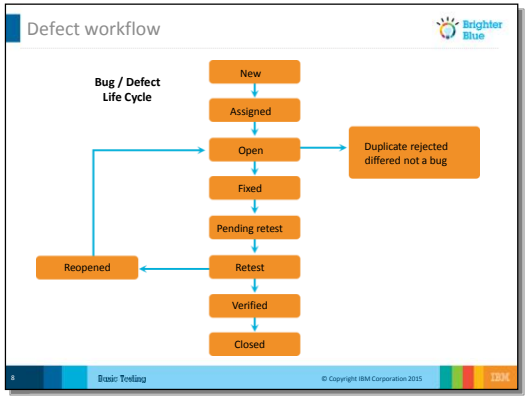
Slide Content	Instructor Guide	Use this space for your notes
Slide 47  	**Purpose:** To define black box and white box testing **Approximate Duration:** 3 mins **What to cover:** • Explain the process of white box testing • Provide an example of when this is applicable	

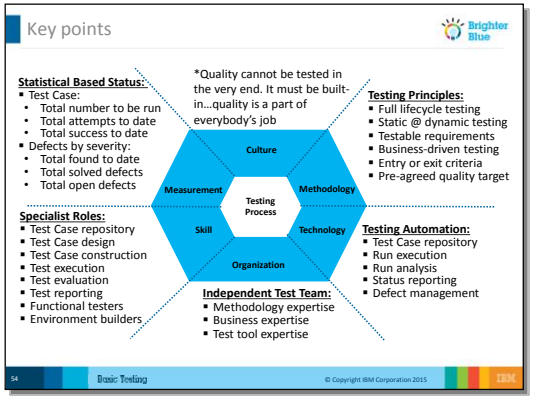
Slide Content	Instructor Guide	Use this space for your notes
Slide 48  	**Purpose:** To explain the types of testing **Approximate Duration:** 3 mins **What to cover:** • Define the types of testing - AIT, UAT, and Structural Testing • Inform the participants about their relevance	

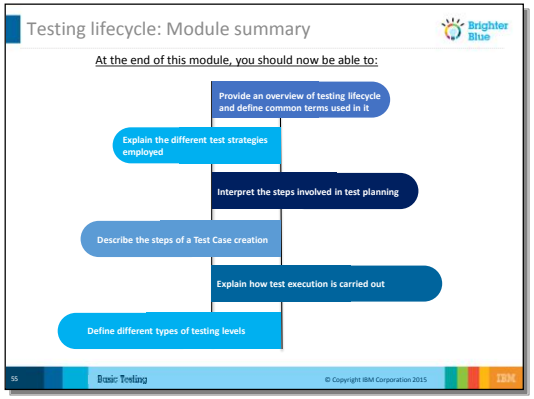
Slide Content	Instructor Guide	Use this space for your notes
Slide 49 	**Purpose:** To explain the types of testing (continued from last slide) **Approximate Duration:** 3 mins **What to cover:** • Define the types of testing – Conversion, Cutover, and Security Testing • Inform the participants about their relevance	

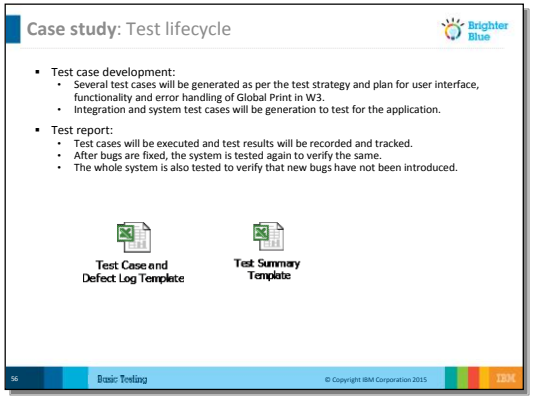
Slide Content	Instructor Guide	Use this space for your notes
Slide 50  **AIT, UAT, Structural in one** **Conversion, cutover, security in second ----- phase wise.** **It is execution of technical application, system, and database tests, such as, backup and recovery testing.** **Documentation tool: Excel**	**Purpose:** To explain the defect workflow **Approximate Duration:** 5 mins **What to cover:** • Describe a typical bug or defect lifecycle	

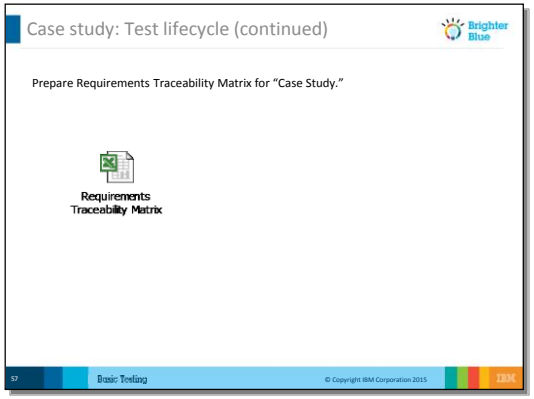
Slide Content	Instructor Guide	Use this space for your notes
Slide 51  **AIT, UAT, Structural in one** **Conversion, cutover, security in second ----- phase wise.**	**Purpose:** To describe the key points of a testing process **Approximate Duration:** 8 mins **What to cover:** • Describe the essential components of a testing process. • Inform the participants about typical examples of the components	

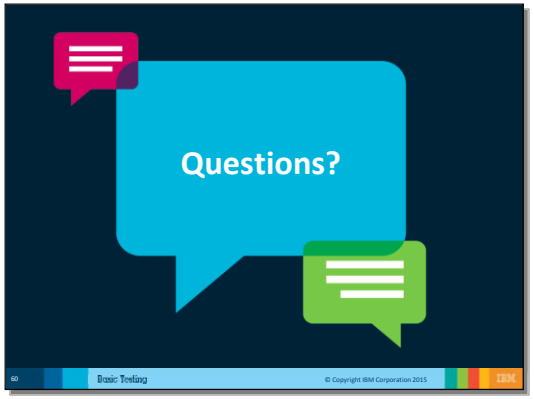
Slide Content	Instructor Guide	Use this space for your notes
Slide 52  **On top is a workflow of a typical defect. The highlighted boxes outline the normal defect workflow, which equals defect status in Quality Centre. The yellow boxes reflect the exception defect workflow. The dashed boxes identify the state of the defect (open or closed).**	**Purpose:** To summarize the take away points of this module **Approximate Duration:** 2 mins **What to cover:** • Inform the participants about the take away points of this module by summarizing the pain points	

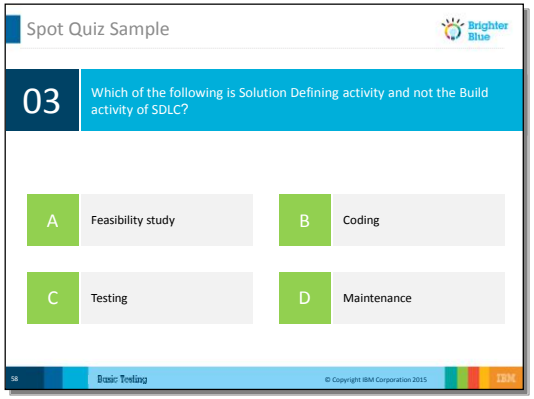
Slide Content	Instructor Guide	Use this space for your notes
Slide 53 	**Purpose:** To understand the test case development and reports in a test lifecycle **Approximate Duration:** 3 mins **What to cover:** • Explain the test case and defect log template and test summary template	

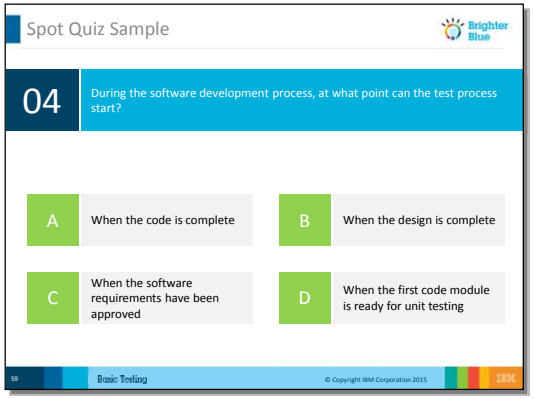
Slide Content	Instructor Guide	Use this space for your notes
Slide 54  	**Purpose:** To understand how to prepare requirements traceability matrix in a test lifecycle **Approximate Duration:** 3 mins **What to cover:** • Explain how to prepare requirements traceability matrix for “Case Study.”	

Slide Content	Instructor Guide	Use this space for your notes
Slide 55  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** • First, ask the participants to refer to the Supporting Documents_Day 1 folder for the embedded files**.** • Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. • The correct answer is **A. Feasibility study**	

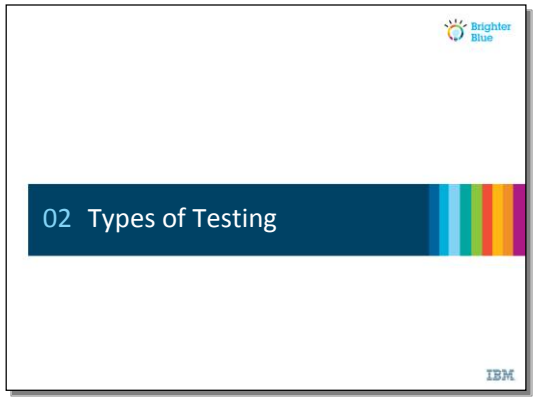
Slide Content	Instructor Guide	Use this space for your notes
Slide 56  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** • First, ask the participants to refer to the Supporting Documents_Day 1 folder for the embedded files. • Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. • The correct answer is **C. When the software requirements have been approved**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 57 	**Purpose:** To learn about the different types of Functional testing **Approximate Duration:** 5 mins **What to cover:** ▪ Enumerate the different types of Functional testing.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 58  **The correct answer is A. Feasibility study**	**Approximate Duration:** 2 mins **What to cover:** • Ask the participants if they have any questions. • Include any questions that will be addressed later in the course as parking lot items. • Use this activity to recap the key takeaways from this module	

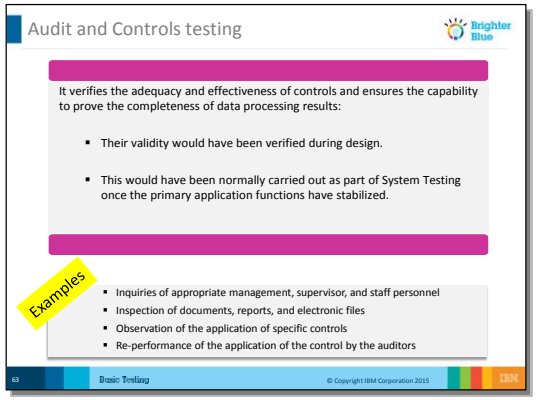
Slide Content	Instructor Guide	Use this space for your notes
Slide 59  **The correct answer is C. When the software requirements have been approved**	**Purpose:** To cover the objectives of this module **Approximate Duration:** 2 mins **What to cover:** ▪ Description, types, and examples of Functional testing ▪ Description, types, and examples of Structural testing	

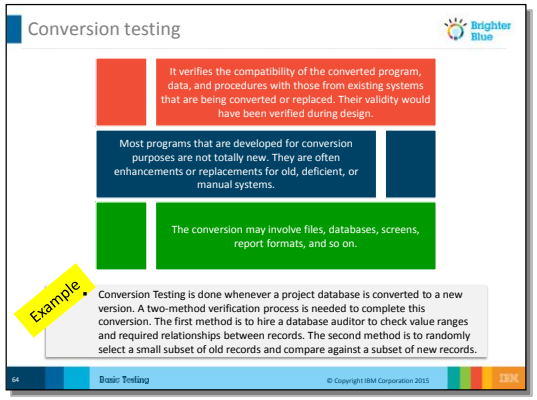
Module 2: Types and Levels of Testing

Slide Content	Instructor Guide	Use this space for your notes
Slide 60  **At the end of this module, you should be able to:** ▪ **Describe the different types of Functional testing** ▪ **Outline the different types of Structural testing** ▪ **Use examples of each type for better explanation and clarity of understanding**	**Purpose:** To define Audit and Controls testing **Approximate Duration:** 3 mins **What to cover:** ▪ Provide definition, purpose, and examples of Audit and Controls testing. ▪ List the examples (in the notes section of the slide) on a flipchart in the classroom.	

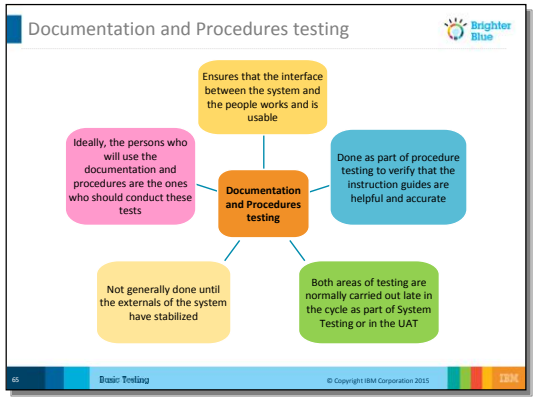
Slide Content	Instructor Guide	Use this space for your notes

Slide Content	Instructor Guide	Use this space for your notes
Slide 61  	**Purpose:** To define Conversion testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Conversion testing. ▪ Share the example provided in the notes section of the slide.	

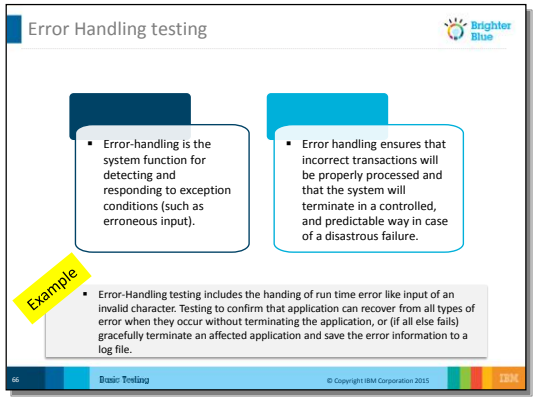
Slide Content	Instructor Guide	Use this space for your notes
Slide 62  	**Purpose:** To define Documentation and Procedures testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Documentation and Procedures testing. ▪ Share examples of Documentation and Procedures testing provided in the notes section of the slide.	

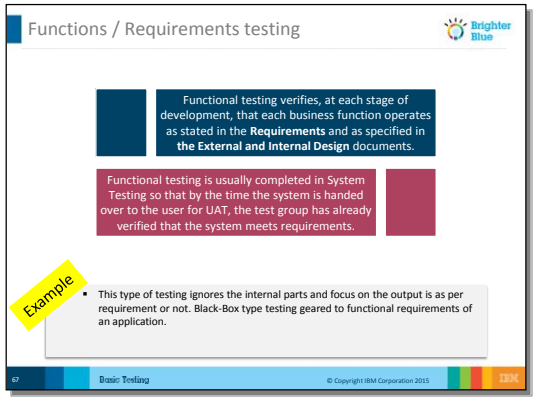
Slide Content	Instructor Guide	Use this space for your notes
Slide 63  **Example: Conversion Testing is done whenever a project database is converted to a new version. The First verification method involves the use of a database auditor that must be built by the development group. When ran against the converted database, the database auditor will check value ranges within a record and the required relationships between records. The second verification method involves the random selection of a small subset of old records and then a direct**	**Purpose:** To define Error Handling testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Error Handling testing. ▪ Use examples to give a better idea to the participants.	

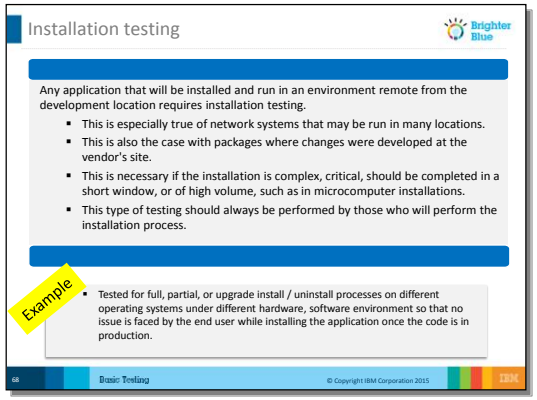
Slide Content	Instructor Guide	Use this space for your notes
comparison against a corresponding subset of the new records.		

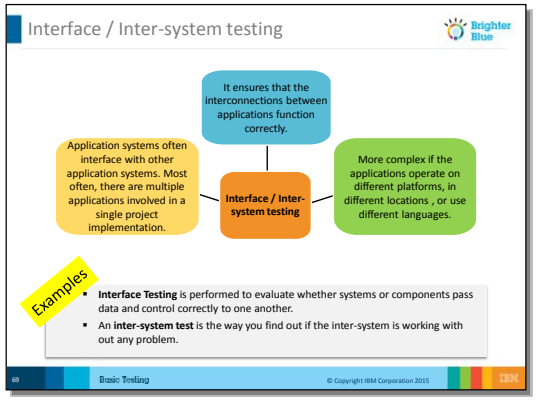
Slide Content	Instructor Guide	Use this space for your notes
Slide 64  **Example: Documentation is as important to a product's success as the product itself. If the documentation is poor, non-existent, or wrong, it reflects on the quality of the product. Document testing includes any type of document verification—test case specification, test incident report, test log, test plan, test procedure, and test report.**	**Purpose:** To define Functions / Requirements testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Functions / Requirements testing. ▪Transfer the knowledge of practical applicability by sharing an example.	

Slide Content	Instructor Guide	Use this space for your notes

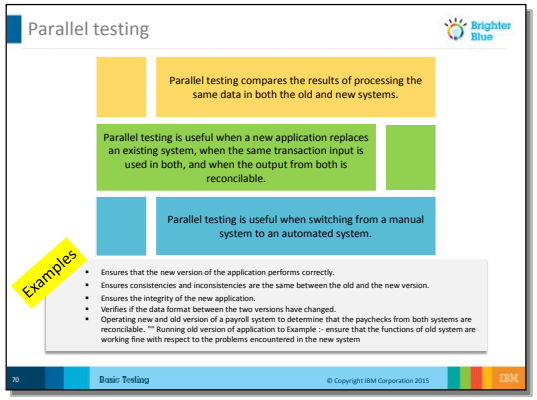
Slide Content	Instructor Guide	Use this space for your notes
Slide 65  	**Purpose:** To define Installation testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Installation testing. ▪ Use examples of Installation testing to give the participants a better understanding.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 66  	**Purpose:** To define Interface / Inter-system testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Interface or Inter-system testing. ▪ Use examples to give the participants a better understanding of practical scenarios.	

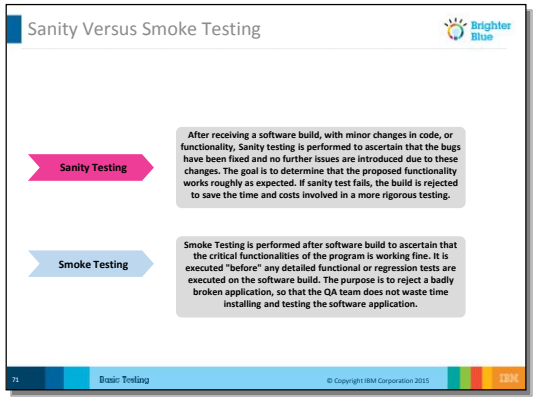
Slide Content	Instructor Guide	Use this space for your notes
Slide 67  	**Purpose:** To define Parallel testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of parallel testing. ▪ Use examples to give the participants a better understanding.	

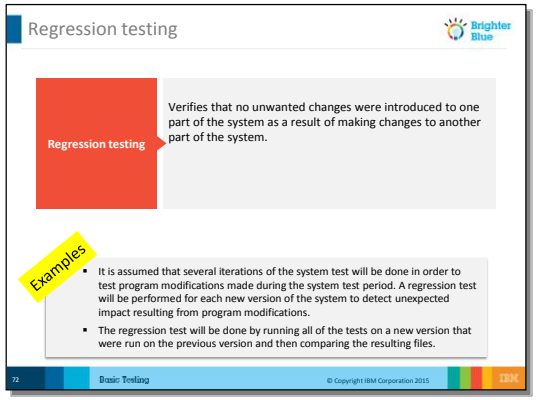
Slide Content	Instructor Guide	Use this space for your notes
Slide 68  **Example: Interface Testing is performed to evaluate whether systems or components pass data and control correctly to one another. It is to verify if all the interactions between these modules are working properly and errors are handled properly.** **An inter-system test, for example, could be a memory map, showing what is in memory at that moment. It basically is the way you find out if the inter-system is working without any problem, and is**	**Purpose:** To define Regression testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Regression testing. ▪ Use examples to give the participants a better understanding of practical scenarios	

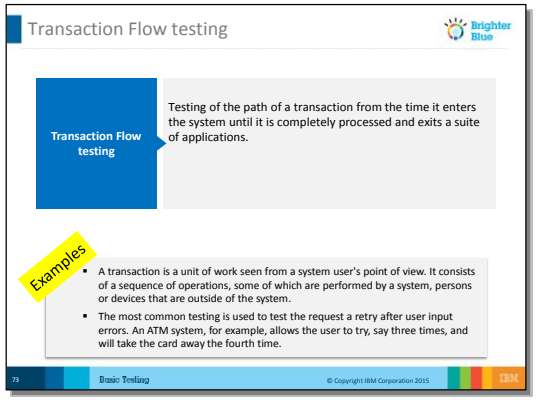
Slide Content	Instructor Guide	Use this space for your notes
a big help when you do have a problem in finding a fix for it.		

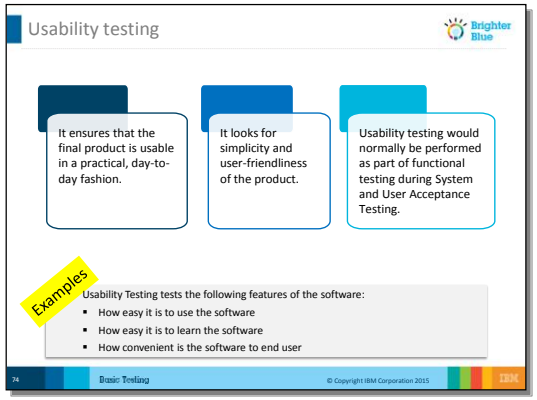
Slide Content	Instructor Guide	Use this space for your notes
Slide 69  **Example: Parallel testing is a testing technique in which the same inputs are entered in two different versions of the application and reporting the anomalies.** ▪ **Ensures that the new version of the application performs correctly.** ▪ **Ensures consistencies and inconsistencies are the same between the old and the new version.** ▪ **Ensures the integrity of the new application.**	**Purpose:** To define Transaction Flow testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Transaction Flow testing. ▪ Transfer the knowledge of practical applicability by sharing an example.	

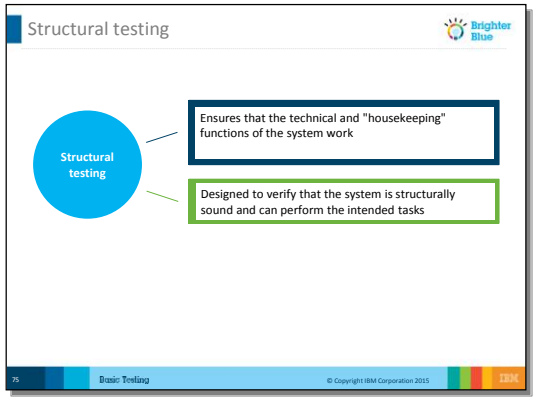
Slide Content	Instructor Guide	Use this space for your notes
▪ Verifies if the data format between the two versions have changed. ▪ Operating new and old version of a payroll system to determine that the paychecks from both systems are reconcilable. “Running old version of application to Example: - ensure that the functions of old system are working fine with respect to the problems encountered in the new system.		

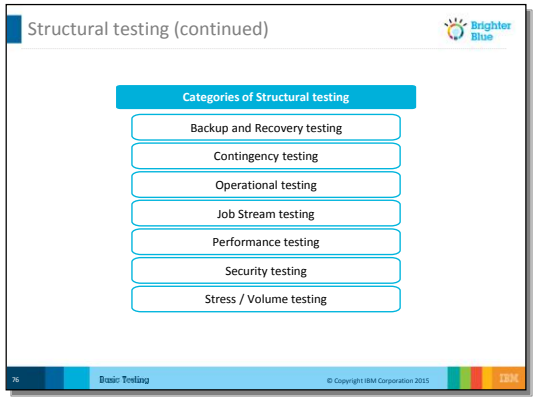
Slide Content	Instructor Guide	Use this space for your notes
Slide 70  	**Purpose:** To define Usability testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition and purpose of Usability testing. ▪ Use examples to give the participants a better understanding of practical scenarios	

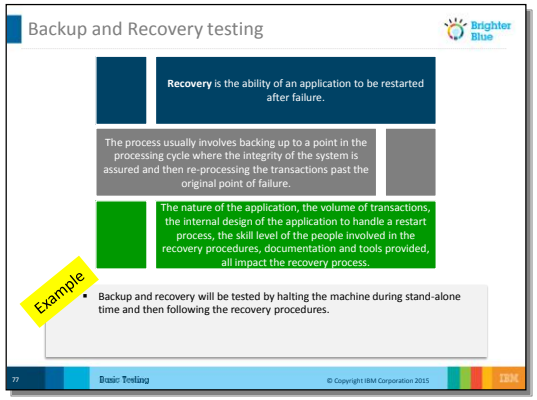
Slide Content	Instructor Guide	Use this space for your notes
Slide 71  	**Purpose:** To understand the features of Structural testing **Approximate Duration:** 5 mins **What to cover:** ▪ Explain the characteristics of Structural testing. ▪ Identify the categories of Structural testing.	

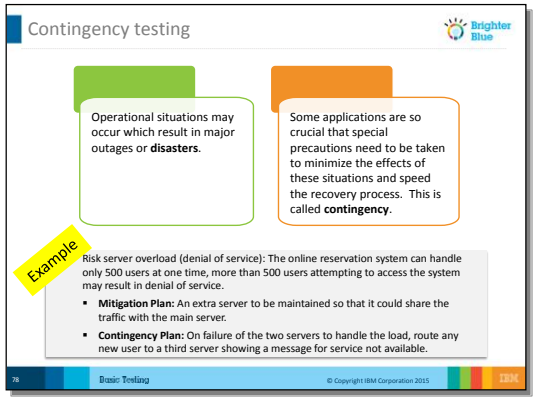
Slide Content	Instructor Guide	Use this space for your notes
Slide 72  	**Purpose:** To define Backup and Recovery testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Backup and Recovery testing.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 73  	**Purpose:** To define Contingency testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Contingency testing.	

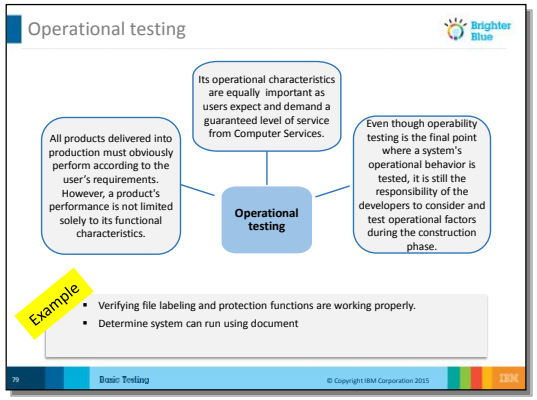
Slide Content	Instructor Guide	Use this space for your notes
Slide 74  	**Purpose:** To define Operational testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Operational testing.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 75  	**Purpose:** To define Job Steam testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Job Stream testing.	

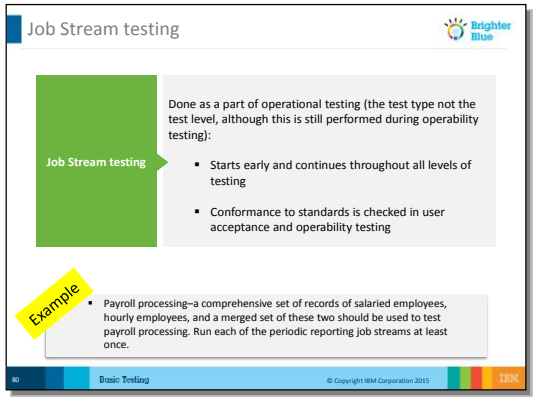
Slide Content	Instructor Guide	Use this space for your notes
Slide 76  	**Purpose:** To define Performance testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Performance testing.	

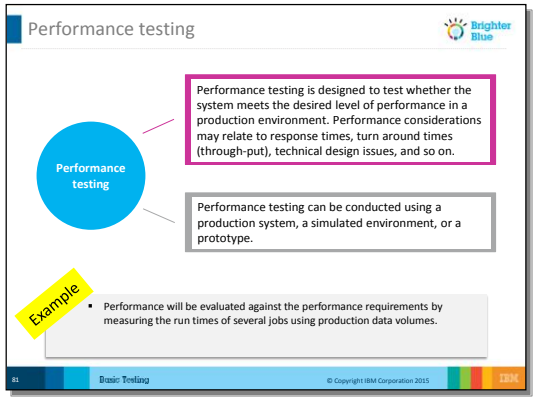
Slide Content	Instructor Guide	Use this space for your notes
Slide 77  **Example: The process of testing that the disaster recovery plans (also known as system contingency plans) and the associated business continuity plans work as specified.** **Risk server overload (denial of service): The online reservation system can handle only 500 users at one time, more than 500 users attempting to access the system may result in denial of service.**	**Purpose:** To define Security testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Security testing.	

Slide Content	Instructor Guide	Use this space for your notes
Mitigation Plan: An extra server to be maintained so that it could share the traffic with the main server. **Contingency Plan: On failure of the two servers to handle the load, route any new user to a third server showing a message for service not available.**		

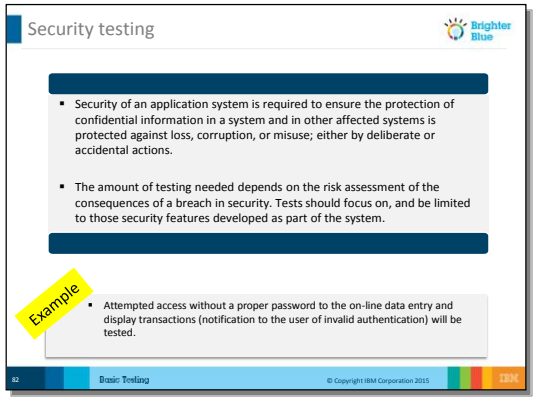
Slide Content	Instructor Guide	Use this space for your notes
Slide 78  **Example: To verify that before application goes to production the operating procedures and staff can properly execute the application. Evaluation of process and execution of process.** ▪ **Verifying file labeling and protection functions are working properly.** ▪ **Determine system can run using document**	**Purpose:** To define Stress / Volume testing **Approximate Duration:** 5 mins **What to cover:** ▪ Provide definition, purpose, and examples of Stress / Volume testing.	

Slide Content	Instructor Guide	Use this space for your notes

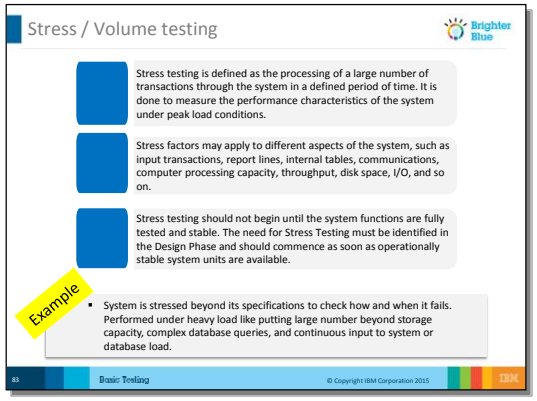
Slide Content	Instructor Guide	Use this space for your notes
Slide 79  	**Purpose:** To understand Bottom-up testing **Approximate Duration:** 7 mins **What to cover:** ▪ Provide definition and purpose of Bottom-up testing. ▪ Enumerate and discuss the advantages and disadvantages of Bottom-up testing. ▪ Cite relevant examples.	

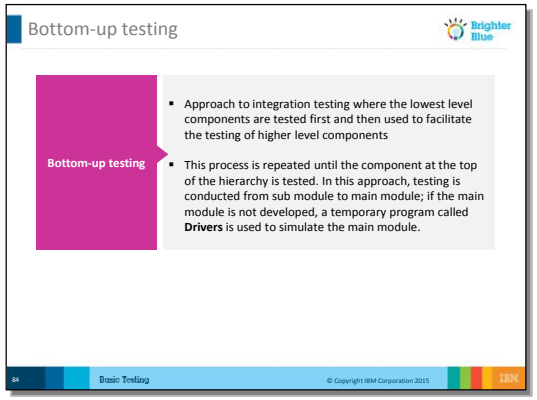
Slide Content	Instructor Guide	Use this space for your notes
Slide 80  	**Purpose:** To apply knowledge of the different types of testing **Approximate Duration:** 10 mins **Additional Materials or Pre-session prep tasks**: ▪ Easel sheets/paper ▪ Pens ▪ Instructor should have several situations/scenarios on hand to present to the class **What to cover:** ▪ Identify what type or kind of testing should be used in the following situations provided by the instructor **Instructions for the participant:** ▪ Break the class into small groups (2-3 depending on the number of participants)	

Slide Content	Instructor Guide	Use this space for your notes
	▪ The instructor will provide different testing scenarios and each group of participants should identify the type of testing to be used. **Debrief:** ▪ Discuss why each correct testing type corresponds to a particular situation.	

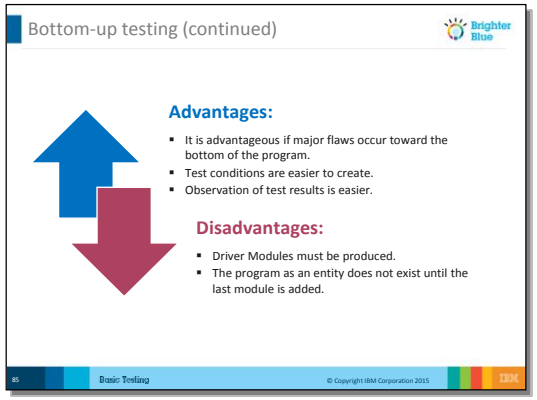
Slide Content	Instructor Guide	Use this space for your notes
Slide 81  **Example: Attempted access without a proper password to the on-line data entry and display transactions (notification to the user of invalid authentication) will be tested.** **Can system be penetrated by any hacking way? Test how well the system protects against unauthorized internal or external access. Check if system database is safe from external attacks.**	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items. ▪ Use this activity to recap the key takeaways from this module. ▪ You are the end of this module. The participants should now be able to: ○ Describe the different types of Functional Testing ○ Outline the different types of Structural Testing	

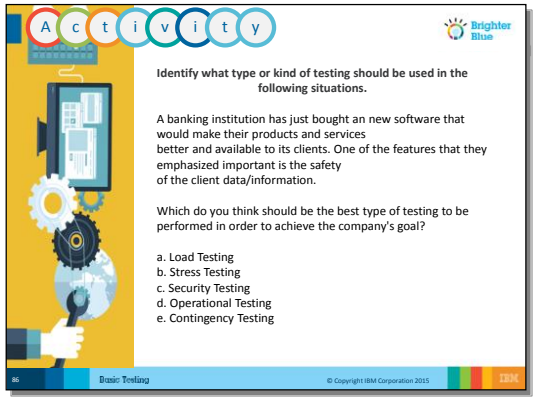
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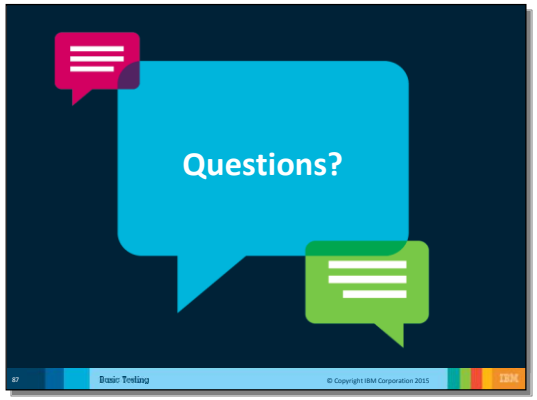
Slide Content	Instructor Guide	Use this space for your notes
Slide 82  	**Purpose:** To check knowledge of the participants for the topic covered **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll–Ask them to raise their hands based on the answer. ▪ The correct answer is **a) Static and Dynamic testing** ▪ Explain the answer using the following explanation:	

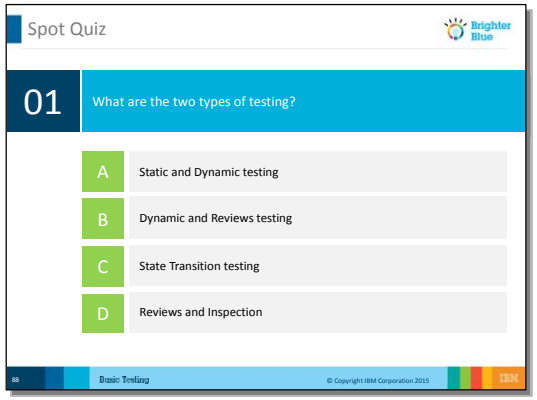
Slide Content	Instructor Guide	Use this space for your notes
Slide 83  	**Purpose:** To check knowledge of the participants for the topic covered **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll – Ask them to raise their hands based on the answer. ▪ The correct answer is **C.** To check that no unwanted changes were introduced to one part of the system as a result of making changes to another part of the system ▪ Explain the answer using the following explanation: As per definition--it is a type of software testing that seeks to uncover new software bugs, or regressions, in existing functional and non-	

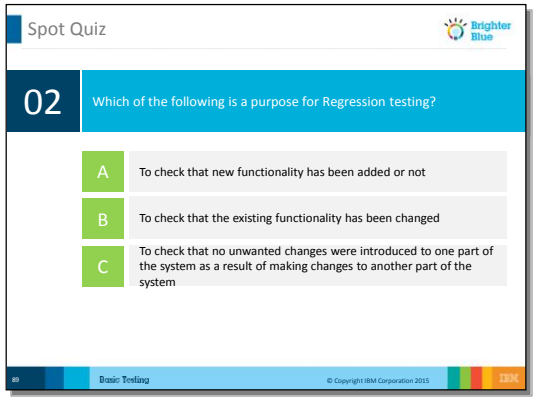
Slide Content	Instructor Guide	Use this space for your notes
	functional areas of a system after changes such as enhancements, patches or configuration changes, have been made to them.	

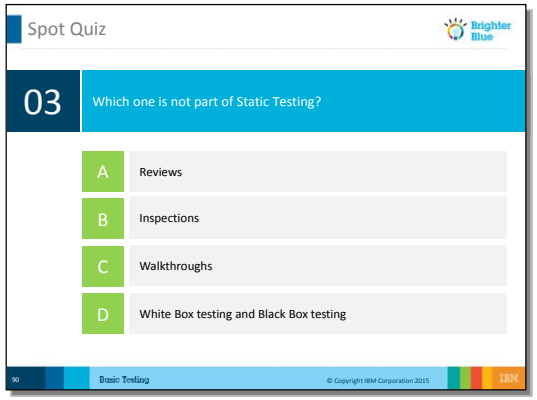
Slide Content	Instructor Guide	Use this space for your notes
Slide 84  **Example: For Unit Testing of "Sales Order Printing" program, a Driver program will have the code which will create Sales Order records using hardcoded data and then call "Sales Order Printing" program.**	**Purpose:** To check knowledge of the participants for the topic covered **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll – Ask them to raise their hands based on the answer. ▪ The correct answer is **D.** White Box testing and Black Box testing	

Slide Content	Instructor Guide	Use this space for your notes
Slide 85  **Identify what type or kind of testing should be used in the following situations provided by the instructor:** ▪ **Is the example given / provided require Functional or Structural testing?** **The correct answer is C. Security Testing**		

Slide Content	Instructor Guide	Use this space for your notes
Slide 86 		

Slide Content	Instructor Guide	Use this space for your notes
Slide 87  		

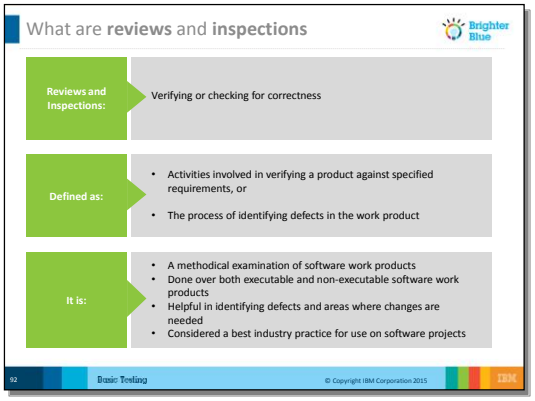
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Slide 88 		

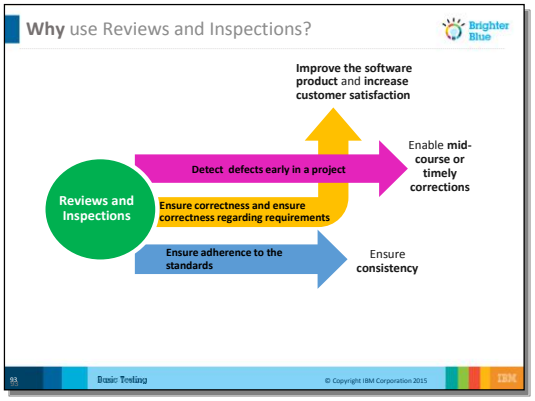
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Slide 89  		

Module 3: Types and Levels of Testing

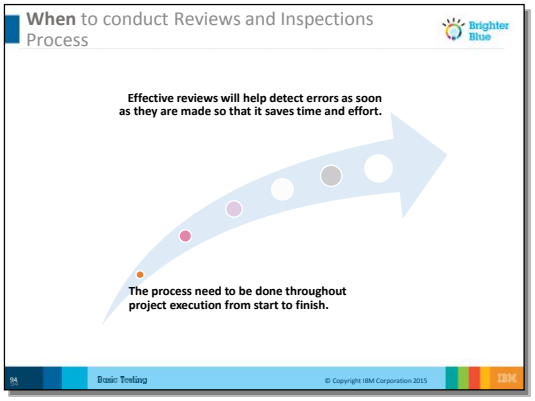
Slide Content	Instructor Guide	Use this space for your notes
Slide 90  **At the end of this module, you should be able to:** ▪ **Recognize the features of reviews and inspections** ▪ **Recognize the benefits of reviews and inspections** ▪ **Recognize the steps in the reviews and inspection processes**	▪ **Purpose:** To give an overview of Reviews and Inspections Process **Approximate Duration**: 2 mins **What to cover:** ▪ Tell the participants about definition of Reviews and Inspections, when and how reviews and inspections are conducted. ▪ Introduce two real world scenarios ▪ Highlight how reviews and inspections are proactive methods of ensuring the success of projects.	

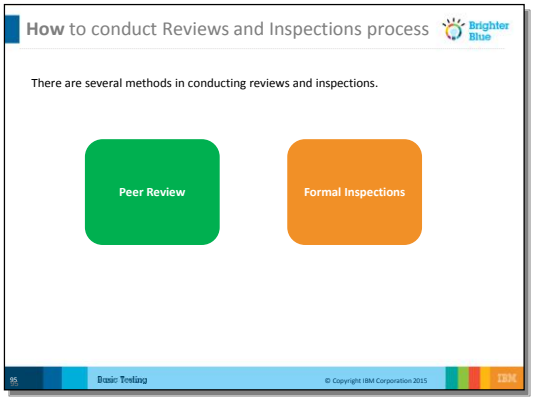
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▪ Show the importance of reviews and inspection through some real-life incidents ▪ Compare peer and facilitation reviews ▪ Recognize the key features of defect classification ▪ Explain how Return on Investment (ROI) helps to justify the cost of reviews		

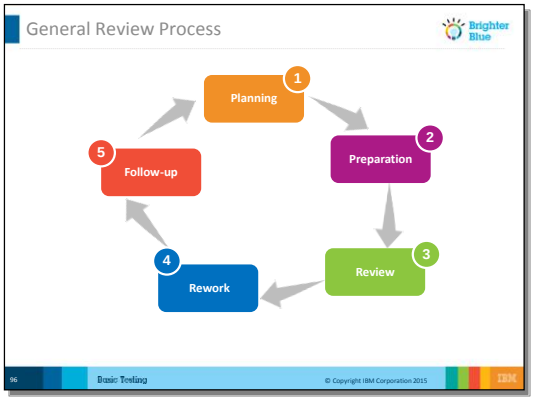
Slide Content	Instructor Guide	Use this space for your notes
Slide 91  **A methodical examination of software work products: Reviews and Inspections is a methodical examination of software work products generated at different phases of software development life cycle or modified during maintenance (for example, code change for a bug in an existing software).**	**Purpose:** To define and explain reviews and inspections **Approximate Duration:** 5 mins **What to cover:** ▪ Define reviews and inspections. ▪ Explain the features of reviews and inspections: ○ Methodical software work product examination ○ Both executable and non-executable software work products covered ○ Helps identify defects and areas needing changes ○ Identified as the best industry practice	

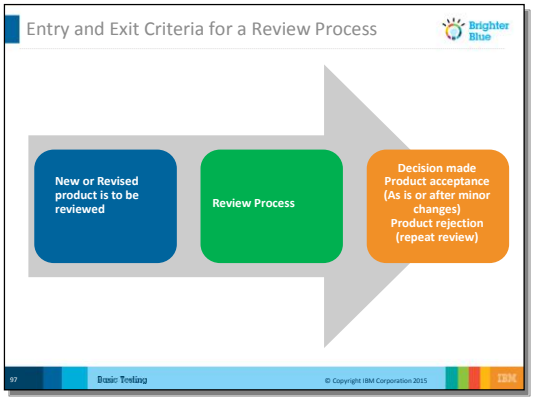
Slide Content	Instructor Guide	Use this space for your notes
Slide 92  **Why?** **It helps to:** • **Detect defects early in a project so as to enable mid-course or timely corrections.** • **Ensure correctness and consistency with respect to requirements, thus improving the software product and increasing customer satisfaction.**	**Purpose** : Explain the benefits of Reviews and Inspections **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants why we need to carry out reviews and inspections: o Early detection of defects o Ensured consistency and adherence to standards o Ensured correctness in all aspects including the requirements ▪ Cover the points detailed in the notes section of the slide.	

Slide Content	Instructor Guide	Use this space for your notes
• Ensure adherence to the standards (set by IBM or customer).		

Slide Content	Instructor Guide	Use this space for your notes
Slide 93  	**Purpose** : Explain when we need to conduct Reviews and Inspections **Approximate Duration:** 2 mins **What to cover:** ▪ Discuss how effective reviews help detect errors at an early stage which saves both effort and time. ▪ Tell the participants about the circumstances where we need to carry out reviews and inspections.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 94  **There are several methods involved in conducting reviews and inspections; for example, peer review and formal inspections.**	**Purpose** : Explain the methods for conducting Reviews and Inspections **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants about the methods of conducting reviews and inspections: o Peer review o Formal inspections	

Slide Content	Instructor Guide	Use this space for your notes
Slide 95  **The following illustrates the general review process.**	**Purpose:** To explain the general review process **Approximate Duration:** 5 mins **What to cover:** ▪ Explain the general review process focusing on the stages: ○ Planning ○ Preparation ○ Review ○ Rework ○ Follow-up ▪ Illustrate the process with the help of a diagram so that the participants can better understand.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 96  **The following diagram illustrates the entry and exit criteria for a review process.**	**Purpose:** To explain the entry and exit criteria for a review process **Approximate Duration:** 2 mins **What to cover:** ▪ Explain the entry and exit criteria for a review process. ▪ Illustrate the process with the help of diagram.	

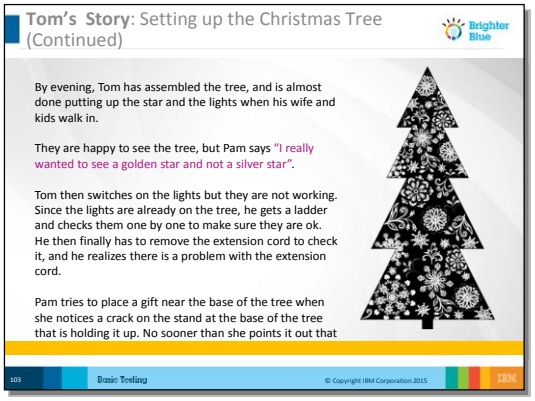
Slide Content	Instructor Guide	Use this space for your notes
Slide 97  	**Purpose:** To explain the importance of reviews and detecting errors early to save time and effort with a real world scenario. **Approximate Duration:** 2 mins **What to cover:** ▪ Start by saying that this is a real life scenario where reviews would have helped ▪ Lead the participants through the conversation slide by slide ▪ Conclude the scenario as on the last slide	

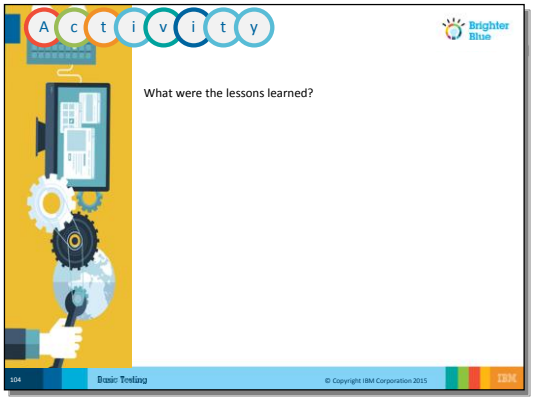
Slide Content	Instructor Guide	Use this space for your notes
Slide 98  	**Purpose:** Use one scenario to let the participants realize that a client's request should be reviewed for a project. **Approximate Duration:** 2 mins **What to cover:** ▪ Use one commonly-seen real life scenario and let the participants realize that a client's requests should be reviewed for a project. ▪ Take up the discussion from the previous slide.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 99  	**Purpose:** Use one scenario to let the participants realize that a client's request should be reviewed for a project. **Approximate Duration:** 2 mins **What to cover:** ▪ Use one commonly-seen real life scenario and let the participants realize that a client's requests should be reviewed for a project. ▪ Take up the discussion from the previous slide. ▪	

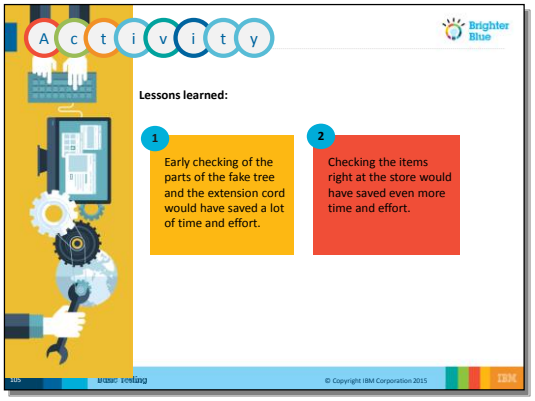
Slide Content	Instructor Guide	Use this space for your notes
Slide 100  	**Purpose:** Use one scenario to let the participants realize that a client's request should be reviewed for a project. **Approximate Duration:** 2 mins **What to cover:** ▪ Use one commonly-seen real life scenario and let the participants realize that a client's requests should be reviewed for a project. ▪ Take up the discussion from the previous slide.	

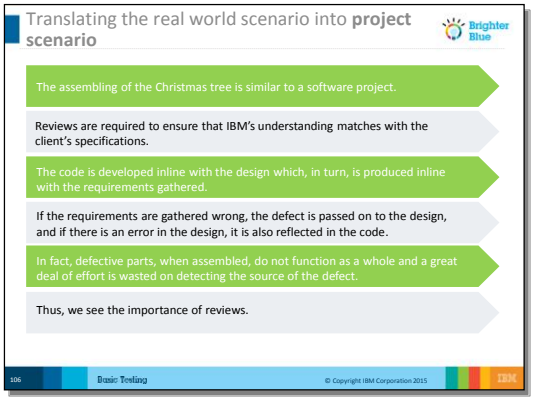
Slide Content	Instructor Guide	Use this space for your notes
Slide 101  	**Purpose:** Use one scenario to let the participants realize that a client's request should be reviewed for a project. **Approximate Duration:** 2 mins **What to cover:** ▪ Use one commonly-seen real life scenario and let the participants realize that a client's requests should be reviewed for a project. ▪ Take up the discussion from the previous slide.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 102  	**Purpose:** Use one scenario to let the participants realize that a client's request should be reviewed for a project. **Approximate Duration:** 2 mins **What to cover:** ▪ Use one commonly-seen real life scenario and let the participants realize that a client's requests should be reviewed for a project. ▪ Take up the discussion from the previous slide. ▪ Conclude the story by pointing out to the problem that crops up.	

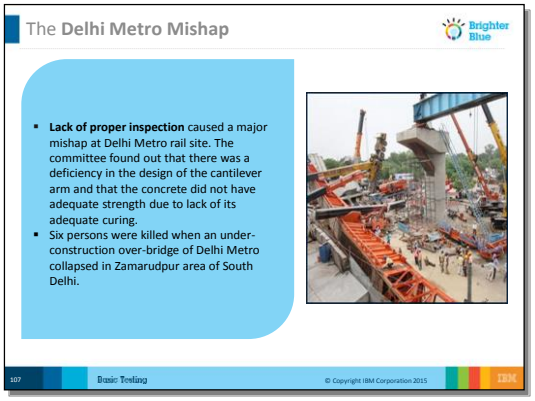
Slide Content	Instructor Guide	Use this space for your notes
Slide 103  **So what were the lessons learned?** **Lessons:** ▪ **Had Tom checked the parts of the fake tree before assembling it and also the extension cord before connecting it, it would have saved him a lot of time and effort. Even then, he would still have to travel all the way back to the store to get it replaced (wasted time).** ▪ **Had Tom checked the parts of the fake tree, the lights, and the extension cord right at the**	**Approximate Duration:** 10 mins **What to cover:** ▪ Ask the question ▪ You can ask the groups to list them on flipcharts provided for each group ▪ Discuss the list from each group ▪ Summarize the lessons learned by showing the second slide **Purpose:** To see if the participants are able identify the lessons learned from the scenario **Instructions for the participants:** ▪ Ask the question to all the participants. ▪ Ask them to raise hand to provide their answers. ▪ Once a participant answers, ask how many of the others agree to him/her by raising hand.	

Slide Content	Instructor Guide	Use this space for your notes
store, he would have saved even more time and effort.	**Debrief:** ▪ Ask the participants about their experience of this activity. ▪ Ask the participants how much the lessons learned will help them in real life situation.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 104  	**Purpose:** To summarize the points discussed in the previous slide **Approximate Duration:** 2 mins **What to cover:** ▪ Discuss the list from each group ▪ Summarize the lessons learned	

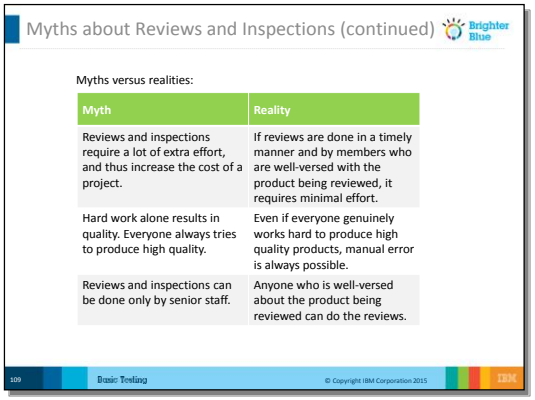
Slide Content	Instructor Guide	Use this space for your notes
Slide 105  ▪ **The assembling of the Christmas tree is similar to a software project. Pam giving directions to Tom is similar to a client specifying his requirements to IBM. Pam is unhappy that Tom bought a silver star instead of a gold one.** ▪ **Similarly, when clients specify requirements, reviews are required to ensure that IBM's understanding matches with the client's specifications.** ▪ **The different tree parts, lights, and extension cord are similar to different blocks of code that**	**Purpose:** To change the real world scenario into project scenario **Approximate Duration:** 5 mins **What to cover:** ▪ Translate the real world scenario into project scenario. ▪ Discuss why conducting reviews and detecting errors at each stage is important to save time and effort.	

Slide Content	Instructor Guide	Use this space for your notes
developers might produce. The code is developed in line with the design which in turn is produced in line with the requirements gathered. ▪ If the requirements are gathered wrong, the defect is passed on to the design, and if there is an error in the design, it is also reflected in the code. So when the code is tested, desired results are not achieved. ▪ In fact, defective parts, when assembled, do not function as a whole, and a great deal of effort is wasted on detecting the source of the defect, that is, is it a defect in the code or a defect in the design or a defect in the requirements gathered? ▪ Thus, we see the importance of reviews and also the importance of detecting errors as soon as they are made.		

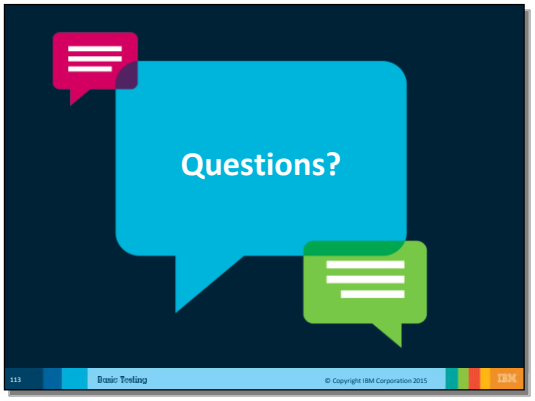
Slide Content	Instructor Guide	Use this space for your notes
Slide 106  	**Purpose:** To explain the importance of reviews and inspections with a real world scenario **Approximate Duration:** 2 mins **What to cover:** ▪ Dramatize the story ▪ Explain the reason and how it could have been avoided ▪ Encourage the learners to participate in a discussion	

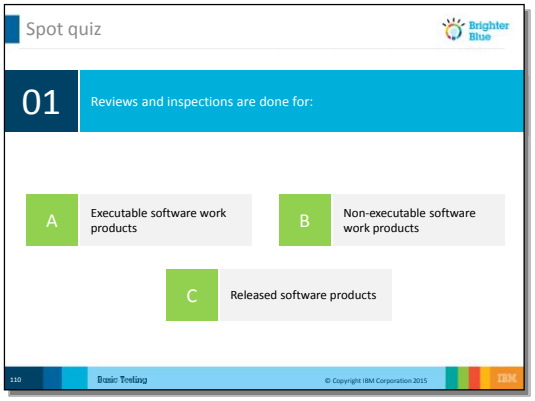
Slide Content	Instructor Guide	Use this space for your notes
Slide 107  ▪ **The main reason of such accidents is lack of supervision, which leads to poor quality construction.** ▪ **There are certain operating procedures which have to be followed at every step of construction and inspection. As these types of projects depend upon team work, it is the moral duty of everyone, either director or laborer, to not compromise on the quality and bypass inspection process.**	**Purpose:** To explain the importance of reviews and inspections with a real world scenario **Approximate Duration:** 3 mins **What to cover:** ▪ Dramatize the story ▪ Explain the reason and how it could have been avoided ▪ Encourage the learners to participate in a discussion	

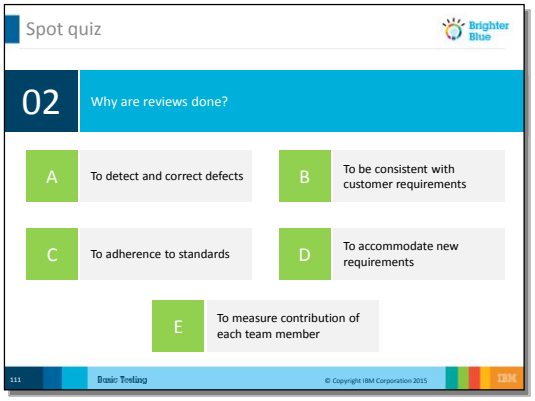
Slide Content	Instructor Guide	Use this space for your notes
▪ **Since Gammon and DMRC had bypassed certain checks and inspection processes, this disaster occurred.**		

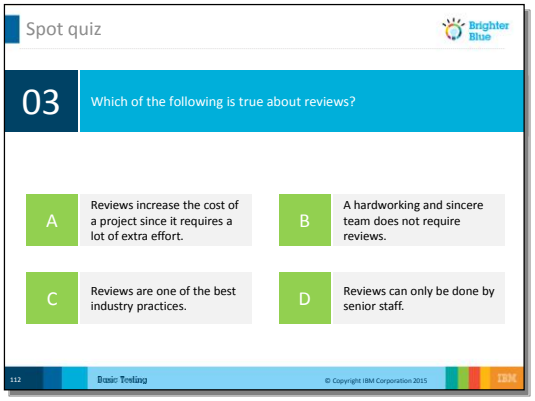
Slide Content	Instructor Guide	Use this space for your notes
Slide 108  **Reality:** ▪ **If reviews are done in a timely manner and by members who are well-versed with the product being reviewed, it requires minimal effort. No reviews result in defects and rework taking even more time and effort to make fixes.** ▪ **Even if everyone genuinely works hard to produce high quality products, manual error is always possible. Also, it is easy to miss defects when we continuously look at the same piece of code. A peer might be able to catch it easily.**	**Purpose:** To see if the participants are able identify the realities about reviews and inspections **Approximate Duration:** 10 mins **What to cover:** ▪ Tell the participants the myths about reviews and inspections. ▪ Ask them to agree or disagree ▪ Compare between the myths and the realities.	

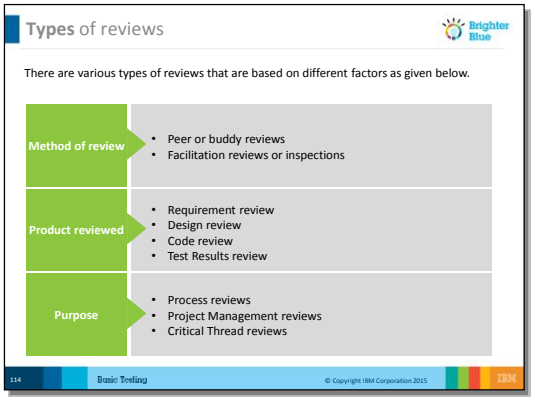
Slide Content	Instructor Guide	Use this space for your notes
▪ **Anyone who is well-versed about the product being reviewed can do the reviews.**		

Slide Content	Instructor Guide	Use this space for your notes
Slide 109 	**Purpose:** To break the monotony with an interactive spot quiz **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **C. Reviews are one of the best industry practices.**	

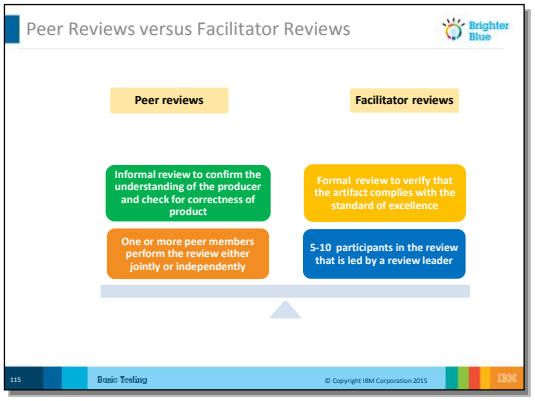
Slide Content	Instructor Guide	Use this space for your notes
Slide 110  	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items. ▪ Use this activity to recap the key takeaways from this module.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 111  	**Purpose:** To break the monotony with an interactive spot quiz **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ This question has multiple correct answers. The correct answers are: • **Executable software work products** • **Non-executable software work products**	

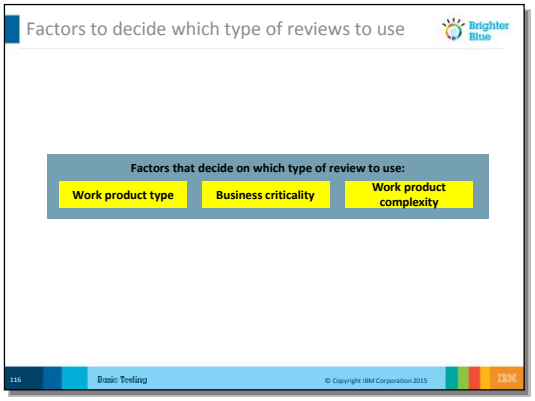
Slide Content	Instructor Guide	Use this space for your notes
Slide 111  	**Purpose:** To break the monotony with an interactive spot quiz **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ This question has multiple correct answers. The correct answers are: **A. Detect and correct defects** **B. Ensure consistency with customer requirements** **C. Ensure adherence to standards**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 113  **There are many types of reviews based on different factors. A few are mentioned here:** **Based on method of review:** • **Peer or buddy reviews** • **Facilitation reviews or inspections** **Based on product reviewed:** • **Requirement review** • **Design review**	**Purpose:** To list and explain the different types of reviews **Approximate Duration:** 5 mins **What to cover:** ▪ Categorize various kinds of reviews based on different factors: ○ Method of review ○ Product reviewed ○ Purpose	

Slide Content	Instructor Guide	Use this space for your notes
• Code review • Test Results review Based on purpose: • Process reviews (to check compliance to organization processes) • Project Management Reviews (to review project health and check risks) • Critical Thread Reviews (meetings to discuss and review topics on people, process, and performance)		

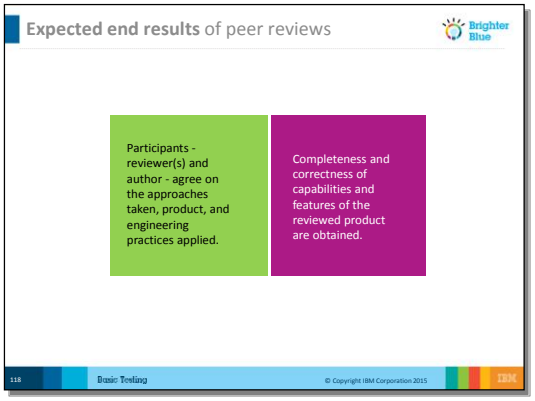
Slide Content	Instructor Guide	Use this space for your notes
Slide 114  ▪ **Peer reviews** ○ **Peer or buddy reviews (walkthroughs) are informal reviews used to confirm the understanding of the producer and check for correctness of product being reviewed.** ○ **One or more peer members (with expertise in the subject of the product to be reviewed)**	**Purpose:** To distinguish between peer reviews and facilitator reviews **Approximate Duration:** 5 mins **What to cover:** ▪ Distinguish between peer reviews and facilitator reviews: ○ Peer reviews are informal while facilitator reviews are formal ○ One or two members can perform peer review while for facilitator reviews 5-10 people are needed ▪ Explain the factors to decide on which type of review to use.	

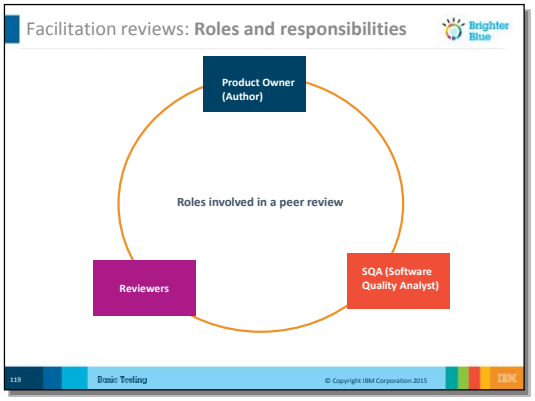
Slide Content	Instructor Guide	Use this space for your notes
perform the review either jointly or independently. ▪ Facilitator Reviews ○ Facilitator reviews (inspection) are formal reviews used to verify that the artifact compiles with the standard of excellence. ○ A team of not less than 5 percent and not more than 10 (each with specific defined roles) participates in the review that is led by a review leader.		

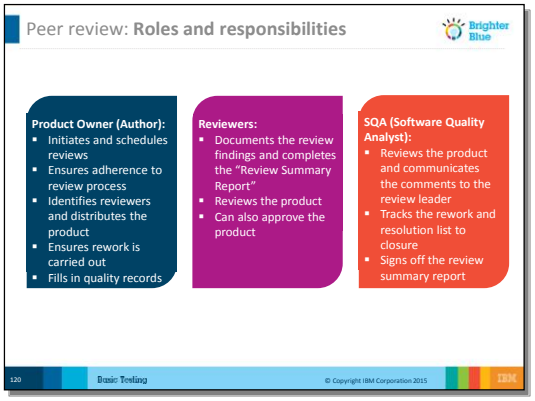
Slide Content	Instructor Guide	Use this space for your notes
Slide 115  **Work product type - Design document, code, or any other**	**Purpose:** Understand the differences between Peer Reviews and Facilitation Reviews **Approximate Duration:** 2 mins **What to cover:** ▪ List the factors that help decide on the type of review to use: o Work product type o Business criticality o Work product complexity	

Slide Content	Instructor Guide	Use this space for your notes
Slide 116  **Peer reviews:** ▪ **This is a less formal verification technique - This is a less formal verification technique in which a technical peer analyzes or improves the quality of the original work product, for example, by finding defects or proposing a more optimal solution.** ▪ **It is the author who initiates the session - Since peer reviews cater to the needs of the producer**	**Purpose:** To explain the characteristics of peer reviews **Approximate Duration:** 5 mins **What to cover:** ▪ Explain the characteristics of peer reviews: ○ Author-initiated session ○ Formal verification technique ○ Several peer reviews in every software development lifecycle activity	

Slide Content	Instructor Guide	Use this space for your notes
or author of the software artifact (in acquiring superior knowledge of all aspects of it), it is the author who initiates the session. ▪ There may be several peer reviews in each software development life cycle activity - There may be several peer reviews in each software development life cycle activity and they yield open issues and action items. While these issues and action items may be tracked to closure, the only measurement taken from these reviews is a count of the number of peer reviews held along with effort spent and process improvements identified.		

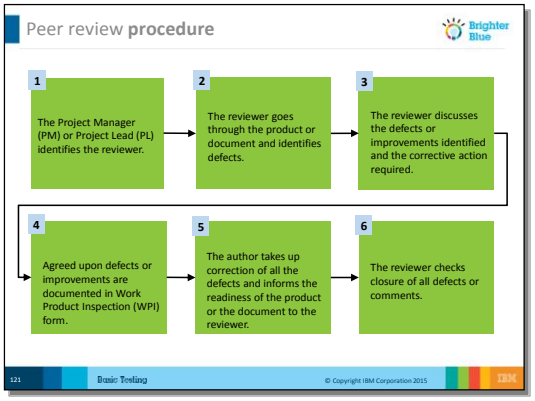
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Slide 117  	**Purpose:** To explain the expected end results of peer reviews **Approximate Duration:** 2 mins **What to cover:** ▪ Explain the expected end results of peer reviews: o Agreement on the taken approaches, engineering practices applied and product o Complete and correct capabilities and features obtained	

Slide Content	Instructor Guide	Use this space for your notes
Slide 118  	**Purpose:** To list the roles involved in a peer review **Approximate Duration:** 2 mins **What to cover:** ▪ List the roles involved in a peer review: ○ Product owner or author ○ SQA or Software Quality Analyst ○ Reviewers	

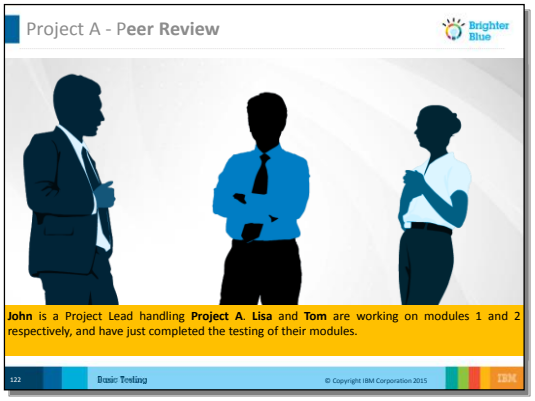
Slide Content	Instructor Guide	Use this space for your notes
Slide 119  **Product Owner (Author):** ▪ **Initiating the review process and scheduling reviews** ▪ **Ensuring the review process is adhered to** ▪ **Identifying reviewers and distributing the product for review. The owner may seek help from his manager in identification.** ▪ **Ensuring any rework as a result of the review is carried out**	**Purpose:** To describe the roles and responsibilities of the product owner, reviewers, and software quality analyst in peer review **Approximate Duration:** 5 mins **What to cover:** ▪ Discuss about the three entities found in peer review: ○ Product owner or author ○ Reviewers ○ SQA ▪ Describe the roles and responsibilities of each of these entities.	

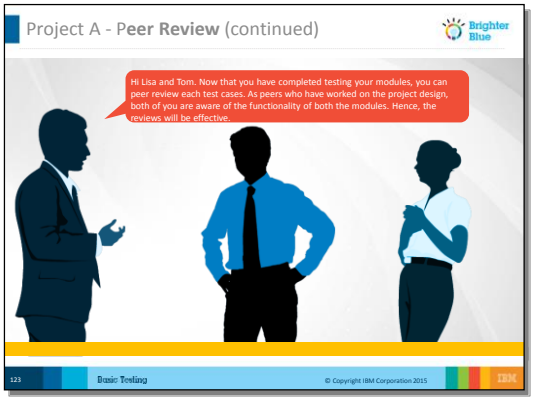
Slide Content	Instructor Guide	Use this space for your notes
▪ Filing the quality records that are generated during the review process as objective evidence of having carried out the review Reviewers: ▪ Documenting the review findings and completing the 'Review Summary Report' ▪ Reviewing the product ▪ Can also approve the product SQA (Software Quality Analyst): ▪ Where applicable, reviewing the product and communicating the comments to the review leader ▪ Where applicable, tracking the rework and resolution list to closure ▪ Signing off the review summary report after ensuring the defect / resolutions list is closed and the summary report is complete		

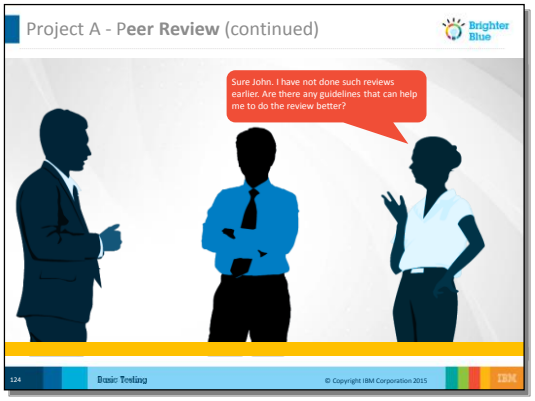
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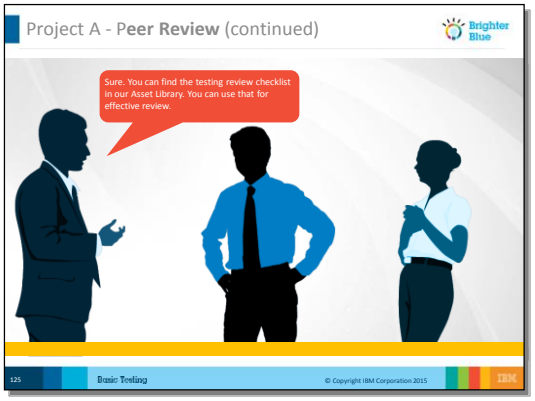
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Slide 120  **•The Project Manager (PM) or Project Lead (PL) identifies the reviewer based on the size, complexity, and importance of the product or document to be reviewed.** **•The reviewer goes through the product or document and identifies defects based on appropriate checklists or specifications or standards.**	**Purpose:** To describe the peer review procedure **Approximate Duration:** 5 mins **What to cover:** ▪ Describe the peer review procedure by focusing on each of the steps: o Identification of reviewer o Review of the product to identify the defects o Discussion of defects or required corrective actions and improvements o Documentation of agreed upon defects and improvements o Correction of defects o Closure of all comments and defects	

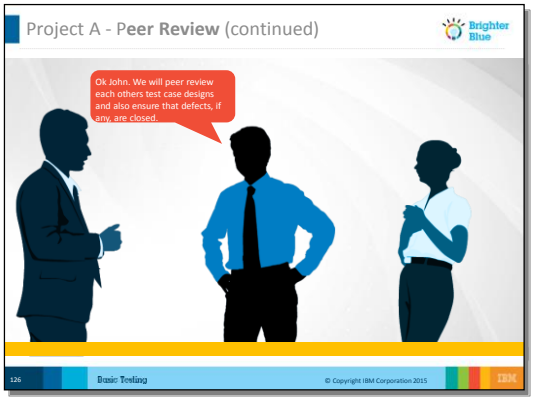
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•The reviewer discusses the defects or improvements identified, the corrective action required, and provides clarifications if any. •Once the defects or improvements are agreed upon, they are documented in the Work Product Inspection (WPI) form. It is the responsibility of the PM or PL to ensure that the reviews are conducted, results documented, and the defect corrections are carried out. •The author takes up the correction of all the defects identified to be corrected and informs the readiness of the product or the document to the reviewer. •If re-review is required, another iteration of review is done, but if the review result requires only verification of the closure of defects or comments, the reviewer checks whether all defects have been closed.		

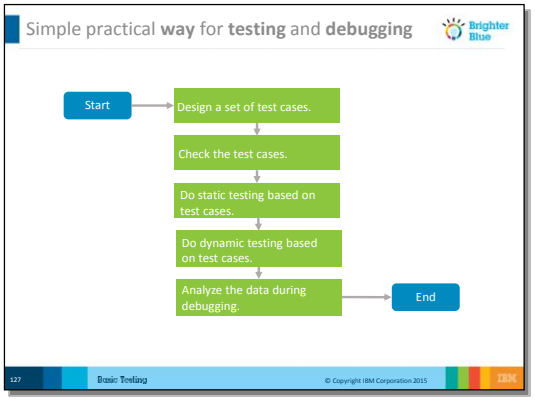
Slide Content	Instructor Guide	Use this space for your notes
Slide 121  	**Purpose:** To explain the importance of peer review with a real world scenario **Approximate Duration:** 1 min **What to cover:** ▪ Explain the scenario. ▪ Discuss the outcome.	

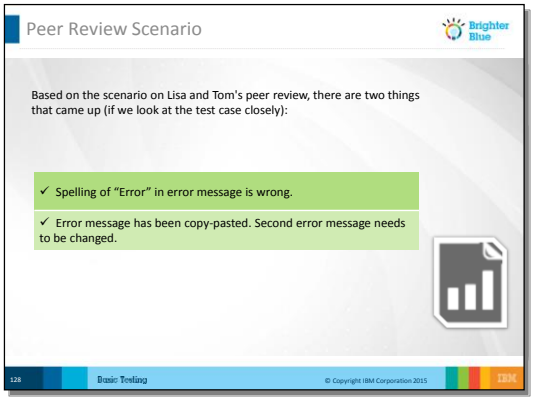
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Slide 122  	**Purpose:** Use one Peer Review scenario to understand how peer review takes place and see the results. **Approximate Duration:** 1 min **What to cover:** ▪ Understand one Peer Review scenario ▪ Find errors and solution	

Slide Content	Instructor Guide	Use this space for your notes
Slide 123  	**Purpose:** Use one Peer Review scenario to understand how peer review takes place and see the results **Approximate Duration:** 1 min **What to cover:** ▪ Understand one Peer Review scenario. ▪ Find errors and solution. ▪ Continue the scenario from the previous slide.	

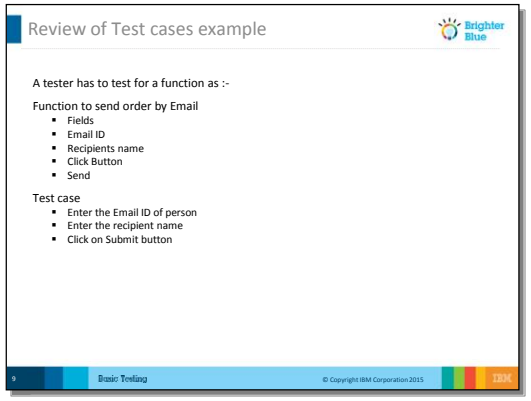
Slide Content	Instructor Guide	Use this space for your notes
Slide 124  	**Purpose:** Use one Peer Review scenario to understand how peer review takes place and see the results **Approximate Duration:** 1 min **What to cover:** ▪ Understand one Peer Review scenario. ▪ Find errors and solution. ▪ Continue the scenario from the previous slide.	

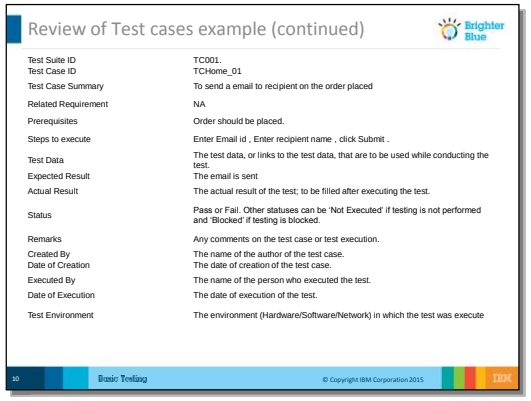
Slide Content	Instructor Guide	Use this space for your notes
Slide 125  	**Purpose:** Use one Peer Review scenario to understand how peer review takes place and see the results **Approximate Duration:** 1 min **What to cover:** ▪ Understand one Peer Review scenario. ▪ Find errors and solution. ▪ Continue the scenario from the previous slide.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 126  	**Purpose:** To explain how to test and debug simply and practically **Approximate Duration:** 2 mins **What to cover:** Show the diagram that points out a simple practical way for testing and debugging.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 127  **Based on the scenario on Lisa and Tom's peer review, there are two things that came up (if we look at the code closely):** • **Spelling of 'Error' in error message is wrong.** • **Error message has been copy-pasted. Second error message needs to be changed.**	**Purpose:** List errors that the participants could find **Approximate Duration:** 2 mins **What to cover:** ▪ List errors that the participants could find: o "Error" spelling wrong o Instead of copy-pasted error message a new error message must be given	

Slide Content	Instructor Guide	Use this space for your notes

Slide Content	Instructor Guide	Use this space for your notes
Slide 128  	**Purpose:** To give an overview of facilitation reviews. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 2 mins **What to cover:** ▪ Give an overview of facilitation reviews: o It is the most rigorous form of review o It is the exit criteria in lifecycle activity	

Slide Content	Instructor Guide	Use this space for your notes
Slide 129 	**Purpose:** To explain facilitation review characteristics. Conduct a walkthrough of the documents provided as supported resources. **Approximate Duration:** 5 mins **What to cover:** ▪ Explain facilitation review characteristics: ○ Done by technical people ○ Consists of well-defined processes and roles ○ Preparation of reviewers in advance ○ Focus on identification of problems ○ Recording of review data to monitor effectiveness	

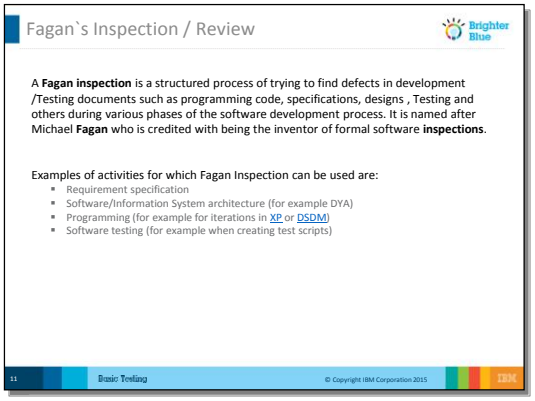
Slide Content	Instructor Guide	Use this space for your notes
Slide 130  	**Purpose:** To list the roles involved in a facilitation review **Approximate Duration:** 2 mins **What to cover:** ▪ List the roles involved in a facilitation review: ○ Author or producer ○ Product owner ○ Recorder ○ Reader ○ Reviewers or inspectors ○ Review leader or inspection leader	

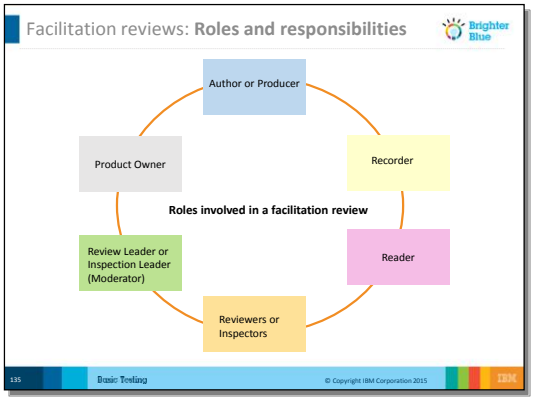
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Slide 131  ▪ **Facilitation Reviews or Inspections (also called Fagan's Review) are the most rigorous form of reviews which involve structured formal reviews done by a group of technical personnel who analyze or improve the quality of the original work product as well as the quality of the method.**	**Purpose:** To explain the responsibilities of the roles involved in facilitation reviews **Approximate Duration:** 5 mins **What to cover:** ▪ Discuss the responsibilities of each involved role in detail including that of: o Author or producer o Product owner o Reader	

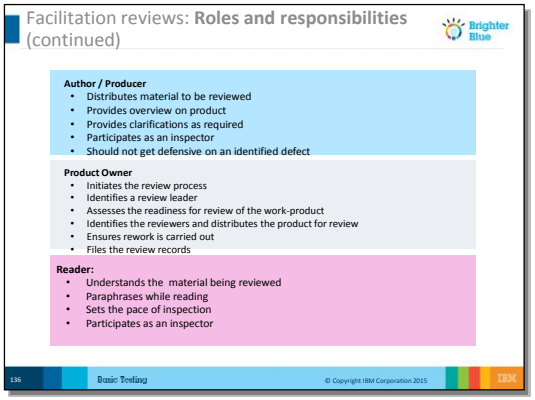
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▪ In a lifecycle activity, the software inspection is the exit criteria or gate that concludes the activity.		

Slide Content	Instructor Guide	Use this space for your notes
Slide 132  **Characteristics of Facilitation reviews** • **They are done by technical people for technical people.** • **They consist of structured, well defined processes with well-defined roles for participants.** • **Reviewers are prepared in advance, and clarifications are obtained before the meeting.**	**Purpose:** To explain the responsibilities of the roles involved in facilitation reviews **Approximate Duration:** 5 mins **What to cover:** ▪ Continue the discussion from the previous slide. ▪ Discuss the responsibilities of each involved role in detail including that of: o Reviewers or inspectors o Review leader or inspection leader o Recorder	

Slide Content	Instructor Guide	Use this space for your notes
• Focus is on identifying problems and not resolving them. • Review data from these reviews is recorded and used for monitoring effectiveness of the review procedure.		

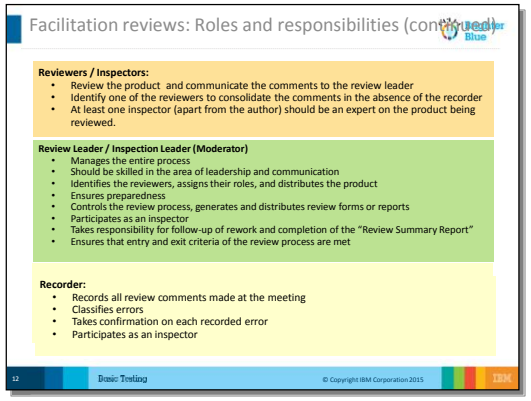
Slide Content	Instructor Guide	Use this space for your notes
Slide 133  	**Purpose:** To describe the facilitation review process **Approximate Duration:** 5 mins **What to cover:** ▪ Describe the facilitation review process steps: ○ Planning ○ Orientation preparation ○ Review meeting ○ Rework ○ Verify ▪ Describe the process with the help of the flow diagram.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 134  	**Purpose:** To explain when to prefer facilitation review over peer review with a real world scenario **Approximate Duration:** 1 min **What to cover:** ▪ Explain the scenario. ▪ Ask the learners if they can relate to the situation. ▪ Encourage them to share their views on why a facilitation review is being preferred here	

Slide Content	Instructor Guide	Use this space for your notes
Slide 135  **Author / Producer:** • **Distributes material to be reviewed** • **Provides overview on product** • **Provides clarifications as required** • **Participates as an inspector** • **Should not get defensive on an identified defect** **Product Owner:** • **Initiates the review process**	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

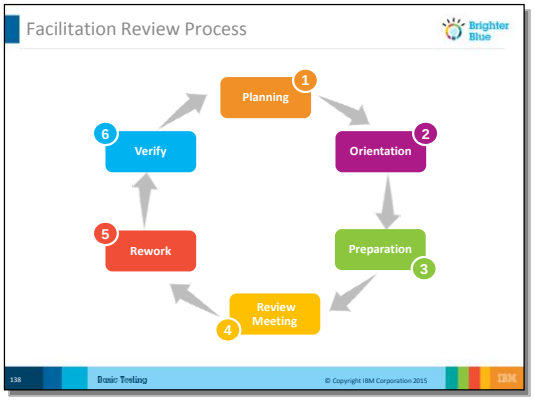
Slide Content	Instructor Guide	Use this space for your notes
• Identifies a review leader; may seek help from his manager in identifying a review leader • Assesses the readiness for review of the work-product along with the review leader • Identifies the reviewers and distributes the product for review along with the review leader • Ensures that any rework as a result of the review is carried out • Files the review records that are generated during the review process as objective evidence of having carried out the review • Review Leader / Inspection Leader (Moderator): • Manages the entire process and ensures that the review process is adhered to • Should be skilled in the area of leadership and communication, and be impartial • Identifies the reviewers and assigns their roles, distributes the product and related documents • Ensures preparedness before the review		

Slide Content	Instructor Guide	Use this space for your notes
• Controls the pace of review meetings, moderates the review, and also generates and distributes review forms or reports • Participates as an inspector • Takes responsibility for follow-up of rework and completion of the 'Review Summary Report' • Ensures that entry and exit criteria of the review process are met		

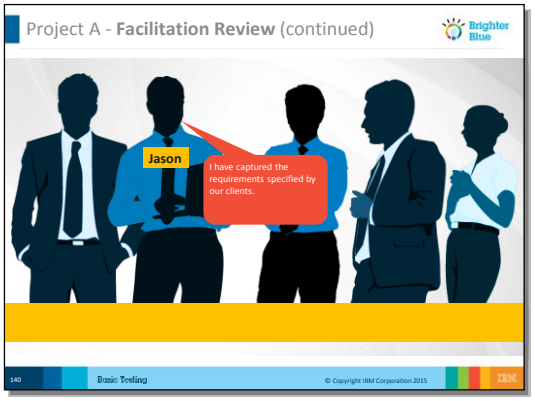
Slide Content	Instructor Guide	Use this space for your notes
Slide 136  **Reviewers / Inspectors:** ▪ **Review the product and communicate the comments to the review leader in the review meeting. This involves:** ○ **Preparing for the inspection meeting** ○ **Understanding the product** ○ **Doing the inspection**	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

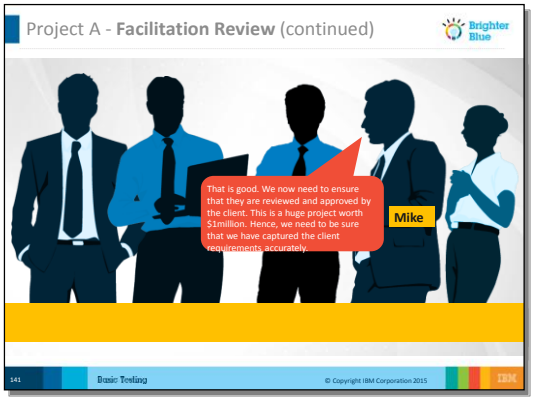
Slide Content	Instructor Guide	Use this space for your notes
○ Focusing on product and NOT the producer ▪ In the absence of the recorder, the review leader is responsible for identifying one of the reviewers to consolidate the comments. ▪ Everybody involved in the facilitation review (leader, reader, recorded, and author) is an inspector, but it is important that at least one inspector (apart from the author) should be an expert on the product being reviewed. Reader: ▪ Understands the material being reviewed ▪ Paraphrases while reading ▪ Sets the pace of inspection ▪ Participates as an inspector Recorder: ▪ Records all review comments made at the meeting (as indicated by leader) ▪ Classifies errors		

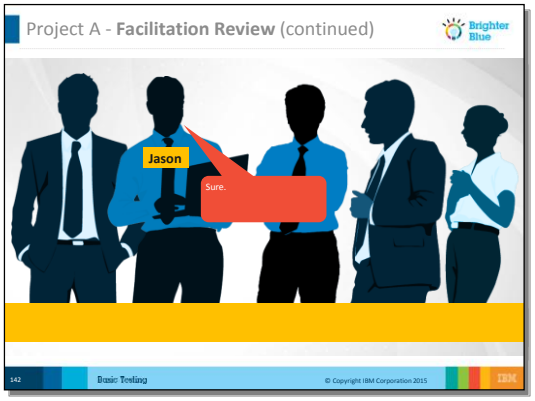
Slide Content	Instructor Guide	Use this space for your notes
▪ Takes confirmation on each recorded error ▪ Participates as an inspector		

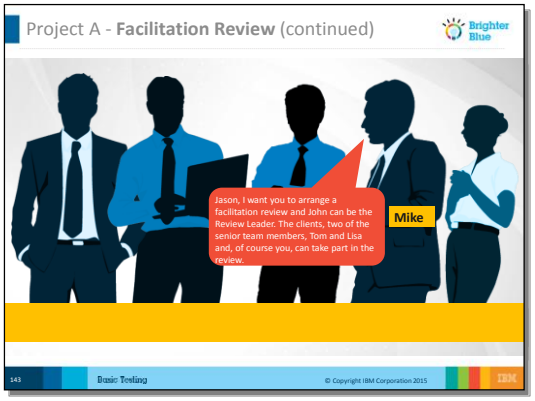
Slide Content	Instructor Guide	Use this space for your notes
Slide 137  ▪ **Planning: Choose team, materials, and dates** ▪ **Orientation: Present product, process and goals (Kick-off)** ▪ **Preparation: Check product, and note issues** ▪ **Review Meeting: Consolidate issues** ▪ **Rework: Correct defects** ▪ **Verify: Verify product / process quality (Follow-up)**	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

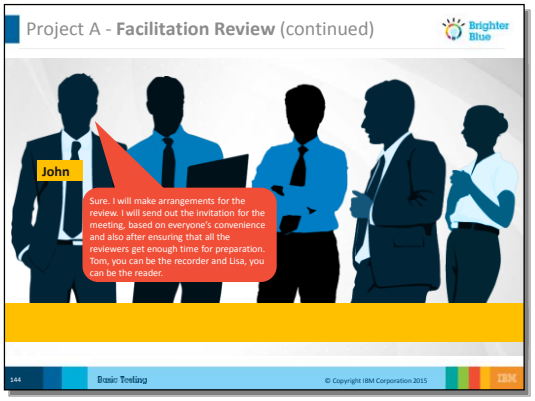
Slide Content	Instructor Guide	Use this space for your notes
Slide 138  **The clients have just specified their requirement to the onsite coordinator, Jason. After which, he went to the site to meet up with the team to discuss it, to ensure that everything as expected is covered and nothing gets missed out.**	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

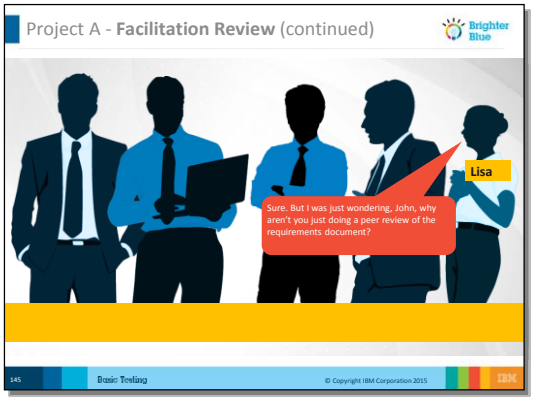
Slide Content	Instructor Guide	Use this space for your notes
Slide 139 	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

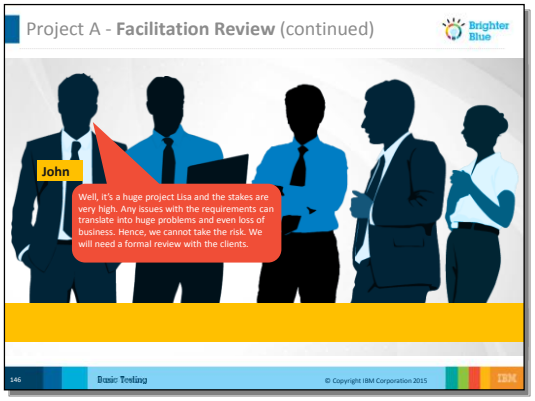
Slide Content	Instructor Guide	Use this space for your notes
Slide 140  	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

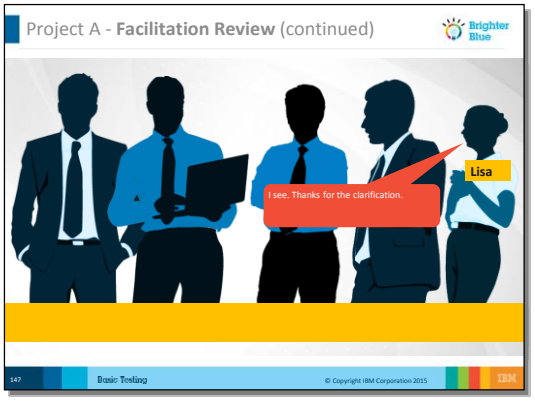
Slide Content	Instructor Guide	Use this space for your notes
Slide 141 	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

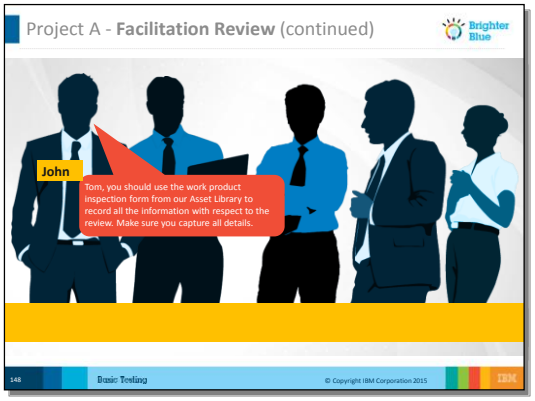
Slide Content	Instructor Guide	Use this space for your notes
Slide 142 	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 143  	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

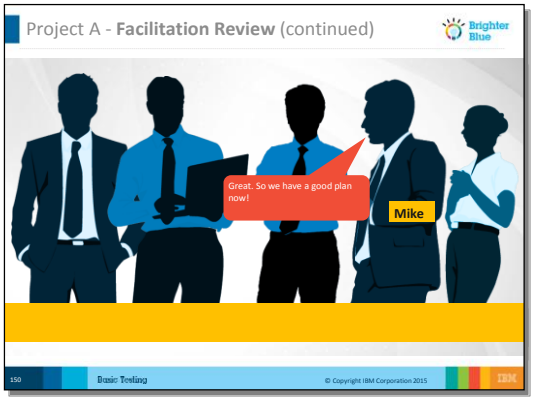
Slide Content	Instructor Guide	Use this space for your notes
Slide 144 	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

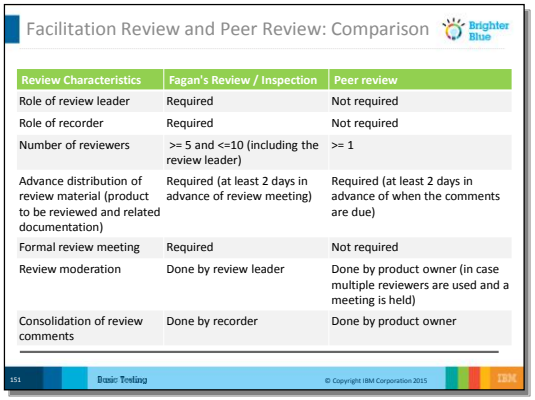
Slide Content	Instructor Guide	Use this space for your notes
Slide 145  	**Purpose:** Use one Facilitation Review scenario to understand how facilitation review is conducted. **Approximate Duration:** 1 min **What to cover:** ▪ Understand how facilitation review is conducted.	

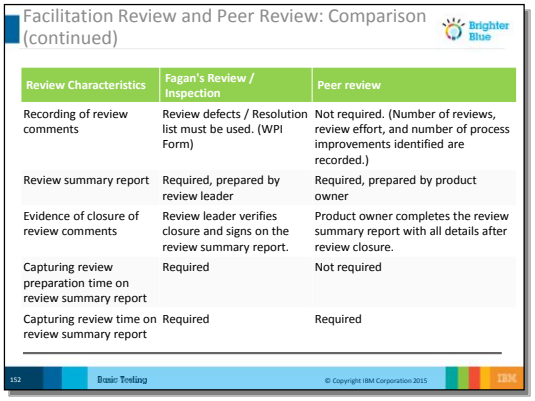
Slide Content	Instructor Guide	Use this space for your notes
Slide 146  	**Purpose:** To compare between facilitation review and peer review **Approximate Duration:** 2 mins **What to cover:** ▪ Compare between facilitation review and peer review based on the review characteristics such as: o Role of review leader o Role of recorder o Number of reviewers o Advance distribution of review material o Formal review meeting o Review moderation o Consolidation of review comments	

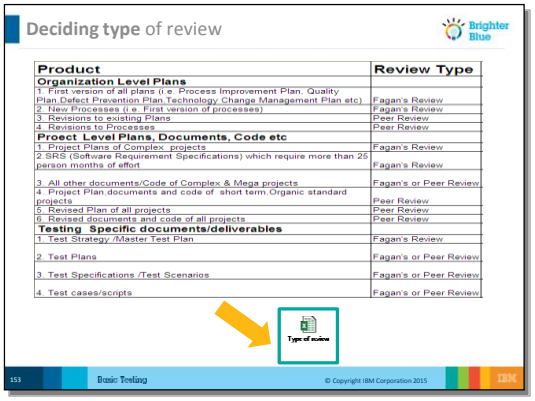
Slide Content	Instructor Guide	Use this space for your notes
Slide 147  	**Purpose:** To compare between facilitation review and peer review **Approximate Duration:** 2 mins **What to cover:** ▪ Compare between facilitation review and peer review based on the review characteristics such as: ○ Recording of review comments ○ Review summary report ○ Evidence of closure of review comments ○ Capturing review preparation time on review summary report ○ Capturing review time on required review summary report	

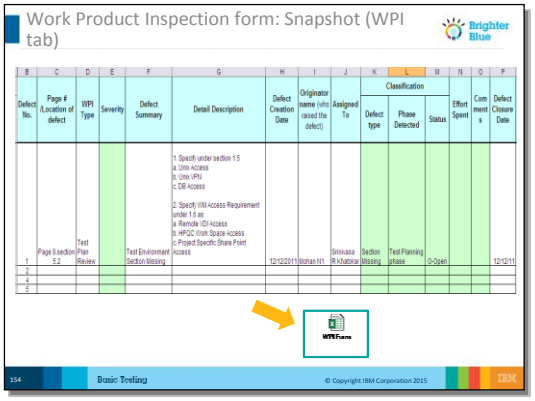
Slide Content	Instructor Guide	Use this space for your notes
Slide 148  	**Purpose:** To give examples of deciding type of review for different cases **Approximate Duration:** 2 mins **What to cover:** ▪ Show the screen capture ▪ Double click on the embedded file icon to open it. ▪ Display it to the participants. ▪ Give examples of deciding type of review for different cases as in the embedded file.	

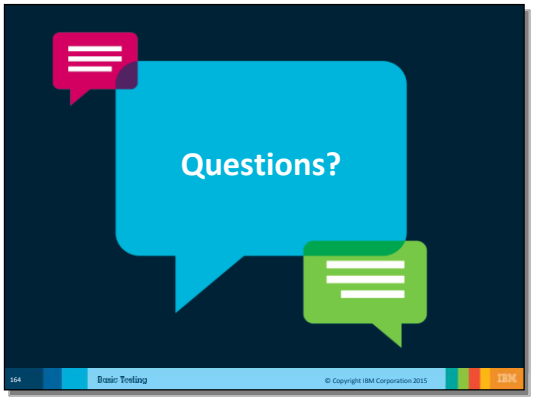
Slide Content	Instructor Guide	Use this space for your notes
Slide 149  	**Purpose:** To give an example of work product inspection form **Approximate Duration:** 2 mins **What to cover:** ▪ Show the screen capture ▪ Double click on the embedded file icon to open it. ▪ Display it to the participants. ▪ Give an example of work product inspection form as in the embedded file.	

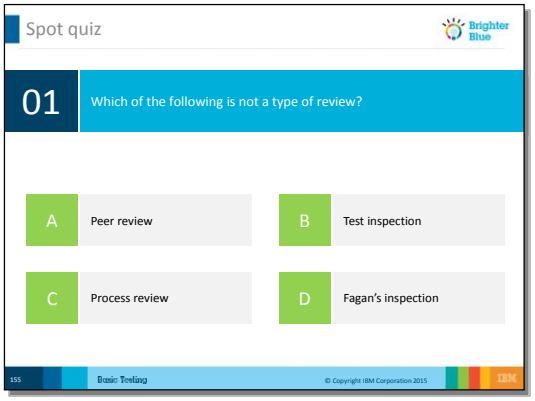
Slide Content	Instructor Guide	Use this space for your notes
Slide 150 	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items. ▪ Use this activity to recap the key takeaways from this module.	

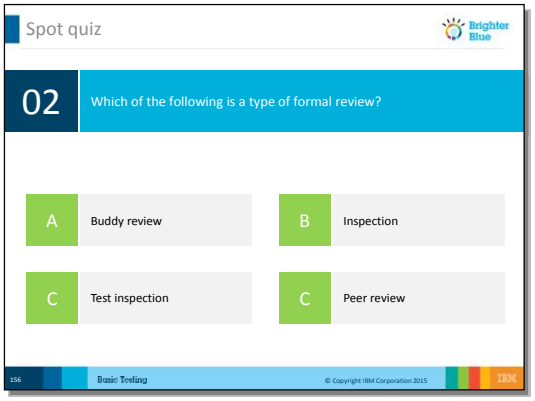
Slide Content	Instructor Guide	Use this space for your notes
Slide 151 	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Test inspection**	

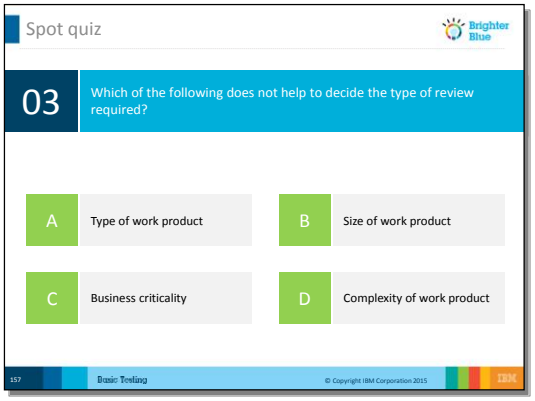
Slide Content	Instructor Guide	Use this space for your notes
Slide 152 	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants to refer to the Supporting Documents_Day 1 folder for the embedded files. ▪ Ask them the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Inspection**	

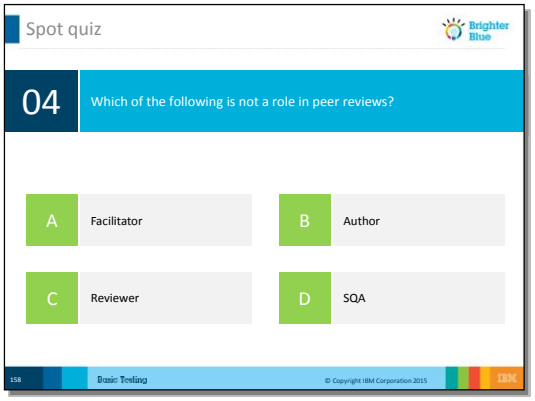
Slide Content	Instructor Guide	Use this space for your notes
Slide 153  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants to refer to the Supporting Documents_Day 1 folder for the embedded files. ▪ Ask them the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Size of work product**	

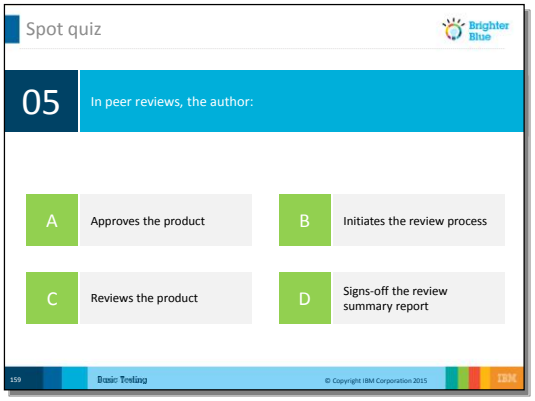
Slide Content	Instructor Guide	Use this space for your notes
Slide 154 	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items. ▪ Use this activity to recap the key takeaways from this module.	

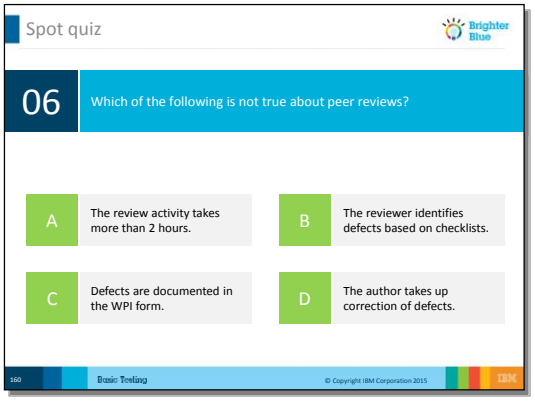
Slide Content	Instructor Guide	Use this space for your notes
Slide 155  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **A. Facilitator**	

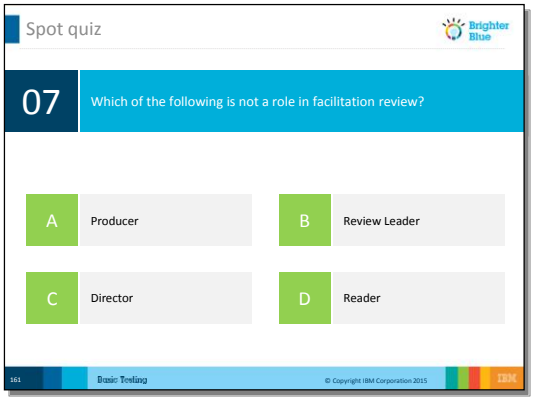
Slide Content	Instructor Guide	Use this space for your notes
Slide 156  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Initiates review process**	

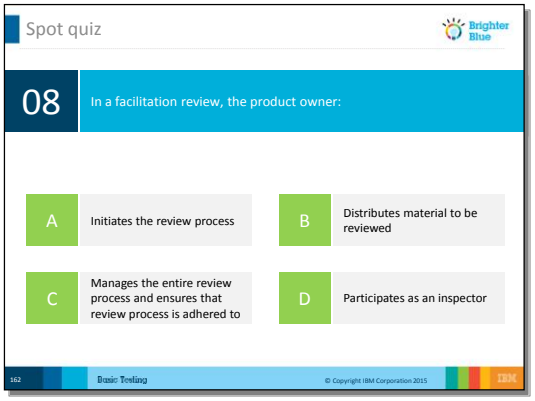
Slide Content	Instructor Guide	Use this space for your notes
Slide 157  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **A. The review activity takes more than 2 hours.**	

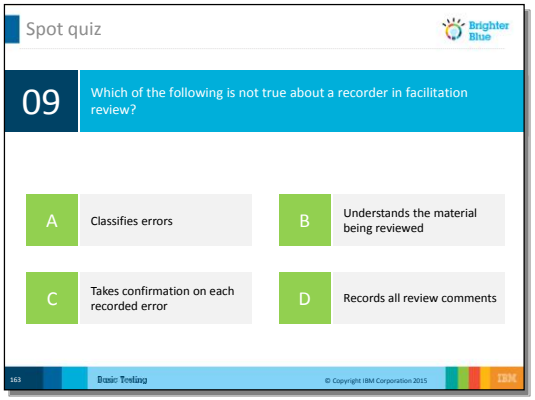
Slide Content	Instructor Guide	Use this space for your notes
Slide 158  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **C. Director**	

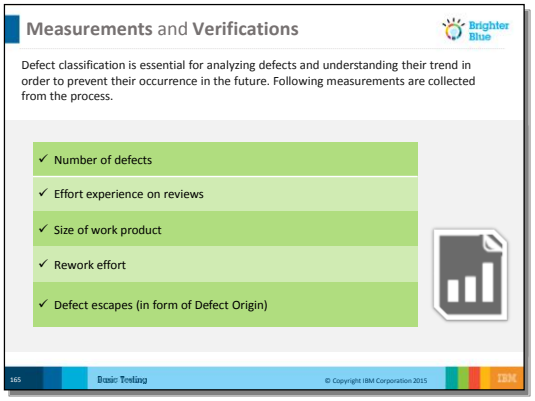
Slide Content	Instructor Guide	Use this space for your notes
Slide 159  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **A. Initiates the review process**	

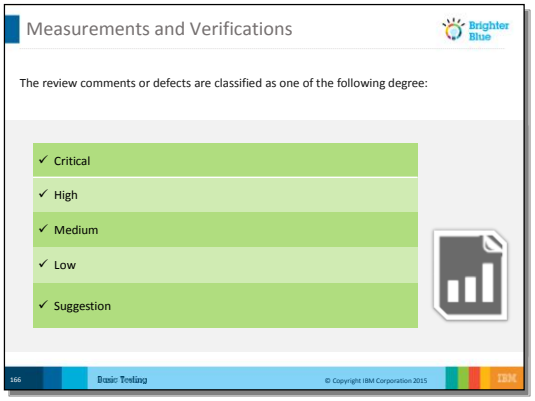
Slide Content	Instructor Guide	Use this space for your notes
Slide 160  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Understands the material being reviewed**	

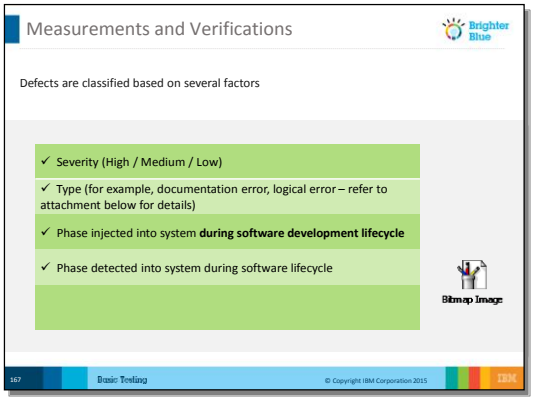
Slide Content	Instructor Guide	Use this space for your notes
Slide 161  	**Purpose:** To explain defect classification **Approximate Duration:** 2 mins **What to cover:** ▪ Explain the measurements which are collected from a process. ▪ Explain the classification of defects. ▪ Explain the factors based on which defects are classified. ▪ Double click on the embedded file icon to open it. ▪ Display it to the participants. ▪ Give examples of the types of defects as in the embedded file.	

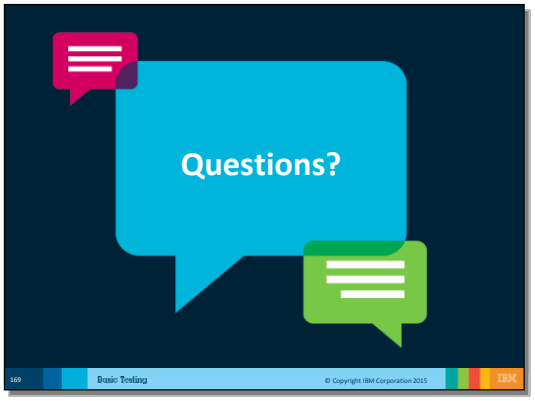
Slide Content	Instructor Guide	Use this space for your notes
Slide 162  	**Purpose:** To explain defect classification **Approximate Duration:** 2 mins **What to cover:** ▪ Explain the measurements which are collected from a process. ▪ Explain the classification of defects. ▪ Explain the factors based on which defects are classified. ▪ Double click on the embedded file icon to open it. ▪ Display it to the participants. ▪ Give examples of the types of defects as in the embedded file.	

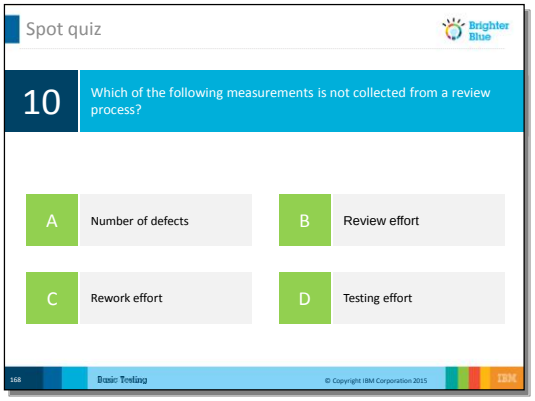
Slide Content	Instructor Guide	Use this space for your notes
Slide 163  	**Purpose:** To explain defect classification **Approximate Duration:** 2 mins **What to cover:** ▪ Explain the measurements which are collected from a process. ▪ Explain the classification of defects. ▪ Explain the factors based on which defects are classified. ▪ Double click on the embedded file icon to open it. ▪ Display it to the participants. ▪ Give examples of the types of defects as in the embedded file.	

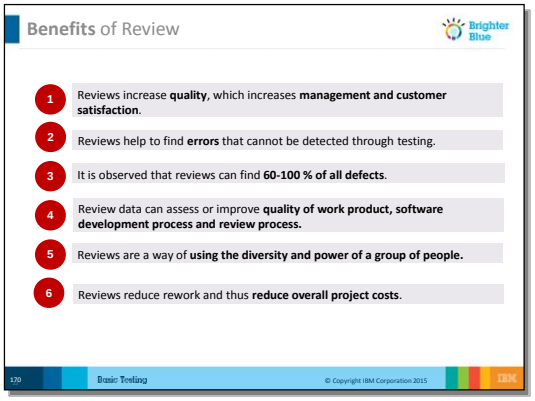
Slide Content	Instructor Guide	Use this space for your notes
Slide 164  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **D. Testing effort**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 165  	**Purpose:** To explain the benefits of reviews **Approximate Duration:** 5 mins **What to cover:** ▪ Explain the benefits of reviews: ○ Increased quality and management and customer satisfaction ○ Errors undetected through testing can be found through reviews ○ 60-100% of defects detected ○ Improved quality of work product, software development process and review process ○ Use of diversity and power of a group of people ○ Reduced rework and overall project costs	

Slide Content	Instructor Guide	Use this space for your notes
Slide 166  	**Purpose:** Describe benefit of review **Approximate Duration:** 2 min **What to cover:** ▪ Describe benefit of review. ▪ Explain how timely reviews help in keeping project cost under control with the help of the graph. ▪ For the bitmap image, ask the participants to refer to Supporting Documents_Day 1 folder.	

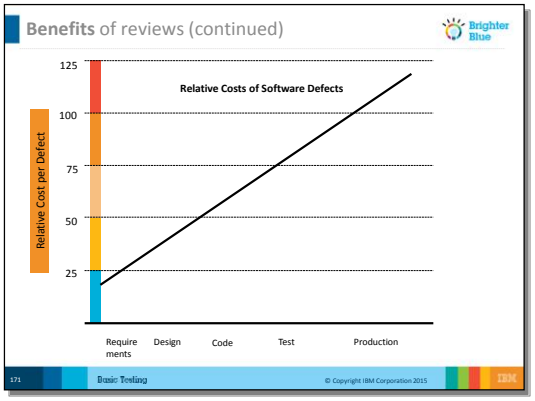
Slide Content	Instructor Guide	Use this space for your notes
Slide 167 	**Purpose:** To discuss the important aspects of Return on Investment **Approximate Duration:** 5 mins **What to cover:** ▪ Discuss the important aspects of Return on Investment. ▪ Explain how Return on Investment is calculated by using the whiteboard	

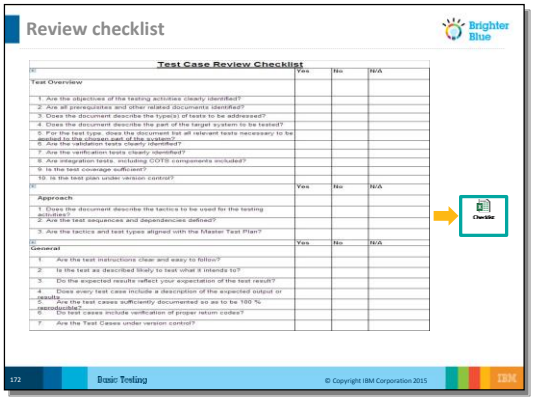
Slide Content	Instructor Guide	Use this space for your notes
Slide 168  	**Purpose:** To describe review checklist **Approximate Duration:** 2 mins **What to cover:** ▪ Show the screen capture ▪ Double click on the embedded file icon to open it. ▪ Display it to the participants. ▪ Describe review checklist as in the embedded file.	

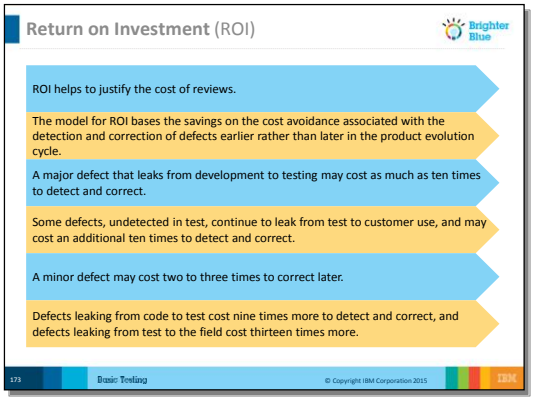
Slide Content	Instructor Guide	Use this space for your notes
Slide 169  ▪ **Reviews increase quality–which increases management and customer satisfaction.** ▪ **Reviews help to find errors that cannot be detected through testing.** ▪ **It is observed that reviews can find 60-100% of all defects.** ▪ **Review data can assess or improve quality of:** ○ **Work product**	**Approximate Duration:** 20 mins **Additional Materials or Pre-session prep tasks**: ▪ Embedded MS Word file **What to cover:** ▪ Explain the scenario. ▪ Tell the participants what they need to do. ▪ Ensure they have the Word file ▪ Review the outcome of the exercise. **Purpose:** To see if the participants are able to carry out a review, and incorporate valid suggestions made in the review **Instructions for the participants:** ▪ Ask participants to form groups of three. ▪ Assign roles to the participants.	

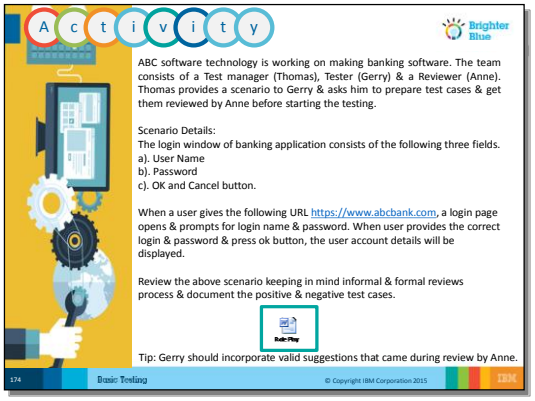
Slide Content	Instructor Guide	Use this space for your notes
○ **Software development process** ○ **Review process** ▪ **Reviews are a way of using the diversity and power of a group of people to:** ○ **Detect defects (errors)** ○ **Remove defects as close to the point of insertion as possible** ○ **Determine product progress or status** ○ **Identify potential improvements** ○ **Produce technical work of a more uniform and predictable quality** ○ **Gain ownership by the project team** ○ **Establish an audit trail from systems requirements allocated to software through the successive phases of development (traceability)** ▪ **Reviews reduce rework and thus reduce overall project cost.**	▪ Tell participants what the desired outcome of the exercise is. **Debrief:** ▪ Ask the participants about their experience of this role play activity. ▪ Ask the participants how much this activity will help them in applying the learned concepts in real life scenario. **Note to SME:** Please validate above instruction.	

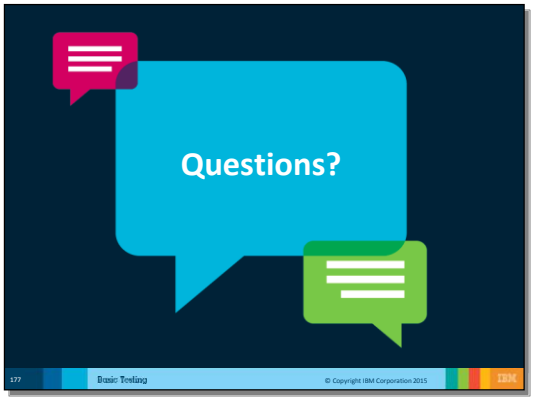
Slide Content	Instructor Guide	Use this space for your notes
○ Rework accounts for 44% of development cost while review accounts for only 15%. ○ The 44% is split as: Requirement (1%), Design (12%), Coding (12%), and Testing (19%).		

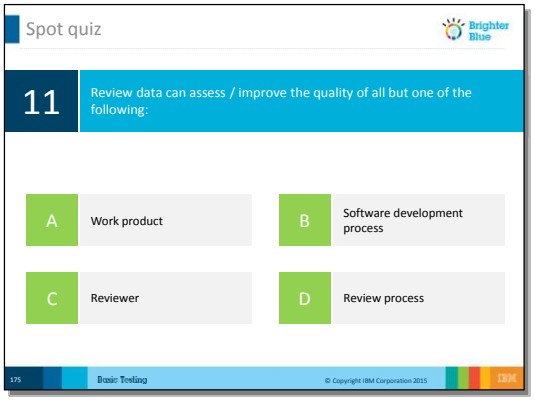
Slide Content	Instructor Guide	Use this space for your notes
Slide 170  **It is observed that the maximum number of defects are injected during requirements phase itself (Requirements – 56%, Design – 27%, Code – 7%, Other – 10%). Upstream defect removal is 10 to 100 times cheaper. Hence, timely reviews help in keeping project cost under control.**	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **C. Reviewer**	

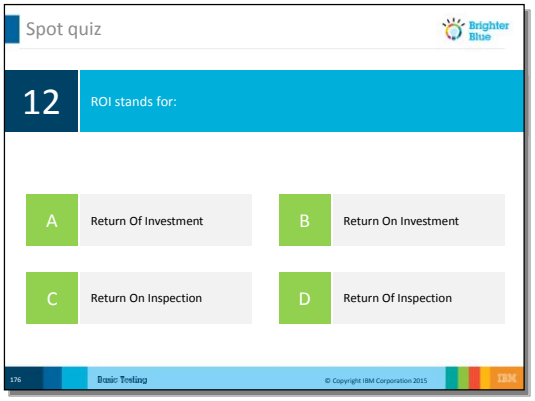
Slide Content	Instructor Guide	Use this space for your notes
Slide 171  	**Purpose:** To check the knowledge of participants and break the monotony **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants to refer to the Supporting Documents_Day 1 folder for the embedded files. ▪ Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Return On Investment**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 172  **Note:** **ROI helps to justify the cost of reviews. It is calculated as follows:** **ROI = Net Savings / Detection Cost** **Where, Net Savings = Cost to Repair – Cost Avoidance** **Detection Cost = Cost of Preparation Effort + Cost of Effort to Conduct Review**	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items. ▪ Use this activity to recap the key takeaways from this module.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 173  	**Purpose:** To go over the objectives of this module. **Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants to refer to the Supporting Documents_Day 1 folder for the embedded files. ▪ Define a test and a test technique. ▪ Describe in details the testing design techniques. ▪ Recognize the test conditions and designing test cases. ▪ List the categories of test design techniques. ▪ Identify the objectives of a good test case. ▪ Recall how to document and design test cases.	

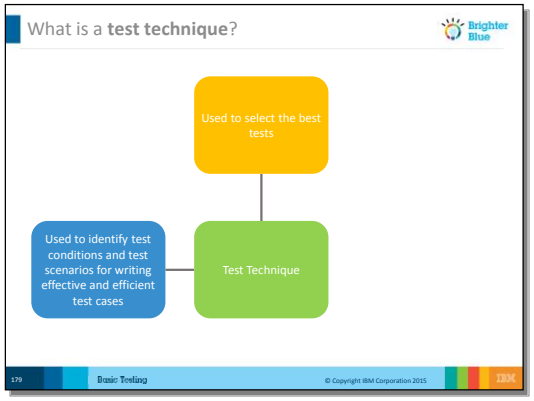
Slide Content	Instructor Guide	Use this space for your notes
Slide 174 	**Purpose:** To discuss the important aspects of Return on Investment **Approximate Duration:** 5 mins **What to cover:** ▪ Discuss the important aspects of Return on Investment. ▪ Explain how Return on Investment is calculated by using the whiteboard	

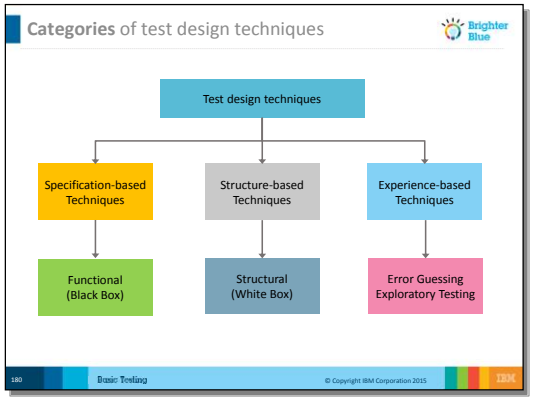
Slide Content	Instructor Guide	Use this space for your notes
Slide 175  		

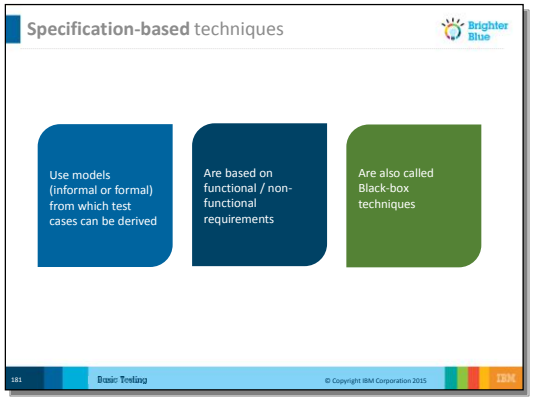
Slide Content	Instructor Guide	Use this space for your notes
Slide 176  		

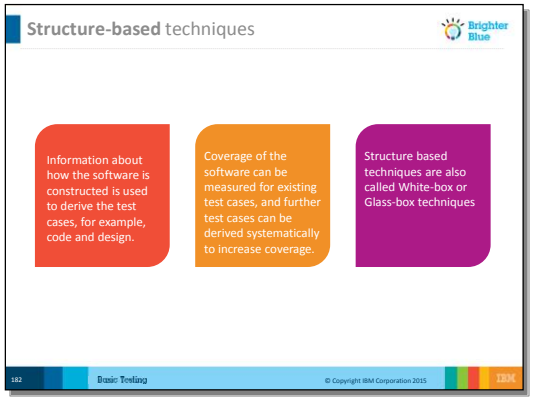
Module 4: Testing Design Techniques and Approach

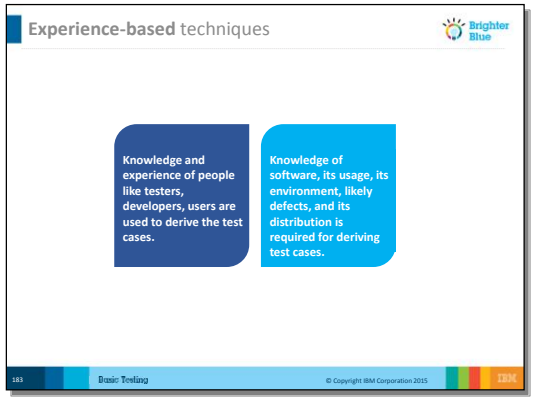
Slide Content	Instructor Guide	Use this space for your notes
Slide 177  **At the end of this module, you should be able to:** ▪ **Identify test conditions for designing test cases** ▪ **Describe categories and types of test design techniques** ▪ **List the components of a test case template**	**Purpose:** To go over the objectives of this module. **Approximate Duration:** 2 mins **What to cover:** ▪ Define a test and a test technique. ▪ Describe in details the testing design techniques. ▪ Recognize the test conditions and designing test cases. ▪ List the categories of test design techniques. ▪ Identify the objectives of a good test case. ▪ Recall how to document and design test cases. techniques	

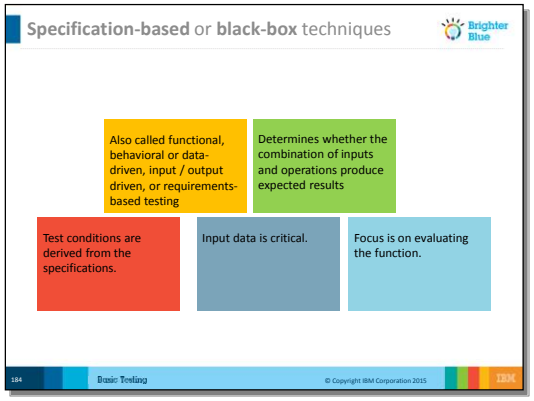
Slide Content	Instructor Guide	Use this space for your notes
Slide 178  ▪ **Test techniques are used to select the best tests from the infinite number of all possible tests for a given system.** ▪ **Test techniques are used to identify test conditions and test scenarios through which effective and efficient test cases can be written.** ▪ **Test techniques help to achieve high test coverage and define tests that will provide insight into the quality of the test object.**	**Purpose:** To explain what is a test technique. **Approximate Duration:** 2 mins **What to cover:** ▪ Define what a test technique is. ▪ Describe its characteristics. ▪ To explain the test techniques better refer to the notes section of the slide (also in the Slide Content Column of this document).	

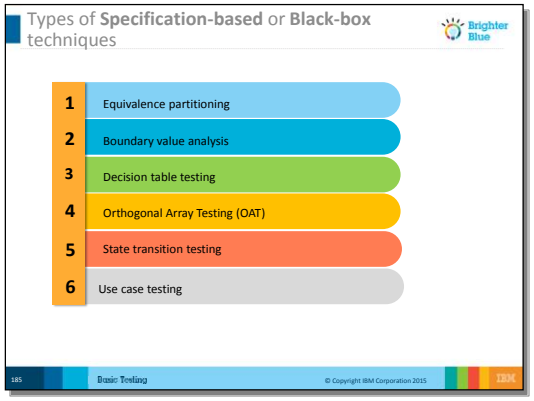
Slide Content	Instructor Guide	Use this space for your notes
Slide 179  	**Purpose:** To point out the categories of test design techniques. **Approximate Duration:** 5 mins **What to cover:** ▪ List the categories of test design techniques for the participants. ▪ Mention to the participants that each of the categories will be explained in details in the following slides. ▪ Tell the participants that we will begin with the Specification-based technique.	

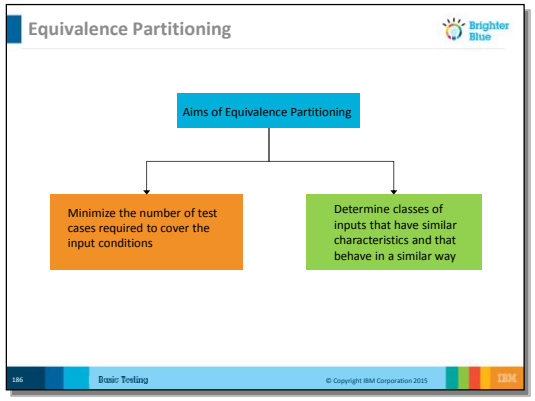
Slide Content	Instructor Guide	Use this space for your notes
Slide 180  	**Purpose:** To explain Specification-based techniques. **Approximate Duration:** 2 mins **What to cover:** Point out the important aspects of the Specification-based techniques.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 181  	**Purpose:** To describe Structure-based techniques **Approximate Duration:** 2 mins **What to cover:** ▪ Explain to the participants the key aspects of the Structure-based techniques.	

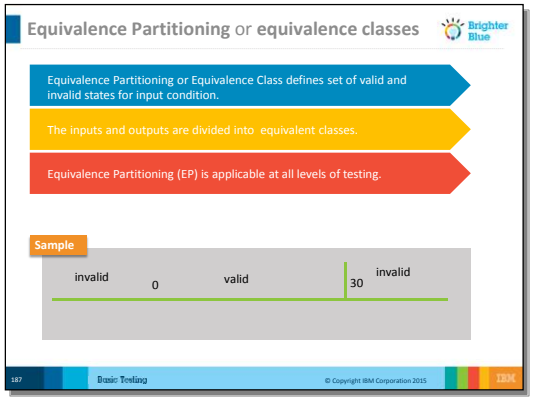
Slide Content	Instructor Guide	Use this space for your notes
Slide 182  	**Purpose:** To define Experience-based techniques **Approximate Duration:** 2 mins **What to cover:** • Tell the participants that Experience-based techniques is the third category. • Use the slide to define this type of technique to them.	

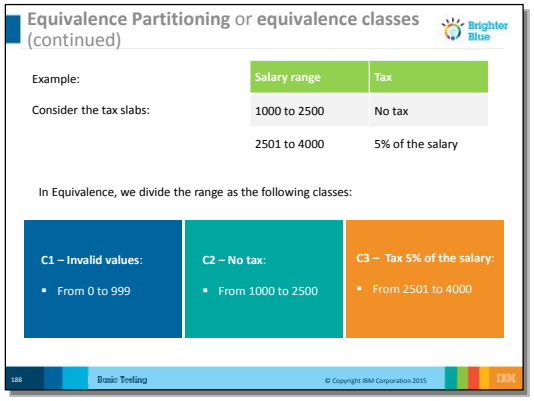
Slide Content	Instructor Guide	Use this space for your notes
Slide 183  	**Purpose:** To explain Specification-based or Black-box techniques **Approximate Duration:** 5 mins **What to cover:** ▪ Mention it to the participants that Specification-based technique is also known as Black-box technique. ▪ There are five primary points to remember while describing this kind of technique. Point these out to the learners from the slide. ▪ Also tell them that in the next slide we will take up the types of Black-box technique.	

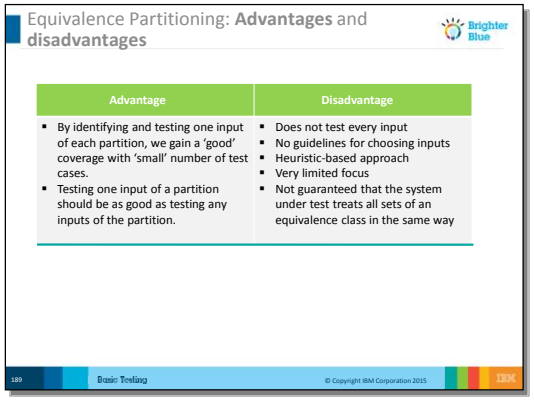
Slide Content	Instructor Guide	Use this space for your notes
Slide 184  	**Purpose:** To identify the types of Specification-based or Black-box techniques. **Approximate Duration:** 2 mins **What to cover:** ▪ There are five types of Black-box techniques. ▪ Point out the list to the participants.	

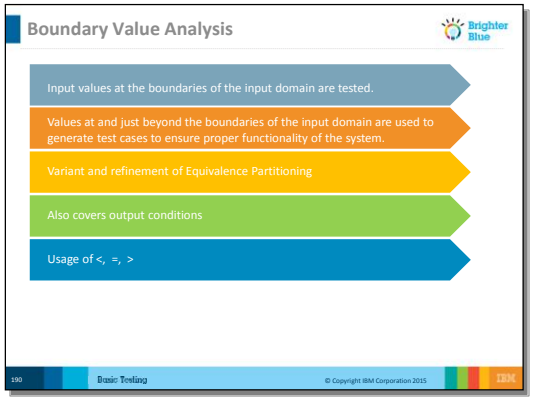
Slide Content	Instructor Guide	Use this space for your notes
Slide 185  **Aims:** ▪ **Minimize the number of test cases required to cover the input conditions** ▪ **Determine classes of inputs that have similar characteristics and that behave in a similar way. Two test cases based on inputs from the same Equivalence Class are likely to reveal same bug.** **Identifying Equivalence Classes (ECs):**	**Purpose:** ▪ To describe Equivalence Partitioning and its aims **Approximate Duration:** 5 mins **What to cover:** ▪ Describe in details what Equivalence Partitioning is. ▪ While describing this refer to the notes section of the slide (also in the Slide Content Column of this document).	

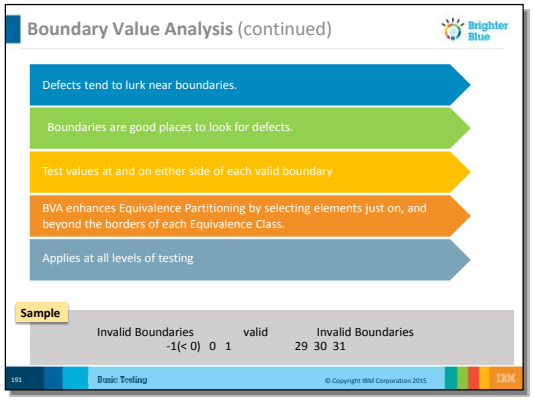
Slide Content	Instructor Guide	Use this space for your notes
▪ If input specifies a RANGE of valid values, define one EC in the range and one outside each end. ▪ If input specifies a NUMBER (N) of valid values, define one valid EC and two invalid ECs (none, and more than N). ▪ If input specifies a SET of valid values, define one valid EC (within) and one invalid EC (not in). ▪ If there is a reason to believe the program handles each valid input differently, each valid input is an EC. ▪ If input specifies Must Be situation, define one valid EC (is) and one invalid EC (is not). ▪ If there is a reason to believe that elements in an EC are not handled in an identical manner, subdivide the EC into smaller ECs.		

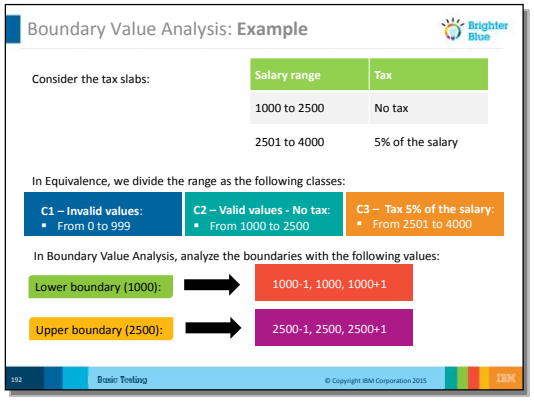
Slide Content	Instructor Guide	Use this space for your notes
Slide 186  	**Purpose:** To explain the key aspects of Equivalence Partitioning or Equivalence Classes **Approximate Duration:** 5 mins **What to cover:** ▪ Point out the three aspects highlighted in the slide to explain the key characteristics of Equivalence Partitioning or Equivalence Classes. ▪ There is a sample is the slide which can be used to explain it better. ▪ Also mention to the participants that we will continue with the concept of Equivalence Partitioning is the next slide.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 187  **To put this in simpler words, since it is practically infeasible to do exhaustive testing, the next best alternative is to check whether the program extends similar behavior or treatment to a certain group of inputs. If such a group of values can be found in the input domain, treat them together as one equivalent class and test one representative from this class.**	**Purpose:** To explain how to divide inputs into equivalence classes with an example. **Approximate Duration:** 5 mins **What to cover:** ▪ Use another example from the slide to describe this concept. ▪ Refer to the notes section of the slide (also in the Slide Content Column of this document) for more input on the equivalence classes.	

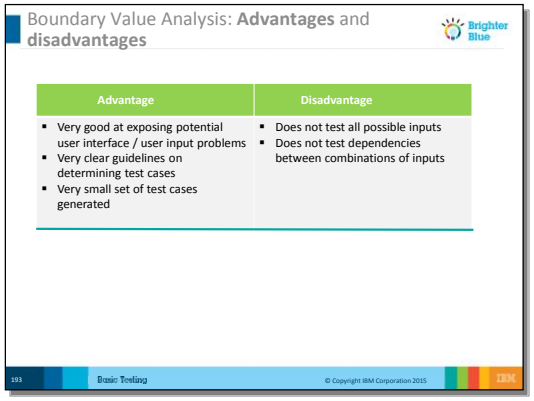
Slide Content	Instructor Guide	Use this space for your notes
Slide 188  	**Purpose:** To discuss the advantages and disadvantages of equivalence partitioning **Approximate Duration:** 5 mins **What to cover:** ▪ Point out the list of advantages and disadvantages of equivalence partitioning to the participants.	

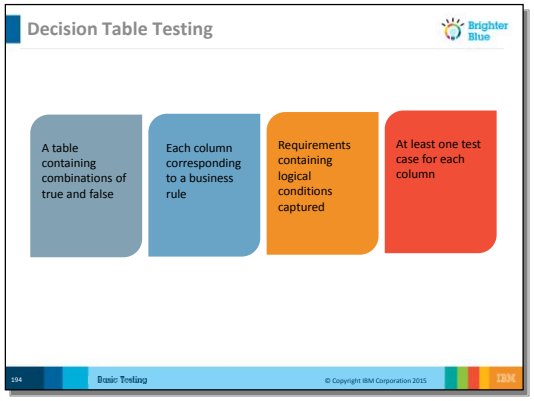
Slide Content	Instructor Guide	Use this space for your notes
Slide 189  	**Purpose:** To describe the key aspects of Boundary Value Analysis. **Approximate Duration:** 5 mins **What to cover:** ▪ Tell the participants that we will browse through the main aspects of Boundary Value Analysis in the next two slides.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 190  	**Purpose:** To describe the key aspects of Boundary Value Analysis. **Approximate Duration:** 5 mins **What to cover:** ▪ As a continuation, keep pointing out the aspects. ▪ There is a sample of Boundary Value Analysis at the end of the slide. Mention it to the participants as you explain it.	

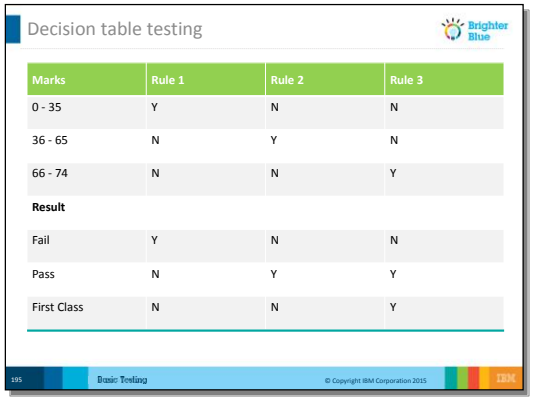
Slide Content	Instructor Guide	Use this space for your notes
Slide 191  **Consider that for salary between 1000-2500----->no tax and from 2501-4000---->tax 5% of their Salary.** **In EQUIVALENCE : we divide the range as the following classes:** **- from 0 to 999 is put together in one class c1- invalid values** **- from 1000 to 2500 in class c2 – no tax- Valid values** **- from 2501 to 4000 in class c3- 5% of their salary**	**Purpose:** To describe how to consider test values for Boundary Value Analysis with an example. **Approximate Duration:** 5 mins **What to cover:** ▪ Tell the participants that this time we will take up an example to describe the Boundary Value Analysis. ▪ Refer to the notes section of the slide (also in the Slide Content Column of this document) to describe it in details to the participants.	

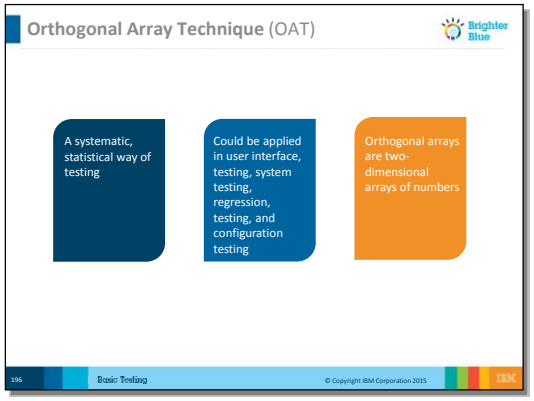
Slide Content	Instructor Guide	Use this space for your notes
Boundary Value: taking the above example We would be analyzing the lower boundary (1000) with values as 1000-1,1000 and 1000+1 Similarly the upper boundary (2500) would be tested with values as 2500-1, 2500 and 2500+1 for the class which is eligible for No Tax. Similarly we can also do the boundary value analysis for the class which is charged with 5% tax.		

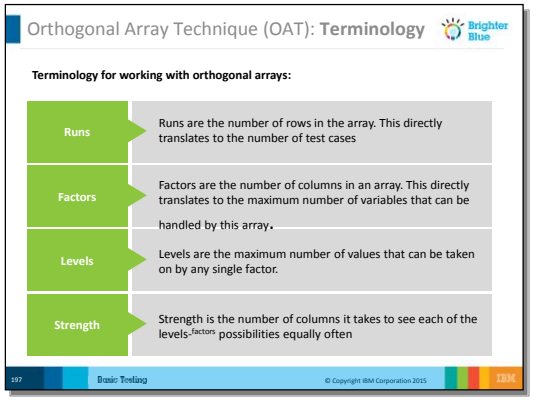
Slide Content	Instructor Guide	Use this space for your notes
Slide 192  	**Purpose:** To discuss the advantages and disadvantages of Boundary Value Analysis **Approximate Duration:** 5 mins **What to cover:** ▪ There are three advantages and two disadvantages in Boundary Value Analysis. ▪ Point these out to the learners.	

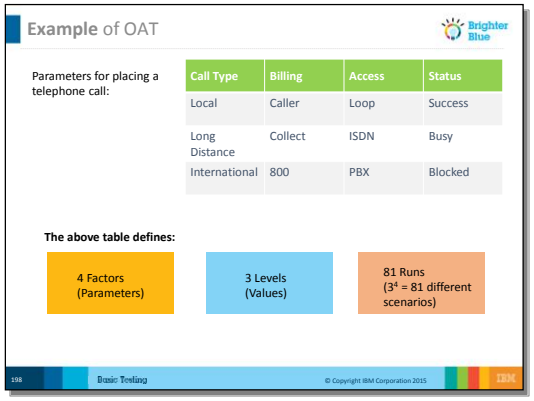
Slide Content	Instructor Guide	Use this space for your notes
Slide 193  **Characteristics of Decision Table Testing:** ▪ **A table contains combinations of true and false for all input conditions and the resulting actions for each.** ▪ **Each column corresponds to a business rule that defines a unique combination of conditions that result in the execution of the actions associated with that rule.** ▪ **Requirements that contain logical conditions can be captured.**	**Purpose:** To describe the characteristics of Decision Table Testing **Approximate Duration:** 5 mins **What to cover:** ▪ Explain the characteristics of Decision Table Testing to the participants. ▪ Refer to the notes section of the slide (also in the Slide Content Column of this document) to describe Decision Table Testing in details to the participants.	

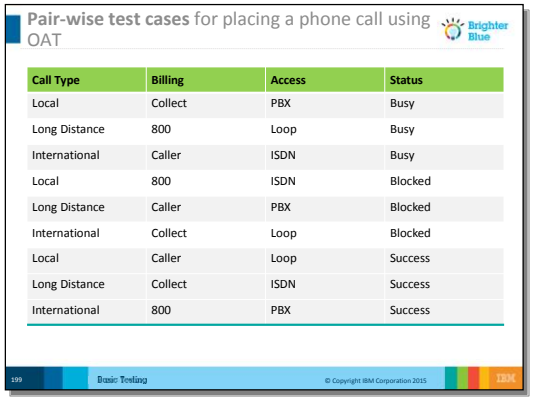
Slide Content	Instructor Guide	Use this space for your notes
▪ At least one test case is written per column.		

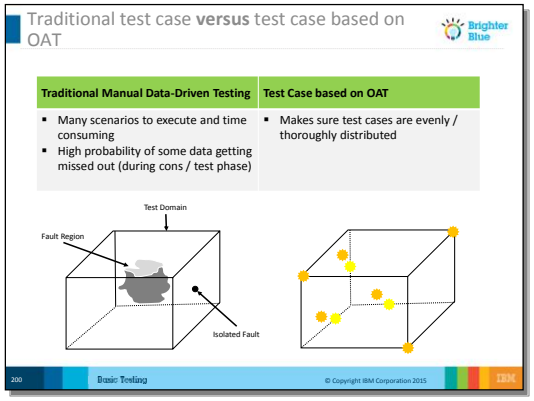
Slide Content	Instructor Guide	Use this space for your notes
Slide 194 	**Purpose:** To discuss Decision Table with the help of an example **Approximate Duration:** 5 mins **What to cover:** ▪ Tell the participants that here is another example to help you understand the concept better. ▪ Describe the example in the tabular format to them.	

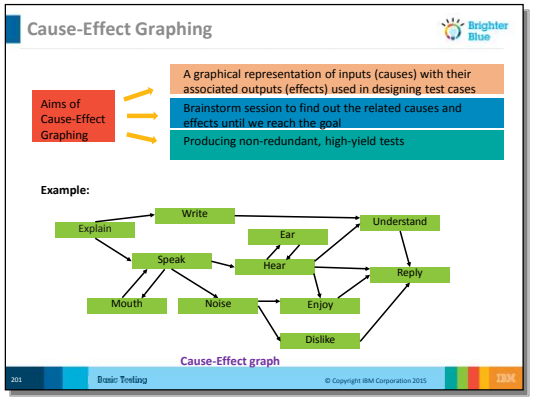
Slide Content	Instructor Guide	Use this space for your notes
Slide 195  **Characteristics of Orthogonal Array Technique:** ▪ **Orthogonal Array Technique / Testing is a systematic, statistical way of testing.** ▪ **Orthogonal arrays could be applied in user interface testing, system testing, regression testing, and configuration testing.** ▪ **Orthogonal arrays are two-dimensional arrays of numbers which possess the interesting quality that by choosing any two columns in the array you receive an even distribution of all the pair-wise combinations of values in the array.**	**Purpose:** To explain the characteristics of Orthogonal Array Technique **Approximate Duration:** 2 mins **What to cover:** ▪ Describe the characteristics of Orthogonal Array Technique. ▪ Refer to the notes section of the slide (also in the Slide Content Column of this document) to describe Orthogonal Array Technique to the participants.	

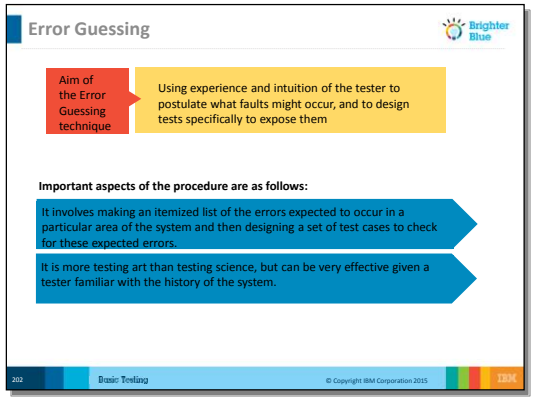
Slide Content	Instructor Guide	Use this space for your notes
Slide 196  	**Purpose:** To identify the key terms that are used in Orthogonal Array Technique (OAT) **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants that there are 4 key terms in Orthogonal Array Technique (OAT). ▪ Name them and then define each of the terms.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 197 	**Purpose:** To explain OAT with the help of an example **Approximate Duration:** 5 mins **What to cover:** ▪ Tell the participants that once more we well take up an example to understand this category of testing technique. ▪ Explain the table and point out the three elements that it defines.	

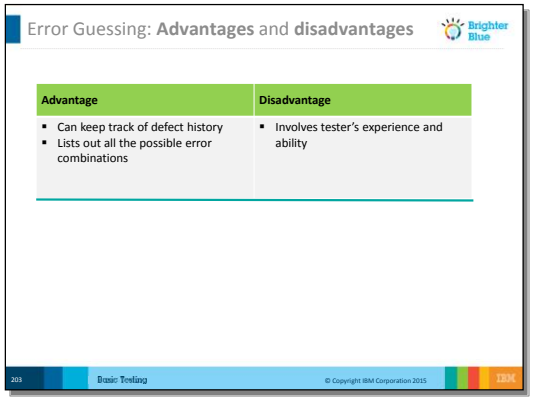
Slide Content	Instructor Guide	Use this space for your notes
Slide 198 	**Purpose:** To explain Pair-wise Test cases **Approximate Duration:** 2 mins **What to cover:** ▪ Pair-wise Test Cases for placing a phone call using OAT is shown in the slide to explain its concept.	

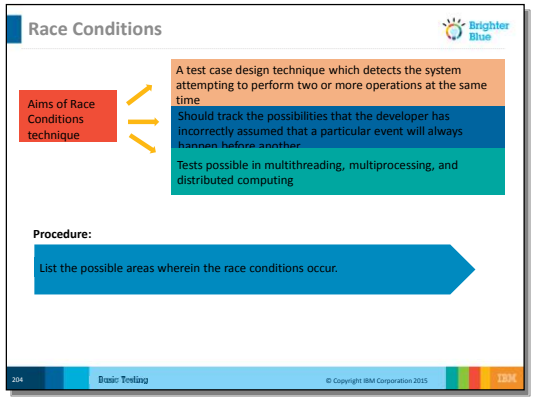
Slide Content	Instructor Guide	Use this space for your notes
Slide 199 	**Purpose:** To compare traditional test cases and test cases based on OAT **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants that we will discuss another aspects of OAT. ▪ Differentiate between traditional test cases and test cases based on OAT.	

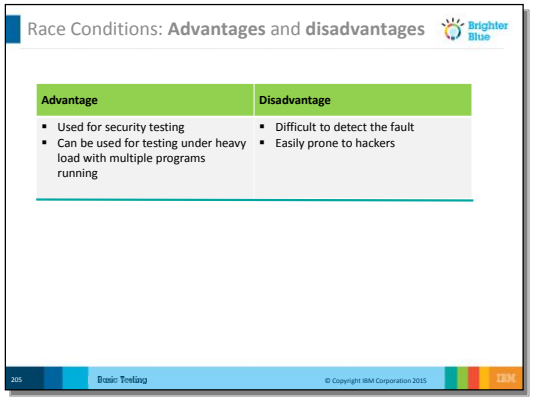
Slide Content	Instructor Guide	Use this space for your notes
Slide 200  	**Purpose:** To describe Cause-Effect Graphing **Approximate Duration:** 5 mins **What to cover:** • Explain to the participants the three aims of the Cause-Effect Graphing. • The diagram is an example that projects the cause and effect graph.	

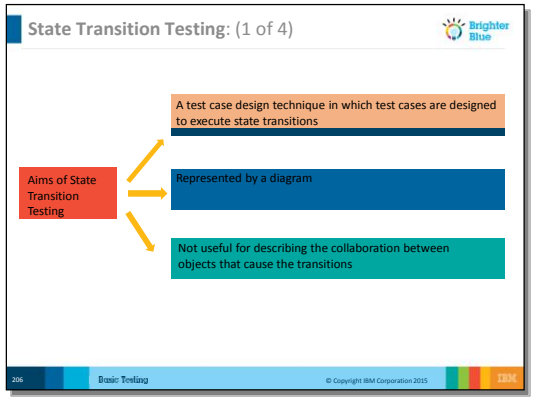
Slide Content	Instructor Guide	Use this space for your notes
Slide 201  **Aim:** ▪ **A test case design technique where the experience and intuition of the tester is used to postulate what faults might occur, and to design tests specifically to expose them** **Procedure:** ▪ **Involves making an itemized list of the errors expected to occur in a particular area of the system and then designing a set of test cases to check for these expected errors**	**Purpose:** To discuss the aim of Error Guessing technique and the key aspects of the procedure **Approximate Duration:** 5 mins **What to cover:** ▪ Begin by explaining the aims of the Error Guessing technique. ▪ Then point out the key aspects of this procedure.	

Slide Content	Instructor Guide	Use this space for your notes
▪ Empty or Null Strings / Lists ▪ Zero Instances / Occurrences ▪ It is more testing art than testing science, but can be very effective given a tester familiar with the history of the system.		

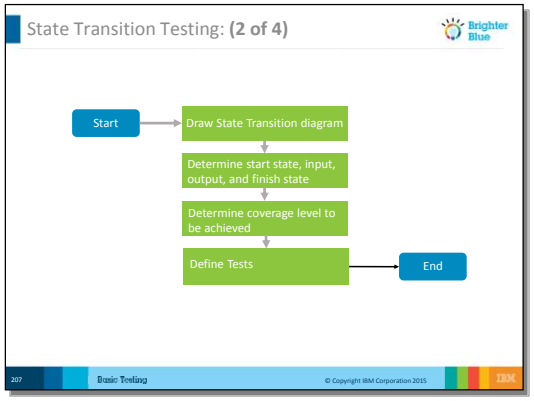
Slide Content	Instructor Guide	Use this space for your notes
Slide 202  	**Purpose:** To point out the advantages and disadvantages of Error Guessing technique **Approximate Duration:** 2 mins **What to cover:** ▪ Tell the participants that like every process, Error Guessing, too has its advantages and disadvantages. ▪ Now point out to list.	

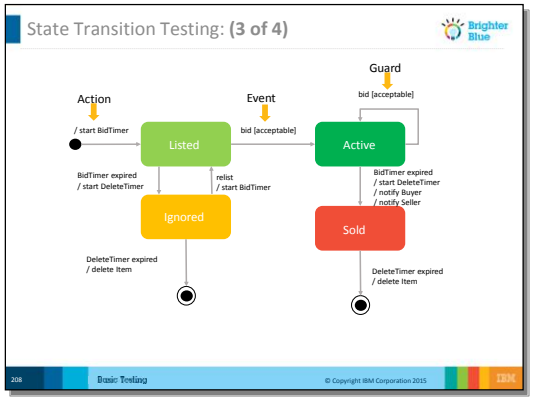
Slide Content	Instructor Guide	Use this space for your notes
Slide 203  **Procedure:** ▪ **List the possible areas wherein the race conditions occur:** ▪ **File open / close** ▪ **Variable checks** ▪ **Signals** ▪ **Access checks** ▪ **Locking**	**Purpose:** To explain Race Condition technique **Approximate Duration:** 5 mins **What to cover:** ▪ Race Condition is another test design technique that the participants will learn about. ▪ Describe its aims and the procedure to the participants. ▪ Refer to the notes section of the slide (also in the Slide Content Column of this document) to describe the procedure further.	

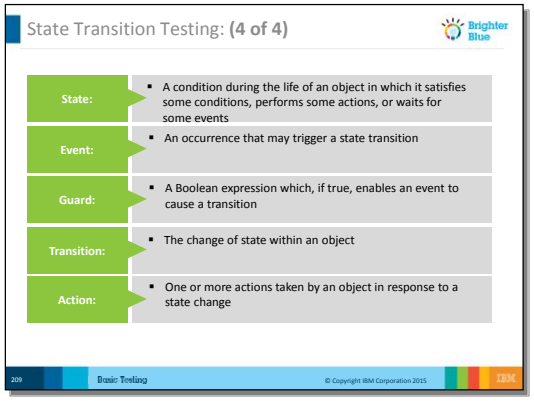
Slide Content	Instructor Guide	Use this space for your notes
Slide 204  	**Purpose:** To explain the advantages and disadvantages of Race Conditions technique **Approximate Duration:** 2 mins **What to cover:** • Point out the advantages and disadvantages of this technique.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 205  **Aim:** ▪ **A test case design technique in which test cases are designed to execute state transitions** ▪ **Represented by a diagram describing:** ▪ **All of the states that an object can have** ▪ **The events under which an object changes state (transitions)** ▪ **The conditions that must be fulfilled before the transition will occur (guards)**	**Purpose:** To define State Transition Testing **Approximate Duration:** 2 mins **What to cover:** ▪ Mention it to the participants that State Transition Testing can be defined with the help of its aims. ▪ Now point out the three main aims of this technique.	

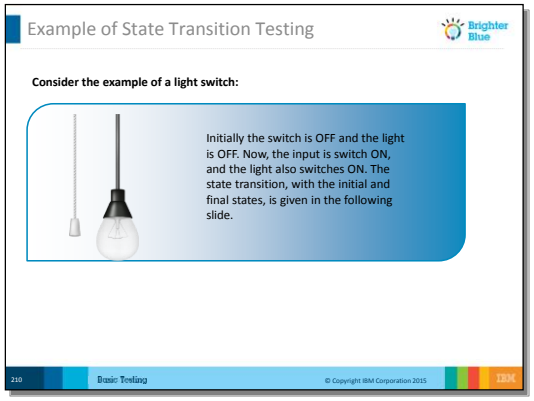
Slide Content	Instructor Guide	Use this space for your notes
▪ The activities undertaken during the life of an object (actions) ▪ Not useful for describing the collaboration between objects that cause the transitions		

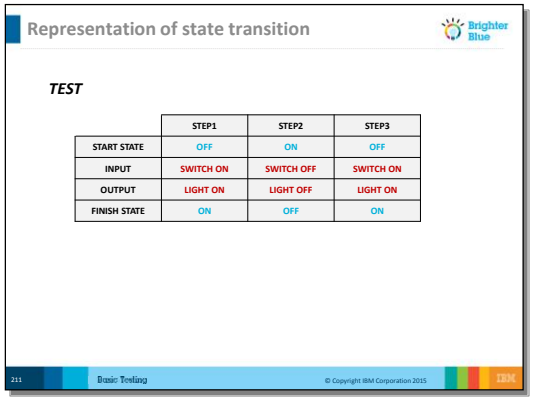
Slide Content	Instructor Guide	Use this space for your notes
Slide 206  	**Purpose:** To describe State Transition Testing procedure **Approximate Duration:** 2 mins **What to cover:** With the help of the diagram briefly explain the procedure of this technique.	

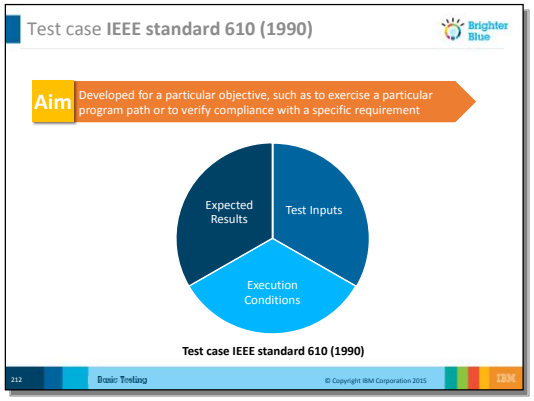
Slide Content	Instructor Guide	Use this space for your notes
Slide 207  	**Purpose:** To describe State Transition Testing diagram **Approximate Duration:** 5 mins **What to cover:** Here is another diagram that will help the participants to understand how State Transition Testing works.	

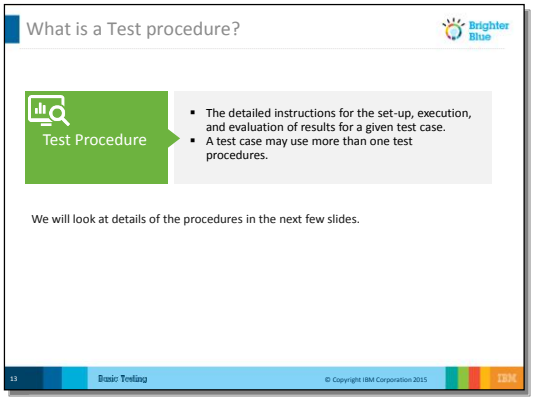
Slide Content	Instructor Guide	Use this space for your notes
Slide 208  Terminology: ▪ Notation: For those not familiar with the notation used for State Transition diagrams, some explanation is in order. ▪ State: A condition during the life of an object in which it satisfies some conditions, performs some actions, or waits for some events ▪ Event: An occurrence that may trigger a state transition. Event types include an explicit signal from outside the system, an invocation from inside the system, the passage of a designated	**Purpose:** To list the key terms used in State Transition Testing **Approximate Duration:** 5 mins **What to cover:** ▪ Tell the participants that once again we are taking up the key terms that are used in a testing design. This time it is for State Transition Testing. ▪ Now describe the terms by following the tabular format.	

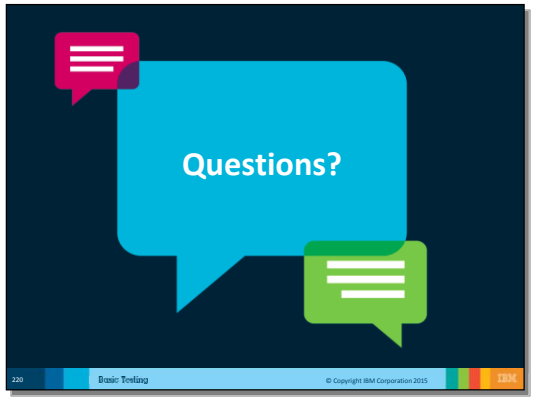
Slide Content	Instructor Guide	Use this space for your notes
period of time, or a designated condition becoming true. ▪ Guard: A Boolean expression which, if true, enables an event to cause a transition ▪ Transition: The change of state within an object ▪ Action: One or more actions taken by an object in response to a state change		

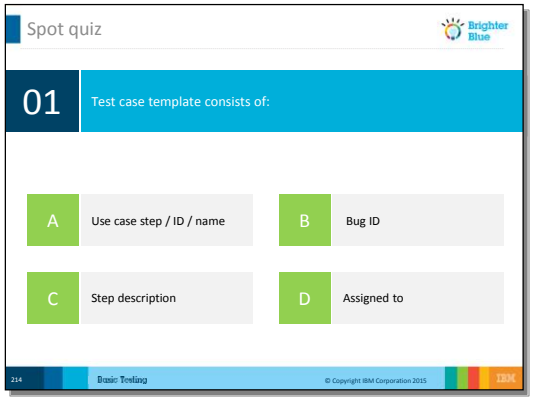
Slide Content	Instructor Guide	Use this space for your notes
Slide 209  	**Purpose:** To explain State Transition Testing with the help of a scenario **Approximate Duration:** 2 mins **What to cover:** Tell the participants that finally here is a scenario that will explain the State Transition Testing better.	

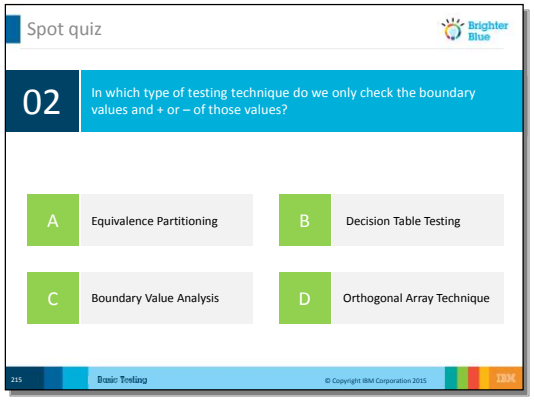
Slide Content	Instructor Guide	Use this space for your notes
Slide 210  	**Purpose:** To point out how State Transition Testing is represented **Approximate Duration:** 2 mins **What to cover:** Show the table to mention how the state transition is represented.	

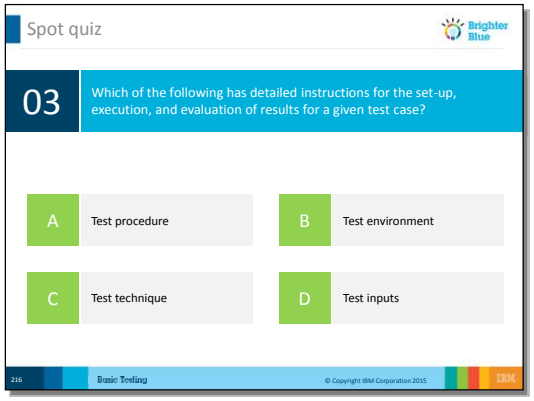
Slide Content	Instructor Guide	Use this space for your notes
Slide 211  **A set of:** ▪ **Test Inputs** ▪ **Execution Conditions** ▪ **Expected Results** **Developed for a particular objective, such as to exercise a particular program path or to verify compliance with a specific requirement**	**Purpose:** To explain Test Case IEEE standard 610 (1990) **Approximate Duration:** 2 mins **What to cover:** ▪ This is an overview for the Test Case IEEE standard 610 (1990). ▪ While explaining it refer to the notes section of the slide (also in the Slide Content Column of this document).	

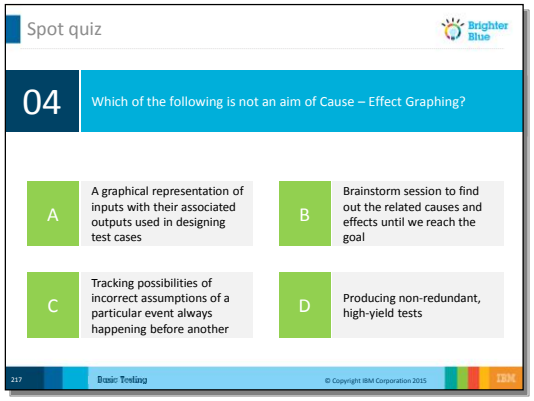
Slide Content	Instructor Guide	Use this space for your notes
Slide 212  	**Purpose:** To explain test procedure **Approximate Duration:** 2 mins **What to cover:** ▪ Introduce the participants to what is test procedure. ▪ This is a new topic. Mention this.	

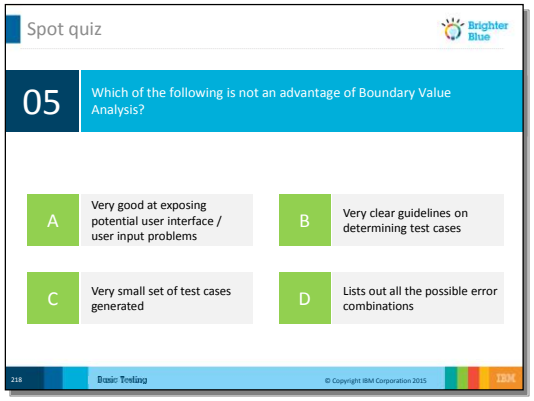
Slide Content	Instructor Guide	Use this space for your notes
Slide 213  **The key takeaways from this module are:** ▪ **Identifying test conditions for designing test cases** ▪ **Describing categories and types of test design techniques** ▪ **Listing the components of a test case template**	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. ▪ Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ This question has multiple correct answers: • The correct answers are:- **A. By identifying and testing one input of each partition, we gain a good coverage with small number of test cases.** **C. Testing one input of a partition should be as good as testing any inputs of the partition.**	

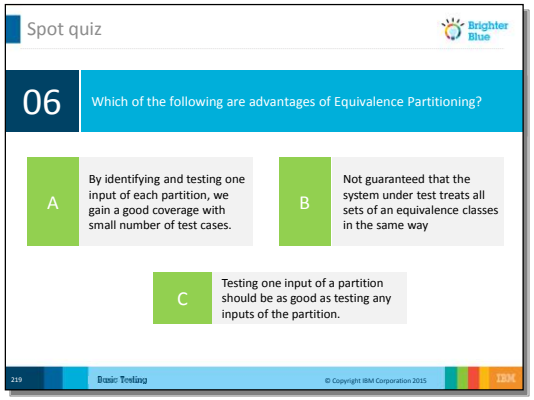
Slide Content	Instructor Guide	Use this space for your notes
Slide 214  	**Approximate Duration:** 2 mins **What to cover:** • Ask the participants if they have any questions. • Include any questions that will be addressed later in the course as parking lot items. • Use this activity to recap the key takeaways from this module.	

Slide Content	Instructor Guide	Use this space for your notes
Slide 215  	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. ▪ Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **B. Selecting the best test.**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 216  	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. ▪ Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **C. Boundary Value Analysis.**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 217  	**Approximate Duration:** 2 mins **What to cover:** • Ask the participants the question on the slide. Treat the question as a poll - Ask them to raise their hands based on their answer. • The correct answer is **A. Test procedure.**	

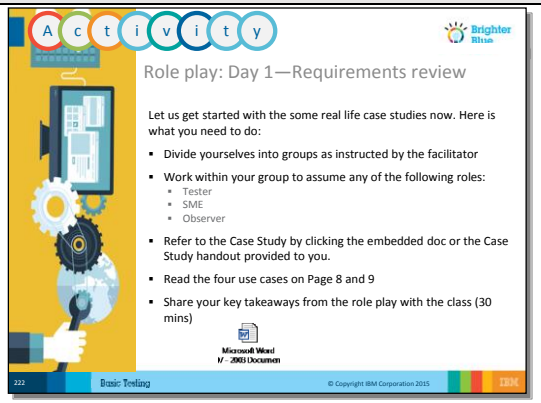
Slide Content	Instructor Guide	Use this space for your notes
Slide 218  	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. ▪ Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **C. Tracking possibilities of incorrect assumptions of a particular event always happening before another.**	

Slide Content	Instructor Guide	Use this space for your notes
Slide 219  	**Approximate Duration:** 2 mins **What to cover:** ▪ Ask the participants the question on the slide. ▪ Treat the question as a poll - Ask them to raise their hands based on their answer. ▪ The correct answer is **D. Lists out all the possible error combinations.**	

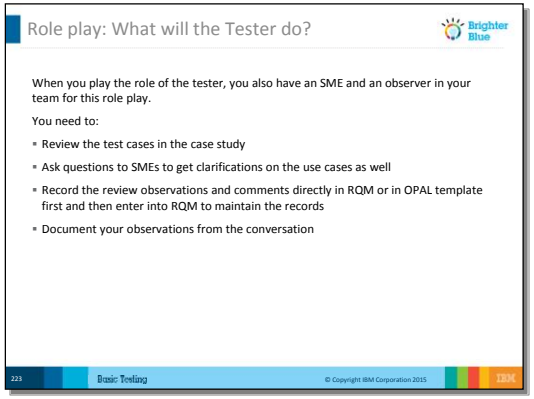
Module 5: Case Study

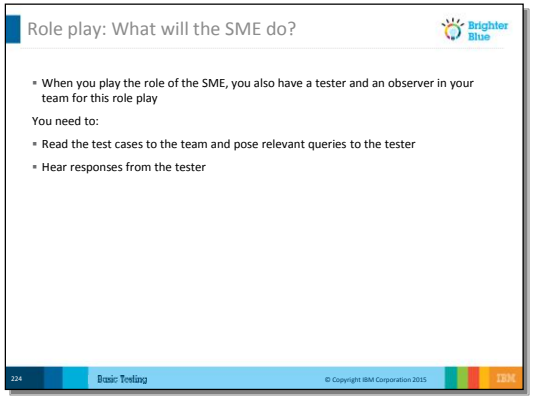
Slide Content	Instructor Guide	Use this space for your notes
Slide 220		

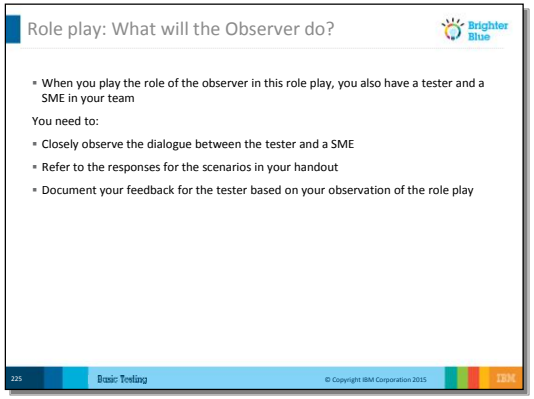
Slide Content	Instructor Guide	Use this space for your notes
 **At the end of this module, you should be able to:** • **Review the test cases in the case study** • **Ask questions to SMEs to get clarifications on the use cases** • **Record the review observations and comments directly in OPAL template first and then enter into RQM to maintain the records**		
Slide 221	**Approximate Duration:** 20 mins	

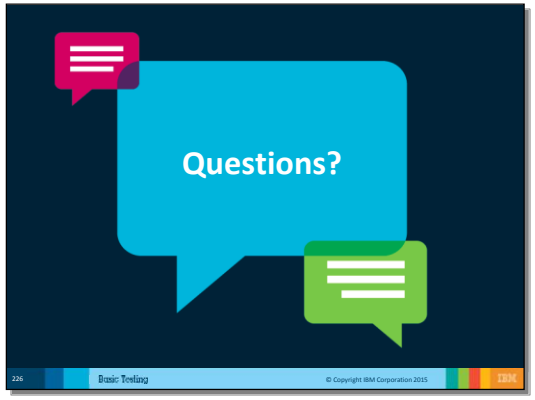
Slide Content	Instructor Guide	Use this space for your notes
	**Additional Materials or Pre-session prep tasks**: • Ensure that PCs have **minimum 2.93 GHz Processor and 4 GB RAM (minimum 2GB), and 100 GB hard disk space** • Ask the participants to refer to the **Supporting Documents_Day 1 folder** for the embedded files. • Software requirements: • Platform: Microsoft Windows • Operating System (OS): Microsoft Windows XP SP3 or higher • Browser: Internet Explorer 8 and / or above • Technologies used: Java/J2EE, JSP, XML, and Tomcat • Software Tools: Rational Quality Manager (RQM) 4.0.3.1 for Test Management • Database: MySQL • Testing: Manual and RFT for Automation	

Slide Content	Instructor Guide	Use this space for your notes
	What to cover: **Purpose:** To elaborate the elaborate the concepts of Reviews and Inspections **Instructions for the participant:** • **Divide yourselves into groups as instructed by the facilitator** • **Work within your group to assume any of the following roles:** • Tester • SME • Observer • **Refer to the Case Study by clicking the embedded doc or the Case Study handout provided to you.** • **Read the four use cases** • **Share your key takeaways from the role play with the class (30 mins)** **Debrief:** • Discuss the key factors required to communicate with the SME in terms of written and verbal.	

Slide Content	Instructor Guide	Use this space for your notes
	• Explain the correct method based on any errors the participants may have introduced to the process of recording the defects / review comments.	
Slide 222  		

Slide Content	Instructor Guide	Use this space for your notes
Slide 223  		

Slide Content	Instructor Guide	Use this space for your notes
Slide 224  		

Slide Content	Instructor Guide	Use this space for your notes
Slide 225 	**Approximate Duration:** 5 mins **What to cover:** • Ask the participants if they have any questions. • Include any questions that will be addressed later in the course as parking lot items • Use this activity to recap the key takeaways from this module	